

Virtuoso® Layout Pro: T7 Module Generator and Floorplanner

Course Version IC 25.1

Lab Manual

Revision 1.0

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Module 1: About This Course

There are no labs in this module.

Module 2: Module Generator

Lab 2-1 Creating a Modgen from Schematic

Objective: To explore the methods to create a Modgen.

Introduction

Module generators are designed to provide a way to generate multiple Pcell instances into a complex, highly matched and structured array. With the Modgen tool, you specify the devices to be arrayed, then specify an interdigitation pattern, insert body contacts, dummy devices, and guard rings. Finally, you control the routing style and generate internal routing geometry.

Module generators can be created from either the schematic or the layout. When you create a Modgen from the schematic, the tool creates a modgen constraint on the schematic that will then be used when the layout is generated using *Generate All From Source* or *Generate Selected From Source*.

This course steps through the MODGEN flow. The objective of this flow is to increase the layout productivity through improved MODGEN functionality and take you through the steps to create a Modgen using different methods.

You will also manipulate the modgen, change its pattern, add features such as body contacts, dummies, guard ring and back annotate dummies.

Finally, you will route it using the Structured Router on a generic 45nm PDK.

In this lab, you:

- ◆ Explore the various methods to create a Modgen.

Opening the Software

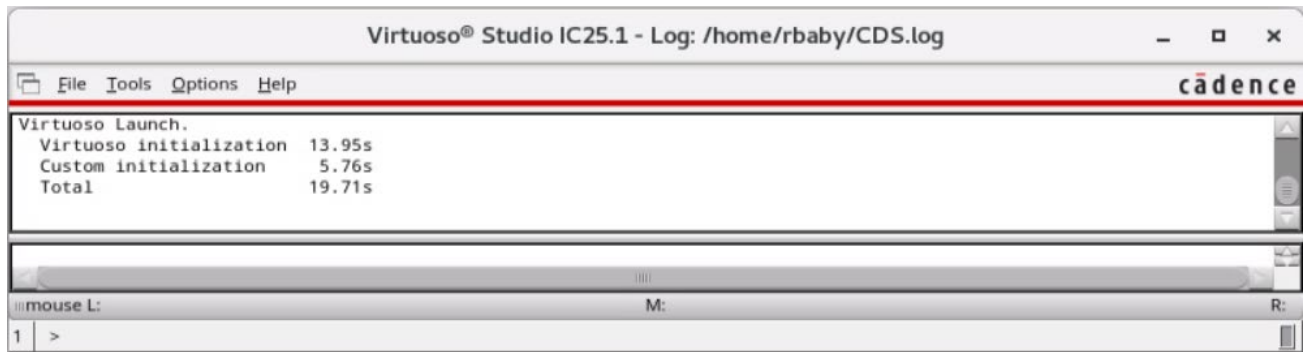
1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in.

```
cd ./VLPT7_IC_25_1/mod02_MODGEN
```

2. To start Cadence® Virtuoso® software, enter this command in the same terminal window.

```
virtuoso &
```

The Command Interpreter Window (CIW) appears.



3. In the **CIW**, choose **File – Open** menu command and enter these values in the form.

Library	ether_adc45n
Cell	adc_cascode_opamp
View	schematic

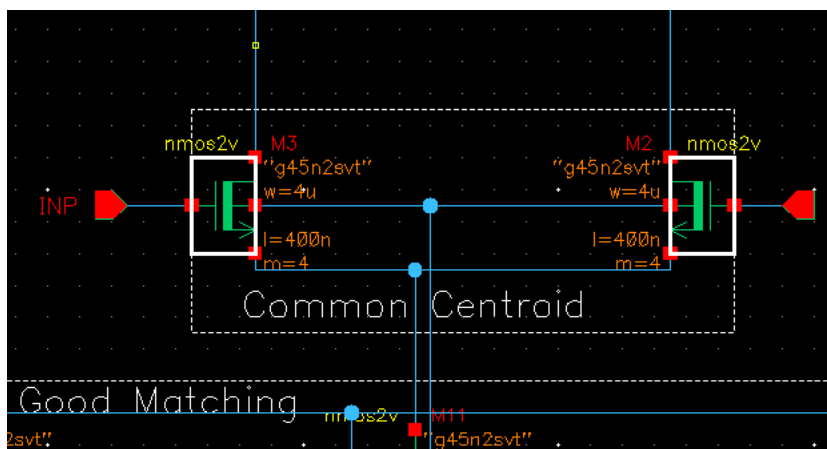
ether_adc45n/adc_cascode_opamp/schematic cellview opens in XL mode for editing.

4. In the **Workspace Configuration**, change the workspace to *Constraints*.

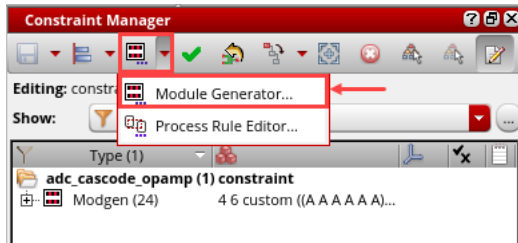
Navigator, Property Editor, and Constraint Manager (CM) assistants are displayed in the schematic window.

Creating a Modgen by Selecting the Instances in the Schematic Canvas

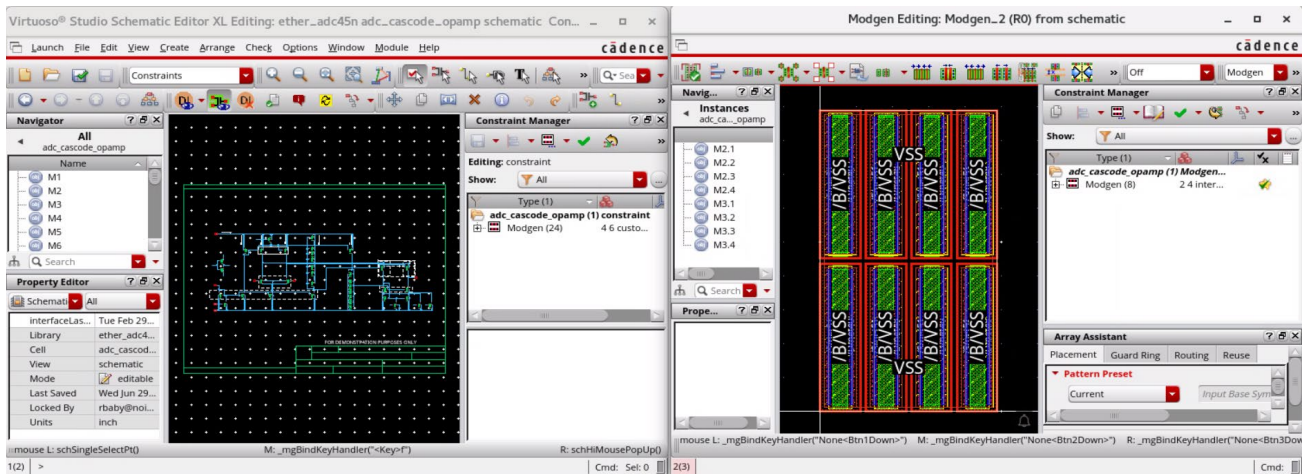
1. Select the instances **M2** and **M3** (use **SHIFT** with the left mouse button) in the Schematic Canvas, as shown here.



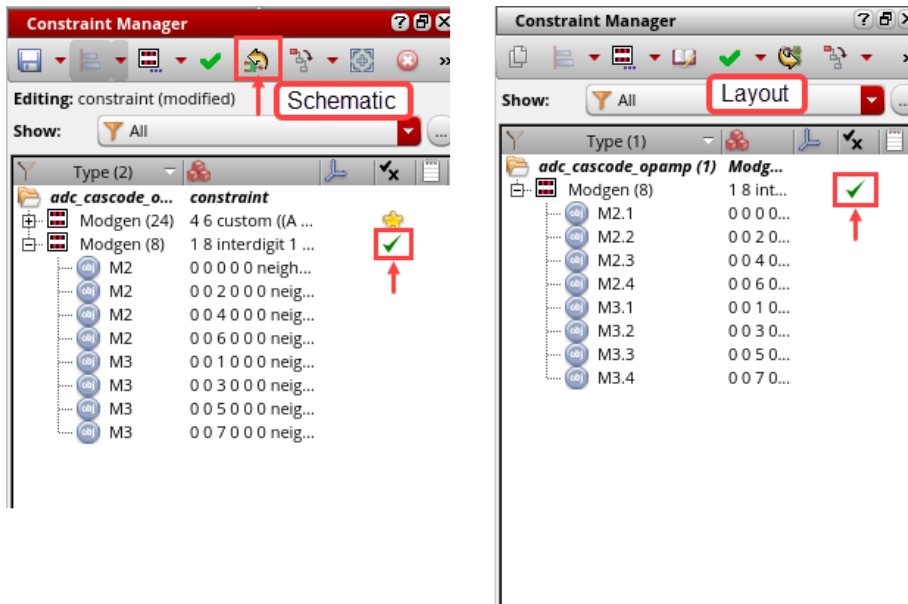
- Click the **Module Generator** icon in the *Modgen* constraint menu, as shown here, to invoke the Modgen Editing window.



The Modgen window appears, as shown here.



- Select the created constraint in Modgen and click the **Update schematic constraints from layout** button in the schematic, as shown here, to update the schematic constraints with layout constraints.



The schematic and layout constraints' **Status** shows as *passed*, as shown here.

Constraint Manager

Editing: constraint (modified)

Show: All

Type (2)

adc_cascode_opamp (2) constraint

Modgen (24) 4 6 custom ({...}

Modgen (8) 2 4 interdigit ...

M2 0 0 0 0 0 neig...

M2 0 0 2 0 0 0 nei...

M2 0 0 1 0 0 0 nei...

M2 0 0 1 2 0 0 0 n...

M3 0 0 1 0 0 0 nei...

M3 0 0 3 0 0 0 nei...

M3 0 0 1 1 0 0 0 n...

M3 0 0 1 3 0 0 0 n...

Constraint (1)

All Parameters

Name	Modgen_2
Owner	ether_adc45n.adc_cascode_opamp:const...
Enabled	true
Notes	
Nb of Rows	2
Nb of Cols	4
router	none
Merge Layers	default

Schematic

Constraint Manager

Show: All

Type (1)

adc_cascode_opam... ModgenTmplayou...

Modgen (8) 2 4 interdigit 1 def...

M2.1 0 0 0 0 0 neighbor...

M2.2 0 0 2 0 0 0 neighb...

M2.3 0 0 1 0 0 0 neighb...

M2.4 0 0 1 2 0 0 0 neigh...

M3.1 0 0 1 0 0 0 neighb...

M3.2 0 0 3 0 0 0 neighb...

M3.3 0 0 1 1 0 0 0 neigh...

M3.4 0 0 1 3 0 0 0 neigh...

Constraint (1)

All Parameters

Name	Modgen_2
Owner	ether_adc45n.adc_cascode_opamp:Modg...
Enabled	true
Status	passed
Notes	
Nb of Rows	2
Nb of Cols	4
router	none
Merge Layers	default

Layout

The Modgen contains **8** members, as shown here, since each instance (**M2** & **M3**) has a *Multiplier* factor of **4**.

Modgen Editing: Modgen_2 (R0) from schematic

cadence

Navigator

adc_cascode_opamp

Modgen_2

M2.1

M2.2

M2.3

M2.4

M3.1

M3.2

M3.3

M3.4

INN

INP

VSS

net40

net48

net51

Search

Property Editor

Constraint Manager

Show: All

Type (1)

adc_cascode_opamp (1) ModgenTmplayout3206797

Modgen (8) 2 4 interdigit 1 default

M2.1 0 0 0 0 0 neighbor 0 -1 0 -1 0 0 ...

M2.2 0 0 2 0 0 0 neighbor 0 -1 0 -1 0 ...

M2.3 0 0 1 0 0 0 neighbor 0 -1 0 -1 0 ...

M2.4 0 0 1 2 0 0 0 neighbor 0 -1 0 -1 ...

M3.1 0 0 1 0 0 0 neighbor 0 -1 0 -1 0 ...

M3.2 0 0 3 0 0 0 neighbor 0 -1 0 -1 0 ...

M3.3 0 0 1 1 0 0 0 neighbor 0 -1 0 -1 ...

M3.4 0 0 1 3 0 0 0 neighbor 0 -1 0 -1 ...

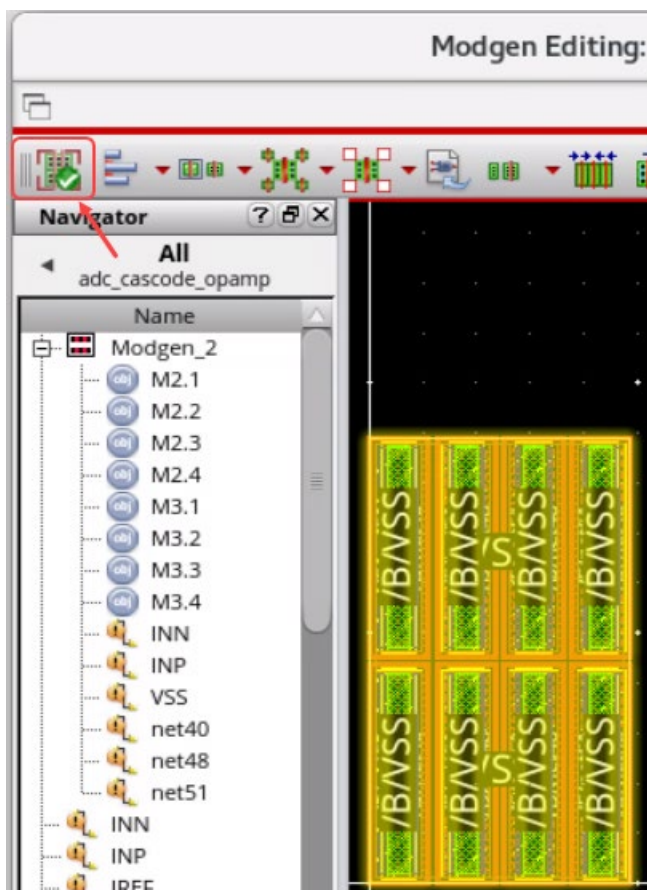
Constraint (1)

All Parameters

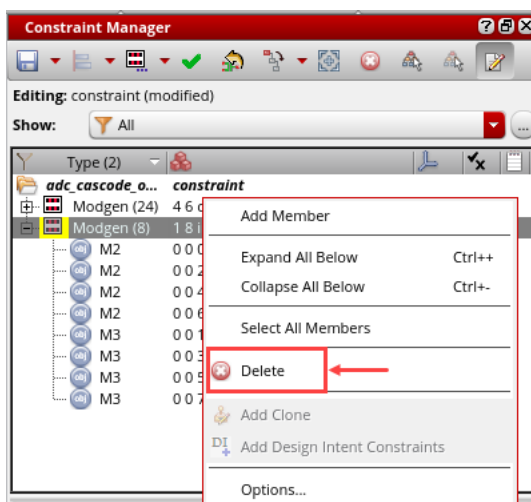
Name	Modgen_2
Owner	ether_adc45n.adc_cascode_opamp:ModgenTmplayout32...
Enabled	true
Status	passed
Notes	
Nb of Rows	2
Nb of Cols	4
router	none
Merge Layers	default

Constraint Editor

4. To exit, click the **Exit** icon in the Modgen window, as shown here.



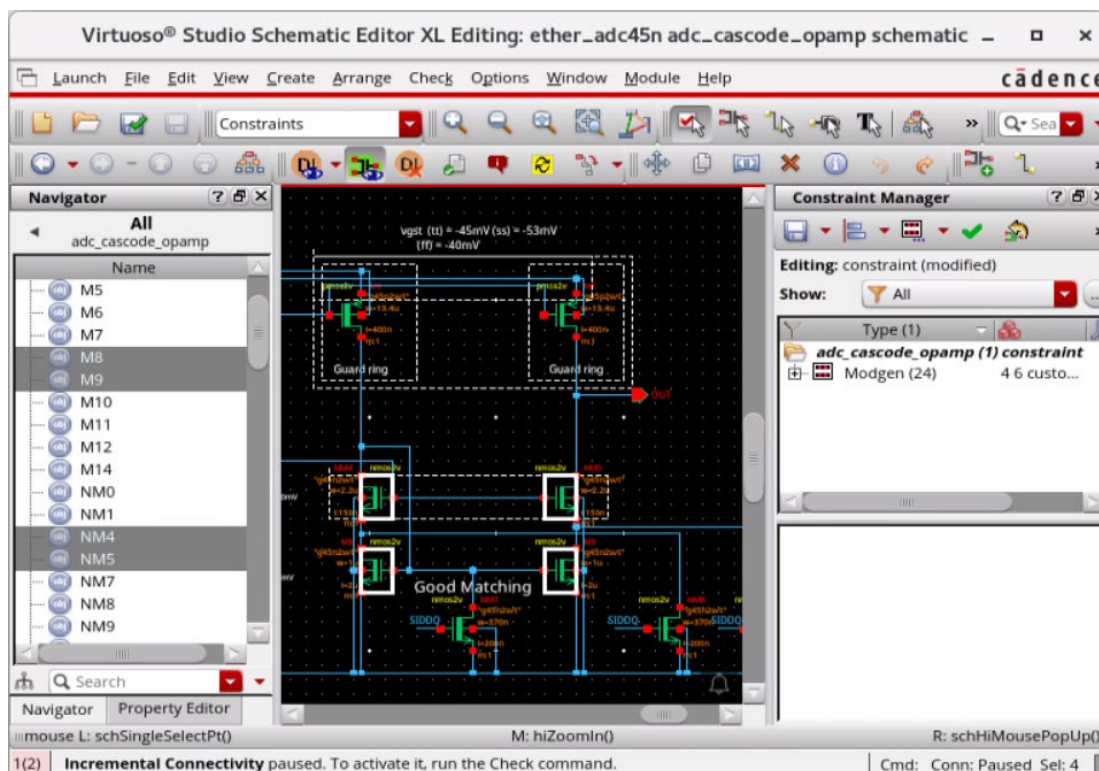
5. To delete the Modgen constraint, select the *constraint*, **right-click**, and select **Delete** in the schematic window, as shown here.



The newly created Modgen constraint is deleted.

Creating a Modgen by Selecting the Instances in the Navigator Assistant

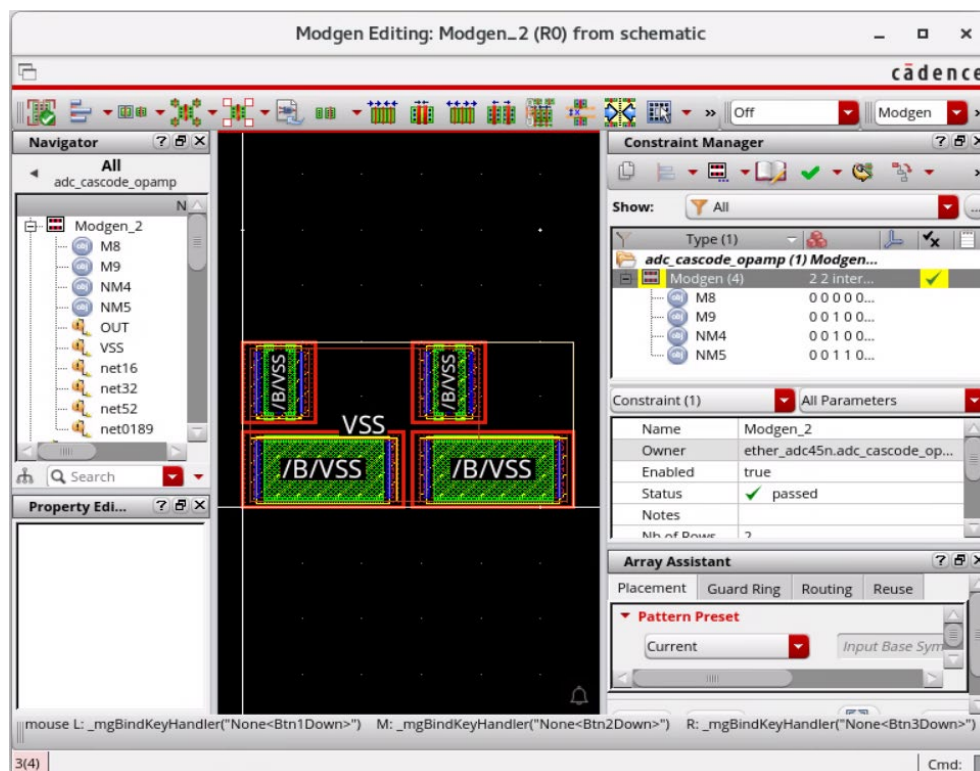
1. Select the instances **M8**, **M9**, **NM4**, **NM5** (use **SHIFT** or **Ctrl** with the left mouse button) in the **Navigator** assistant.



The other steps to create the Modgen constraint are identical to the ones described in the previous method: **A. Creating a Modgen by Selecting the Instances in the Schematic Canvas**.

2. Follow the procedure described in **Steps 2–5** in the previous method to explore how to create the Modgen and delete the Modgen constraint.

You can observe that it contains 4 members since all 4 instances have a *Multiplier* factor of 1.

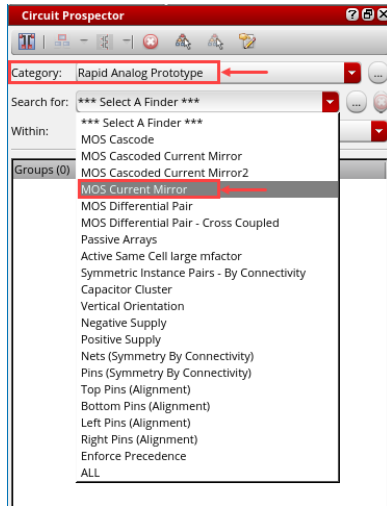


Creating a Modgen Using the Circuit Prospector Assistant for Special Configurations

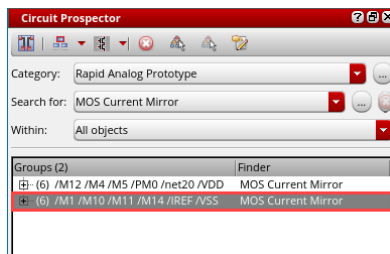
Another method consists of using the *Circuit Prospector*. It helps you find special configurations which are available through the *RAP* flow (*Rapid Analog Prototype*). You can also define your own structures.

1. Choose **Constraint-Helper** workspace in the schematic to launch the *Circuit Prospector* assistant.

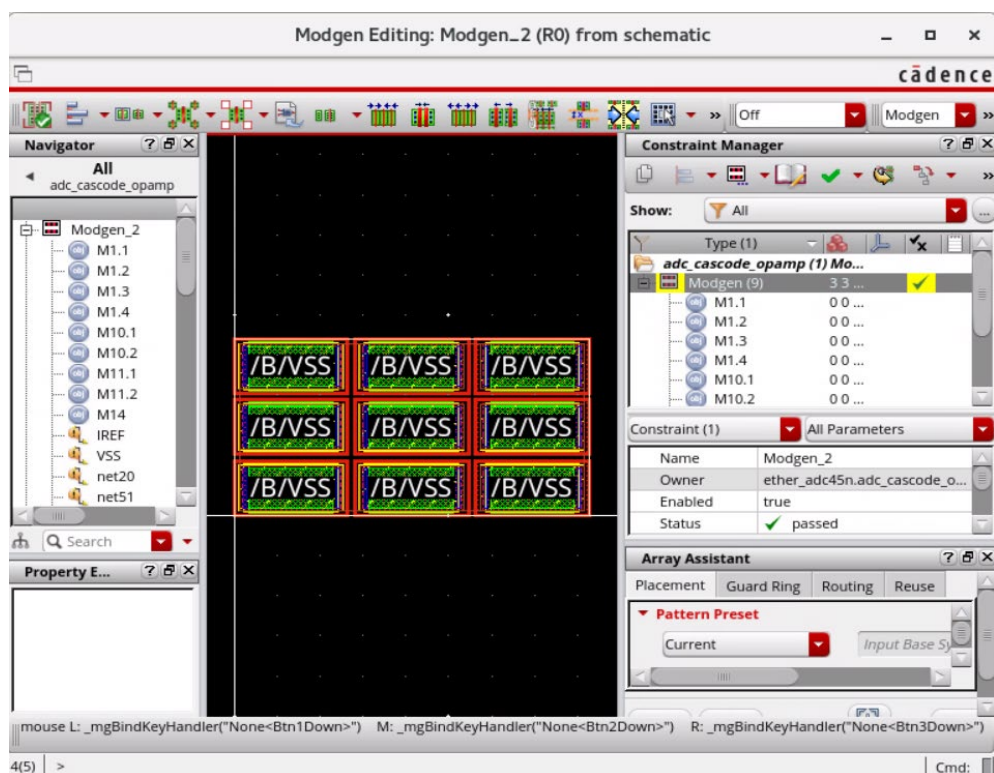
2. In the Circuit Prospector, choose **Category** as *Rapid Analog Prototype*, and in the list of available *Finders*, search for **MOS Current Mirror**.



3. Using the methods, you have learned in the previous sections – **A & B** – create a Modgen constraint for the **second MOS Current Mirror** shown here.



You will obtain the following result.



It contains 9 members since each instance (M1=4, M10=2, M11=2, M14=1) has a *Multiplier* factor, as given here.

- Keep the cellviews open (schematic and modgen) and the Virtuoso session running and continue with the next lab.

Summary

In this lab, you learned:

- ◆ The various methods to create a Modgen.



Lab 2-2 Editing a Modgen

Objective: To modify the default Modgen settings.

In this lab, you:

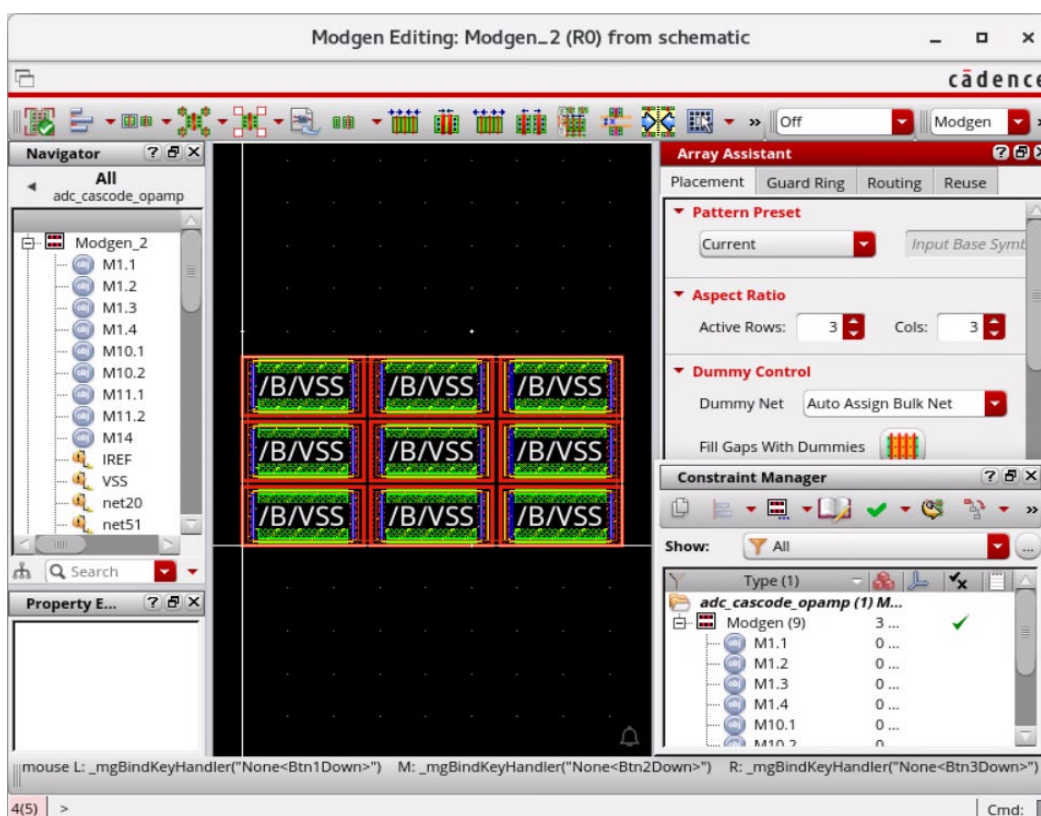
- ♦ Modify the default Modgen settings.

From the previous lab, the cellviews (schematic and modgen) should be opened. If not, open them now.

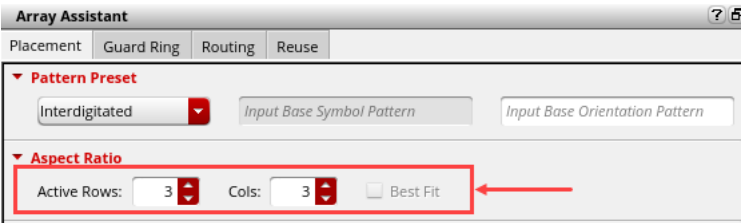
Modifying the Pattern

You will see how to change the pattern.

1. From the Modgen Editing window, choose the **Window > Assistants > Array Assistant** to invoke the Array Assistant if not open already.



The Modgen is created, as shown here.



Also, the instances have been automatically mapped to letters, as shown here.

The screenshot shows the 'Pattern Symbol Mapping' table. The table has columns: Sym, Name, Type, # avail, and # in use. The rows are as follows:

	Sym	Name	Type	# avail	# in use
0	*	*	dummy	-	0
1	A	M1	inst	4	4
2	B	M10	inst	2	2
3	C	M11	inst	2	2
4	D	M14	inst	1	1

Note: To create a grid of specific dimensions, go to the Aspect Ratio section and specify the number of Active Rows and Columns. The engine will automatically adjust the number of columns to fit the instances and create a suitable grid. When you place devices on the grid, they will be automatically interdigitated.

2. The Modgen feature allows you to set patterns for device placement. You can use preset patterns, specify base symbol patterns, or define a base orientation pattern. The available pattern preset are:

- Current
- Clustered
- Interdigitated
- Compact
- Custom
- Zigzag
- Shift
- Common Centroid

Note: For this pattern preset, we used symbols A, B, C, and D as the input base pattern. The table below shows the number of instances available.

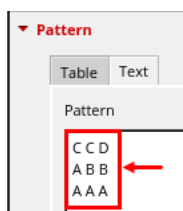
The screenshot shows the 'Pattern Symbol Mapping' table. The table has columns: Sym, Name, Type, # avail, and # in use. The rows are as follows:

	Sym	Name	Type	# avail	# in use
0	*	*	dummy	-	0
1	A	M1	inst	4	4
2	B	M10	inst	2	2
3	C	M11	inst	2	2
4	D	M14	inst	1	1

- a. **Current** Uses the current Modgen pattern.

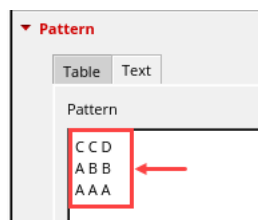
- b. **Clustered** displays a pattern based on the bottom-up approach, with a row-based split.

The Pattern -Text automatically updated as shown below.



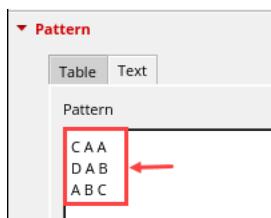
- c. **Interdigitated** Applies an interdigitation of 1 to the devices.
- d. **Compact** Adjusts the pattern to achieve the maximum abutment of devices.

The Pattern -Text automatically updated as shown below.



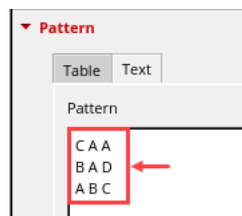
- e. The repeat pattern of **Custom** patterns in each row starts from left to right.

In the example, a custom pattern A B C D is applied. The Pattern -Text automatically updated as shown below.



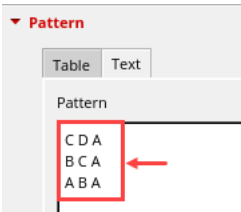
- f. To alternate the array rows, **Zigzag** applies the base symbol pattern in the opposite horizontal direction, starting with left-to-right for row index 0.

In the example, a zigzag pattern A B C D is applied. The pattern starts from left to right for odd rows and right to left for even rows. The Pattern -Text automatically updated as shown below.



- g. **Shift** Shifts the pattern by one position for each consecutive row, starting from the bottom-most row.

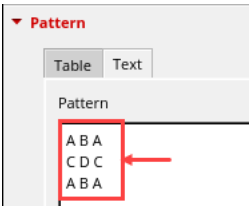
For example, symbol pattern A B C D is generated as:



- h. **Common Centroid** Places devices in a grid following a common centroid pattern. Symbols are ordered in precedence, outward from the center.

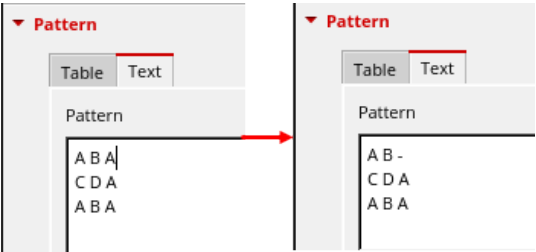
The priority order for placement follows the sequence of available instances starting from the center of the array and moving outward in a circular arrangement. The priority is determined by symbols in ascending order by their number of instances. Symbols with the same number of instances are in ascending order alphabetically.

The Pattern -Text automatically updated as shown below.

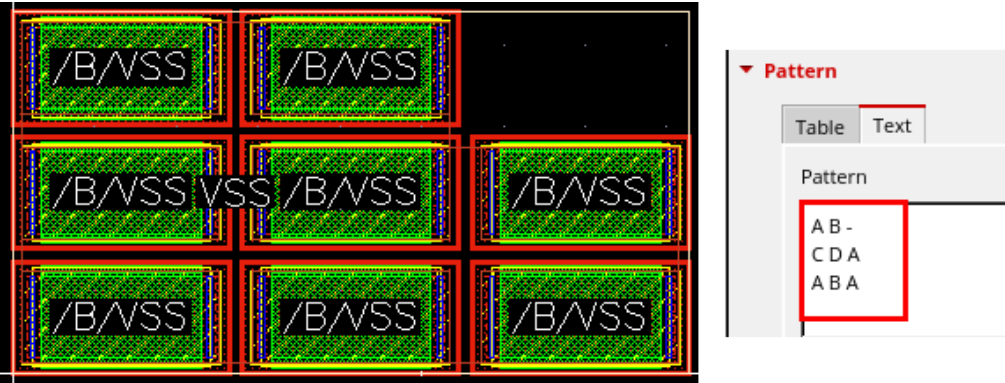


3. Enter letters manually in the field. If the right number of instances is not implemented, then it will automatically update to the right number of instances or dummy will be added (-).

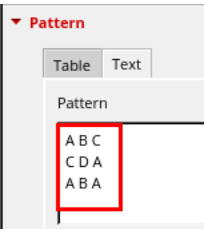
Enter the following pattern and click **Apply**.



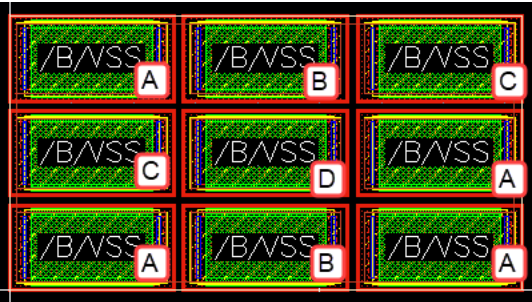
The number of placed A's does not match the number of layout instances. Therefore, the one A is replaced with the dummy.



4. To remove the dummy, replace A with C and then click on Apply.



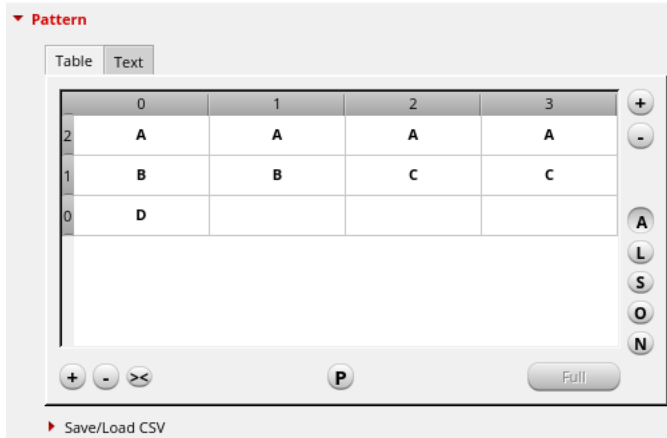
The result looks as in the graphic, as shown here.



Abutting the Devices

You will see how to automatically abut the instances. You would like to abut the instances **A**, **B**, **C** to reduce the Modgen footprint.

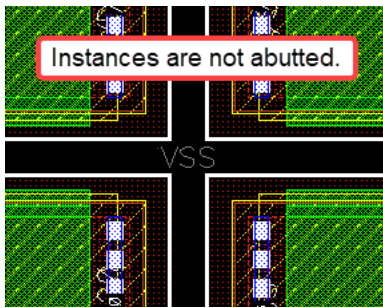
1. Before abutting the instances, first arrange the instances as shown here.



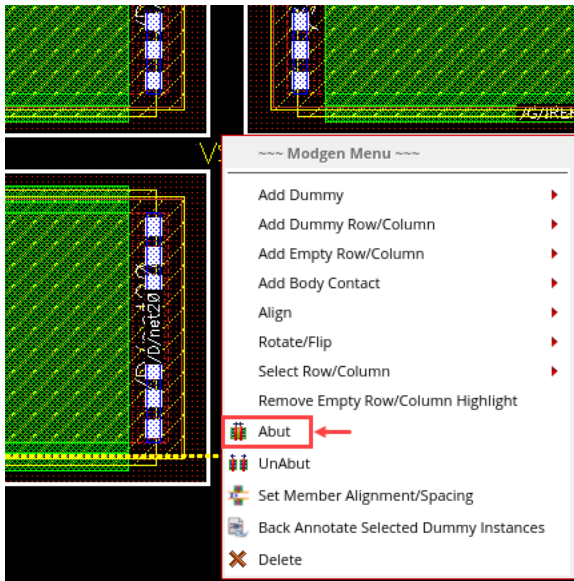
2. Select the cells **A**, **B**, and **C** using window select.
3. You will add the **Dimming** option. For this, **right-click** on the *Modgen Editing Window Banner*, add **Toolbars – Options**, and click on **Dim**.



4. Zoom inside the array to view that the instances are not abutted.



5. Click the **Abut** icon  in the *Modgen Placement* toolbar or **right-click** on the *Modgen Editing Window Banner* and click the **Abut** option from the *Modgen Menu* with the cells selected.



The result is the following, the selected instances are abutted.



- Zoom inside the array to see that the selected instances have been abutted.

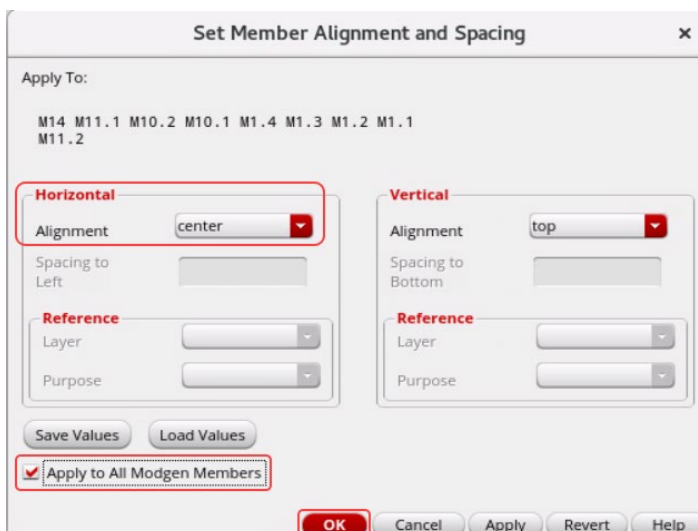


- Press **Ctrl+D** to deselect all the instances.
- Remove the **Dimming** and **Options** menu that you added in **Step 2**.

Aligning the Devices

You will see how to change instance alignment in Modgen. You will try to align the center of instance **D** in the bottom row with the center of instances **A**, **B**, and **C** in the row above. You will use the **Member Alignment/Spacing** command.

- Click the **Member Alignment/Spacing** icon  in the *Modgen Placement* toolbar or **right-click** on the *Modgen Editing Window Banner* and click the **Set Member Alignment/Spacing** option from the *Modgen Menu*.
- Select the following options in the Set Member Alignment and Spacing form.



- Select **Alignment** as *center* in the **Horizontal** section.
- Select **Apply to All Modgen Members** radio button.
- Click **OK**.

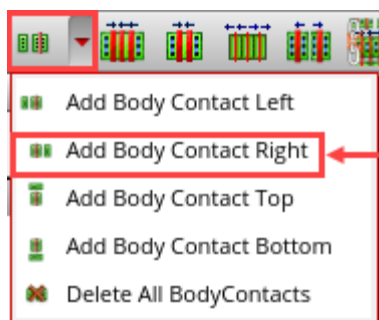
The result is the following alignment. All members are aligned according to the *center* axis.



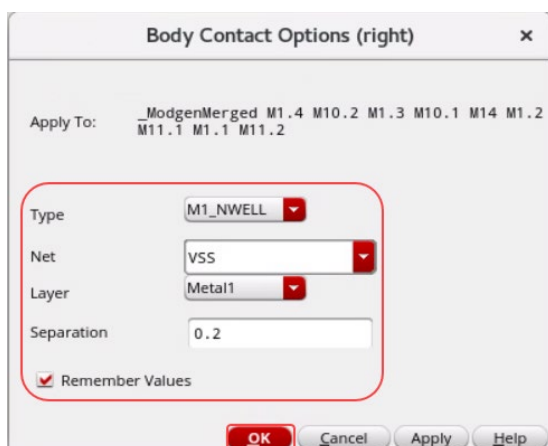
Adding the Body Contacts and Dummies

You will see how to add the body contacts and dummies at a specified location.

- Select all Modgen instances and click the **Body Contacts** icon  in the *Modgen Placement* toolbar or access this command from the *Modgen Menu*.
- Click **Add Body Contact Right** in the pull-down menu that appears.



The Body Contact Options (right) form is displayed.



The dialog box titled "Body Contact Options (right)" contains the following fields and controls:

- Apply To:** A list of module names: _ModgenMerged M1.4 M10.2 M1.3 M10.1 M14 M1.2 M11.1 M1.1 M11.2
- Type:** A dropdown menu set to "M1_NWELL".
- Net:** A dropdown menu set to "VSS".
- Layer:** A dropdown menu set to "Metal1".
- Separation:** A text input field containing "0.2".
- Remember Values:** A checked checkbox.
- Buttons:** OK, Cancel, Apply, and Help.

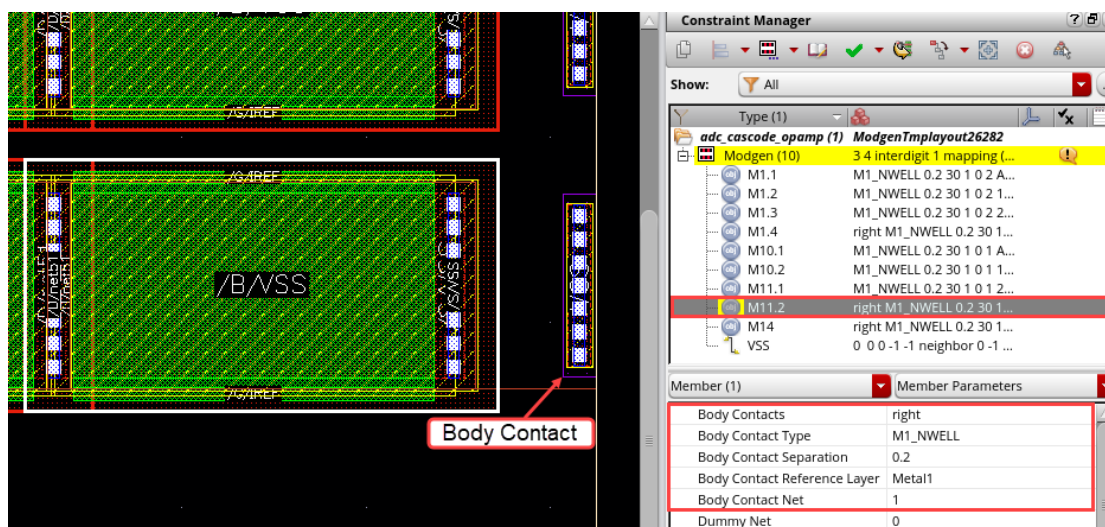
3. Select the following in the Body Contact Options (right) form.

- Type** as *M1_NWELL* (Body Contact from techfile)
- Net** as *VSS* (Body Contact connection)
- Layer** as *Metal1* (Reference Layer)
- Separation** as *0.2* (Spacing)
- Select **Remember Values**
- Click **OK**

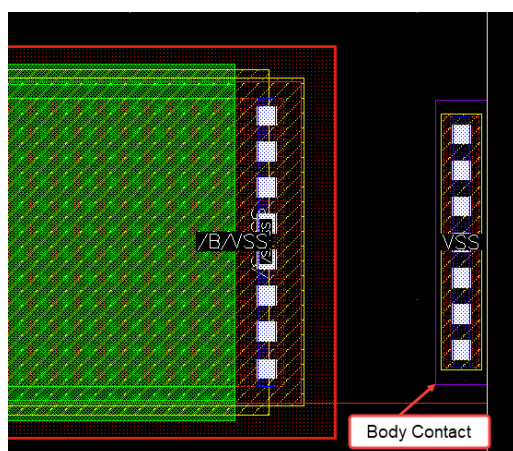
The result is the following. Body Contacts are placed on the right of every member, except the abutted ones.



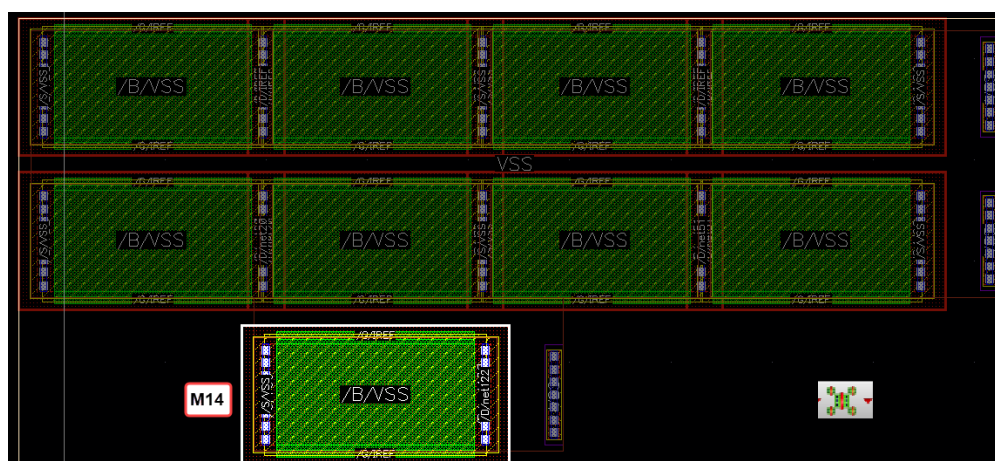
4. Zoom in to the **bottom-right** and select the instance **M11.2** and observe the result.



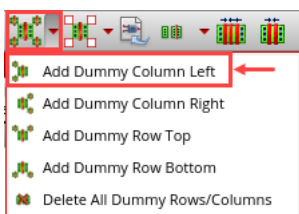
5. Zoom again to see the **Body Contact**.



6. Select the **bottom row** member **M14** and click the **Add Dummy Row/Column** icon in the *Modgen Placement* toolbar or access this command from the *Modgen Menu*.



7. Click **Add Dummy Column Left** in the pull-down menu that appears.



The Dummy Options (left) form is displayed.

8. Select the following in the *Dummy Options (left)* form, as shown here.

A screenshot of the 'Dummy Options (left)' dialog box. The dialog has a title bar 'Dummy Options (left)' and a close button 'x'. Inside, there's a section 'Apply To:' with 'M14' selected. Below that is the 'Dummy Net' section, which has a 'Default' dropdown menu set to 'VSS' (highlighted with a red box) and a checkbox 'Set Individual Terminal Net' which is unchecked. There are four rows of input fields labeled 'B', 'D', 'G', and 'S', each with a dropdown menu set to 'VSS'. Below that is the 'Dummy Type' section, with a 'Type' dropdown set to 'neighbor' and a checkbox 'Specify Parameters' which is unchecked. Under 'Dummy Parameters', there are three input fields: 'Number of Fingers' (set to 'CDF Default'), 'Length' (set to 'CDF Default'), and 'Width' (set to 'Same As Neighbor'). Below that is the 'Default Dummy' section, which has a 'Browse' button and three input fields: 'Dummy Library', 'Dummy Cell', and 'Dummy View'. At the bottom, there is a checkbox 'Remember Values' which is unchecked, and four buttons: 'OK' (highlighted with a red box), 'Cancel', 'Apply', and 'Help'.

- a. Select the net **VSS** in the **Dummy Net Default** cyclic field to connect Drain/Gate/Source.
- b. Keep the other values as defaults and click **OK**.

Dummies have been added to the left of the selected member, as shown here.



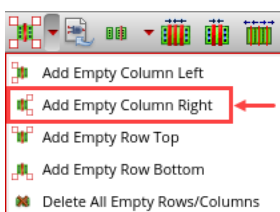
Adding/Removing the Empty Row/Column

You will see how to add an empty row/column.

1. Select the member **M10.2** of the **middle row** and click the **Add Empty Row/Column** icon in the *Modgen Placement* toolbar or **right-click** on the canvas and access this command from the *Modgen Menu*.



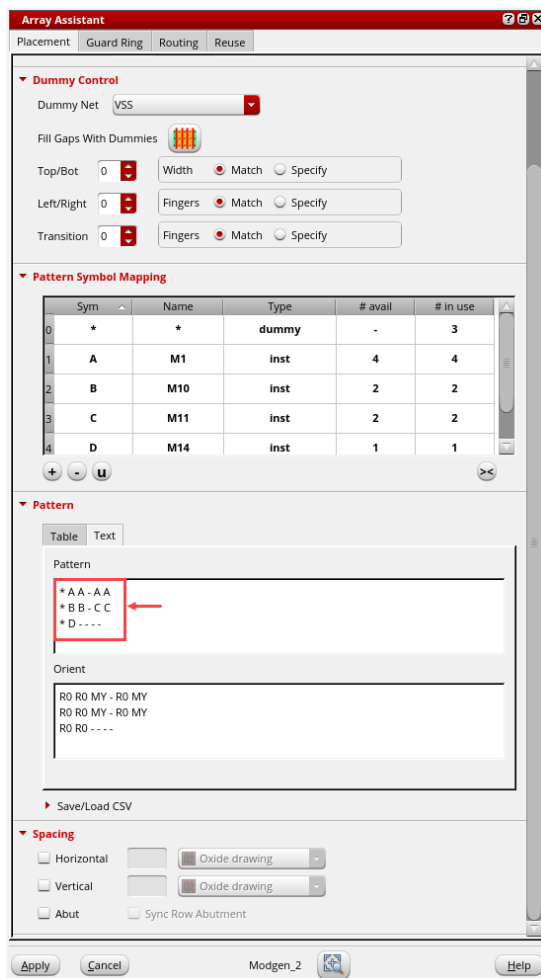
2. Click **Add Empty Column Right** in the pull-down menu that appears.



An empty column is created to the right, as shown here.



- From the Modgen Editing window, choose the **Window > Assistants > Array Assistant** to invoke the Array Assistant.



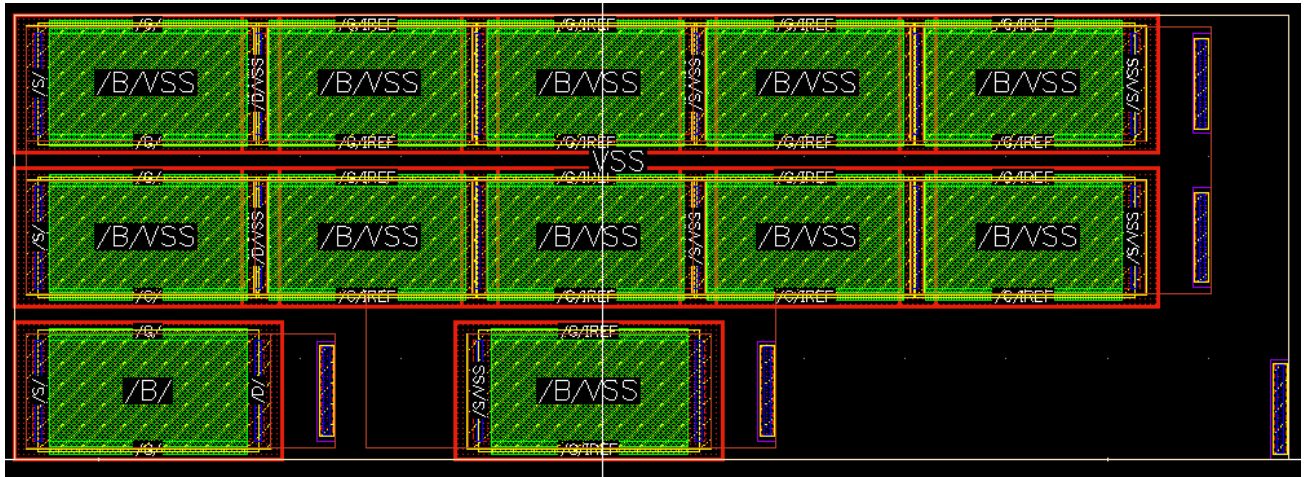
New symbols have been added to the pattern, as shown here.

- “_” for an empty member.
- “*” for a dummy.

- ▼ **Pattern**

Table	Text
Pattern	
* A A A A * B B C C * D - - -	

- The result is shown below. The empty vertical column has been removed.



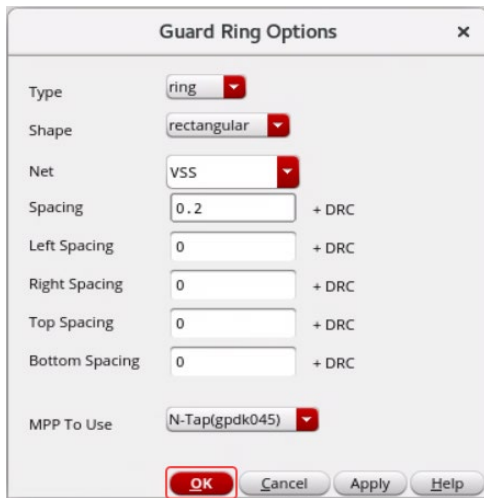
You will see how to add a guard ring.

- 

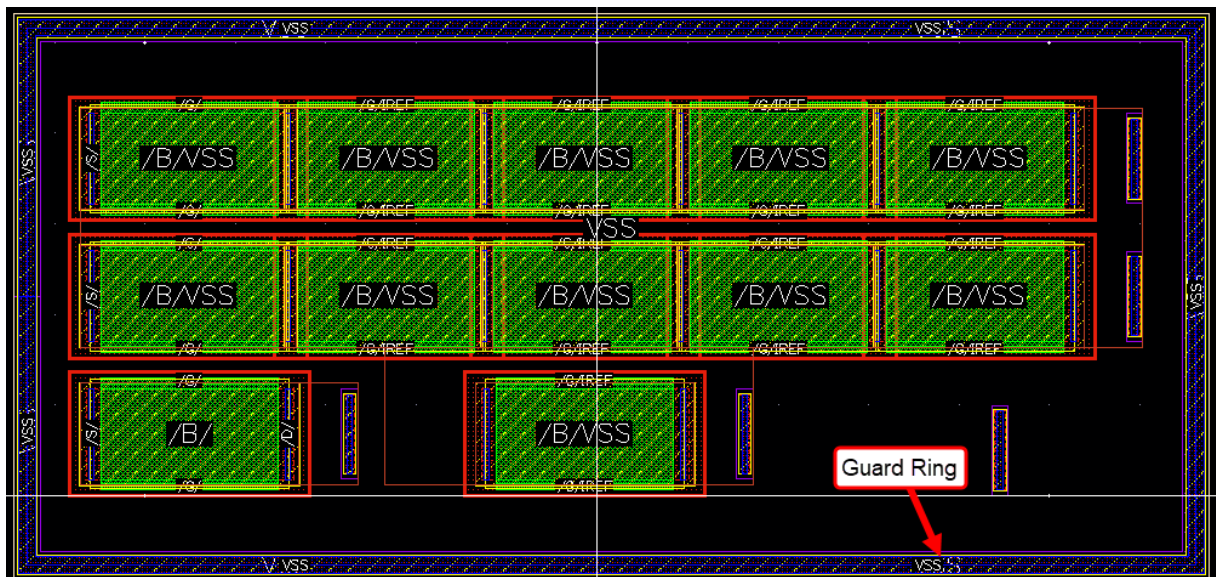
- 
- Guard Ring
- Create Fluid Guard Ring
 - Create MPP Guard Ring
 - Create Identical Guard Ring
 - Remove Guard Ring
 - Enable Fluid Guard Ring Hand Edit Mode

31

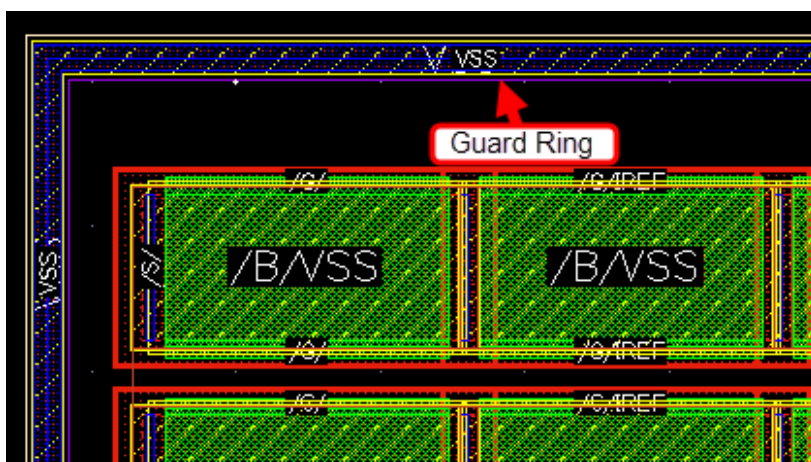
3. Select the options as shown here, and click **OK**.



The Guard Ring is created, as shown here.



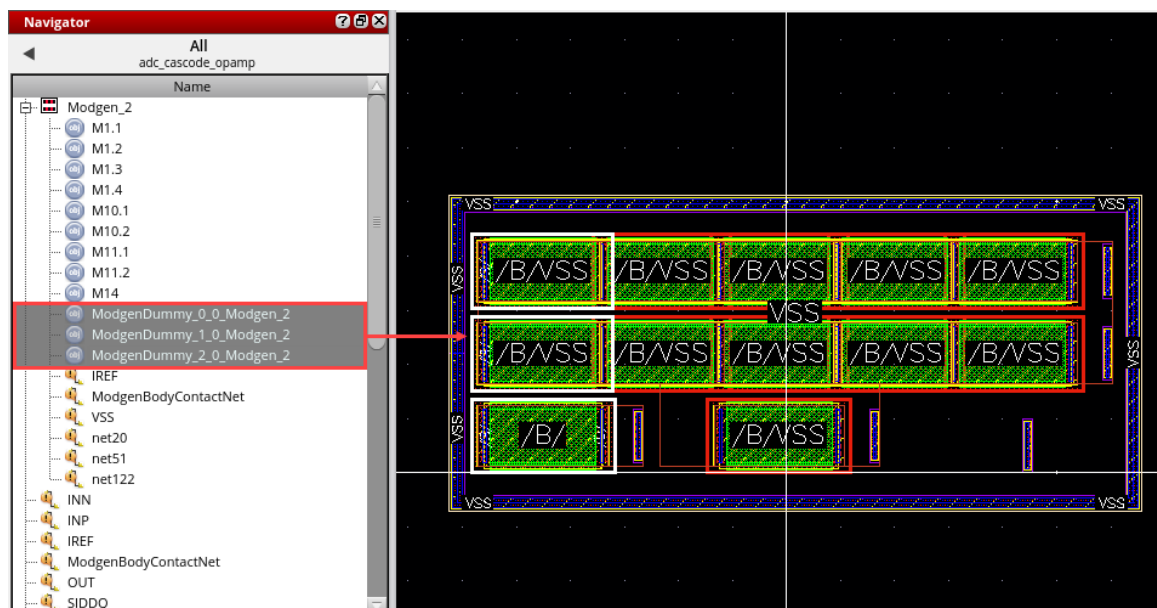
4. Zoom in to see the **Guard Ring**.



Backannotating the Dummy Devices

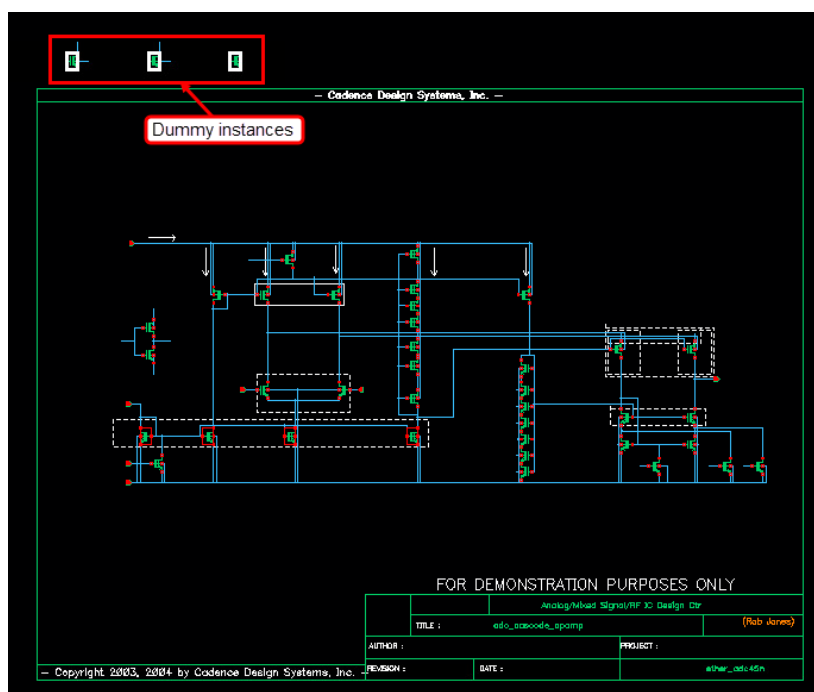
In this section, you will see how to backannotate the dummy instances in the schematic.

1. Select all the dummy members of the Modgen and click the **Backannotate All Active Dummy Instances**  icon from the *Modgen Placement* toolbar.

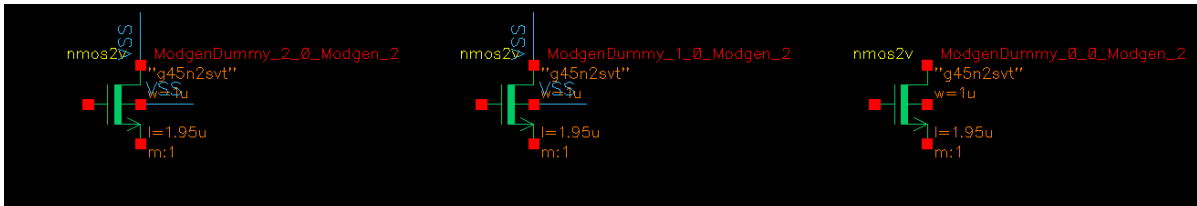


2. Click **OK** in the pop-up window, if any.

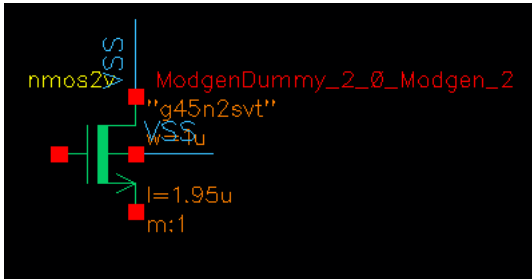
The dummy devices get created in the schematic window, as shown here.



- Zoom to the **top-left** to view the dummy instances.



- Zoom again to the left device.



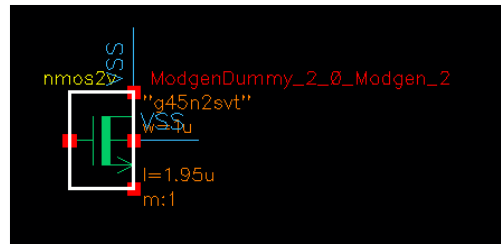
These dummy devices will be considered if you run a simulation.

You can see that Drain and Bulk are connected to the net **VSS**.

- Select the dummy instance and open the *Edit Object Properties* form by choosing the **Edit – Properties – Objects** menu command or use the bindkey **Q** and observe the values.

Property	Value	Display
Library Name	gpdk045	off
Cell Name	nmos2v	value
View Name	syml01	off
Instance Name	ModgenDummy_2_0_Modgen_2	off
Origin	(-2.8125 12.5)	

Pin Direction	Pin Name	Net Name
inputOutput	G	net1
inputOutput	D	VSS
inputOutput	B	VSS
inputOutput	S	net2



- Keep the cellviews open (schematic and modgen) and the Virtuoso session running and continue with the next lab.

Summary

In this lab, you learned:

- How to modify the default Modgen settings.



Lab 2-3 Routing a Modgen

Objective: To route a Modgen using the Structured Router.

In this lab, you:

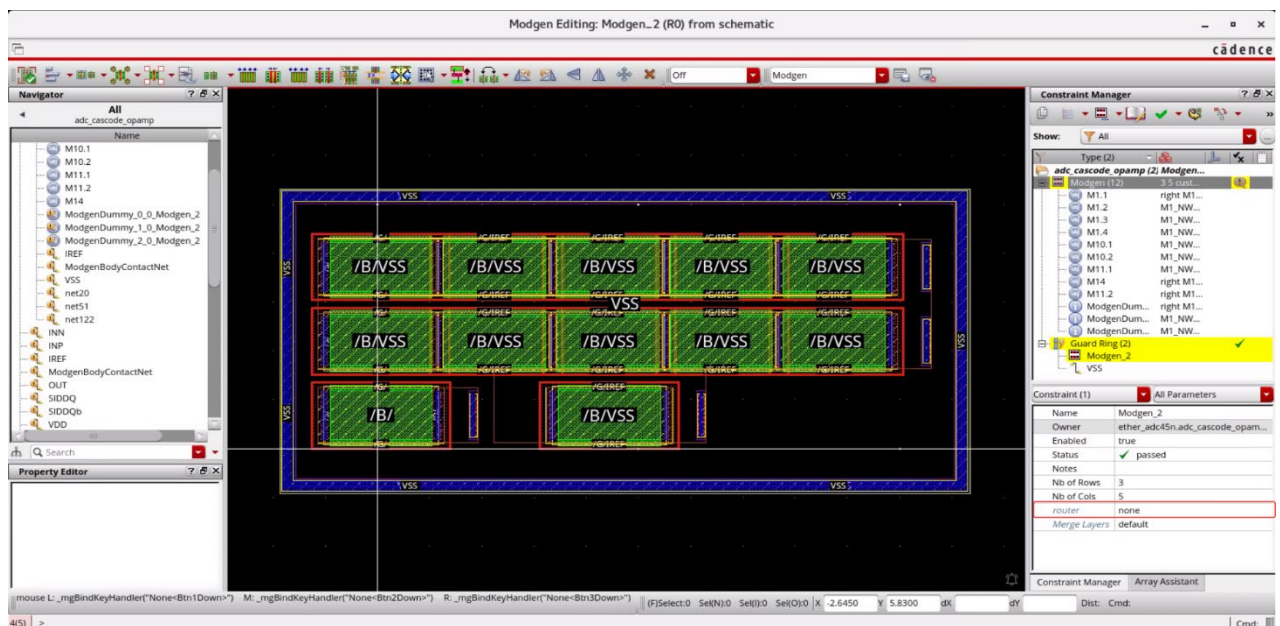
- ◆ Use the *Structured Router* from the *Constraint Parameter Editor* to route a Modgen.

From the previous lab, the cellviews (Schematic and Modgen) should be opened. If not, open them now.

Structured Router

The routing can be performed from the *Constraint Parameter Editor*.

1. In the **Constraint Manager** assistant, select the *Modgen Constraint* to be routed.

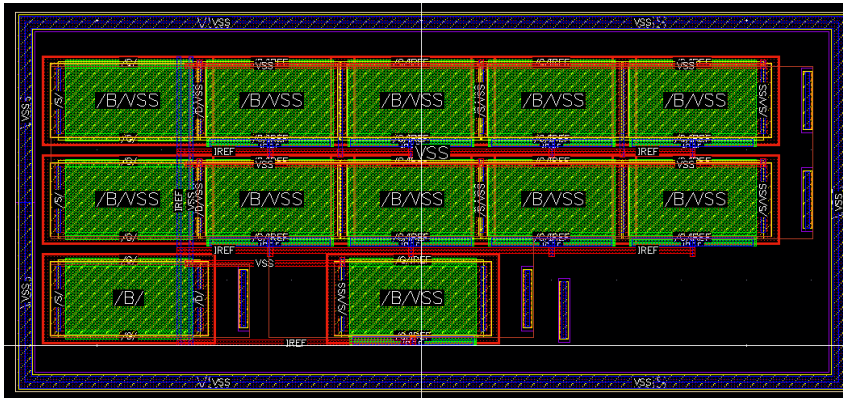


The following pull-down routing options are displayed in the *router* parameter in the *Constraint Parameter Editor*:

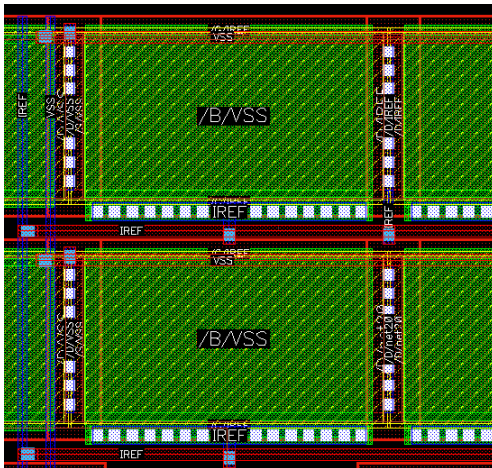


- Click *structured* from the *router* parameter to route the Modgen.

The Modgen is routed using the Structured Router in the design as shown here.



- Zoom to several locations to see the routed Modgen.



- Click *none* from the *router* parameter to delete the routing.
- Close all the cellviews without saving the changes and exit the Virtuoso session.

Summary

In this lab, you learned:

- How to route a Modgen using the Structured Router from the Constraint Editor.



Module 3: Modgen Enhancements

Lab 3-1 Using the Array Assistant

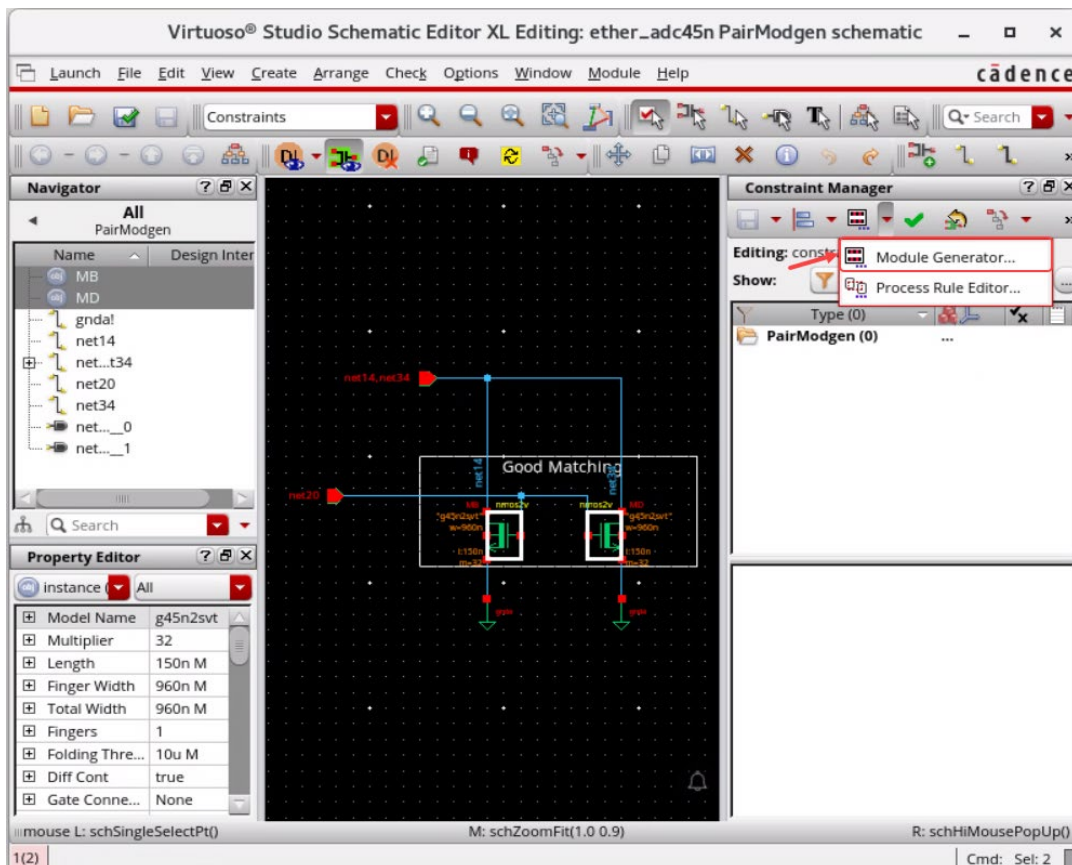
Objective: To explore the tabs in the Array Assistant.

In this lab, you:

- ◆ Explore the tabs introduced from the IC23.1 version, namely, Placement, Guard Ring, and Routing.
1. Start a new Virtuoso® session in the `./VLPT7_IC_25_1/mod03_MODGEN_Enhancements` directory.
 2. Use the Library Manager window to select and open the following in **XL** mode:

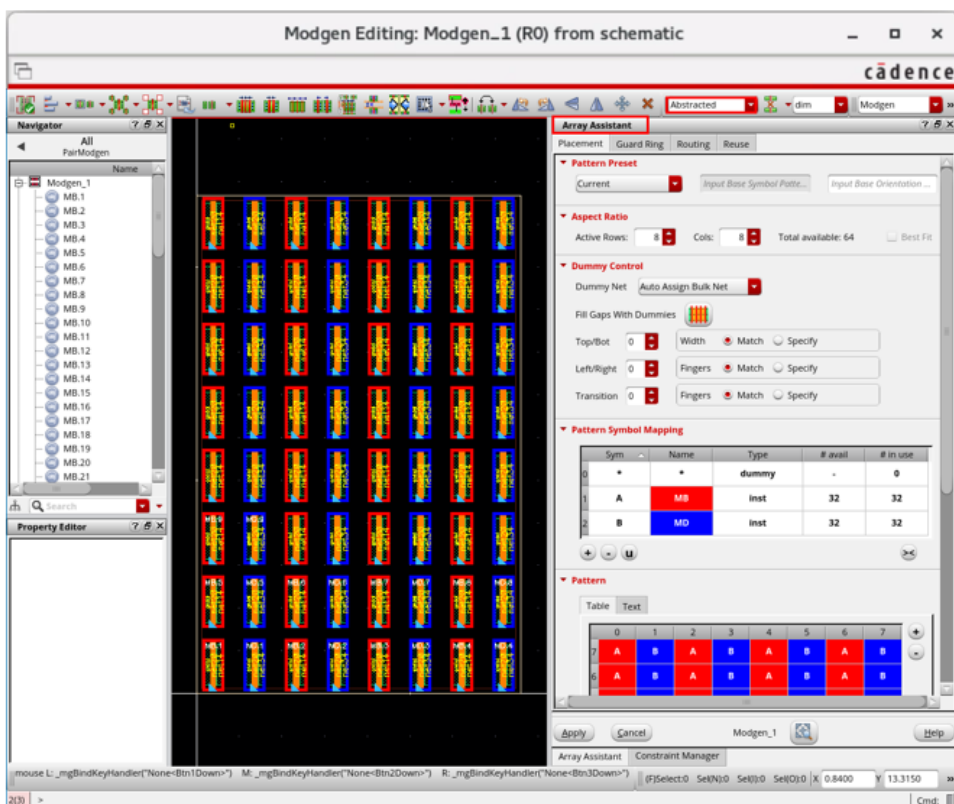
Library	ether_adc45n
Cell Name	PairModgen
View Name	schematic

3. Select the devices **MB** and **MD** and invoke the Module Generator, as shown here.



The Modgen window appears with the Array Assistant.

4. Set the *Smart Display Configuration* to **Abstracted** in the Modgen window.

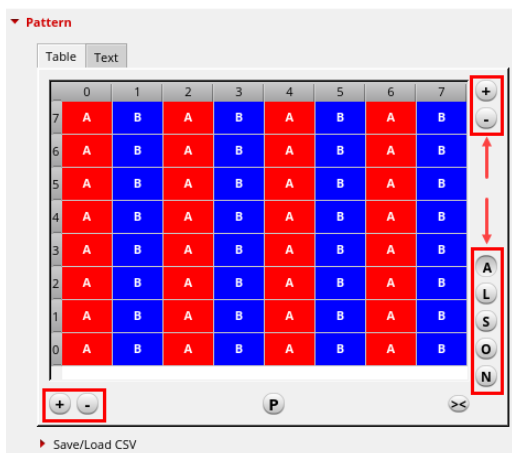


The Modgen pattern is displayed based on the specified preset, active rows, and dummies. This pattern is derived using the specified preset, active rows, and dummies.

Using the Placement Tab

1. Click on the **A/L/S/O/N** radio buttons to view the various names in the pattern.
The Modgen is created on **8** rows and **8** columns. You will change the number of **Rows/Columns** using the Array Assistant.
2. In the **Aspect Ratio**, just type **4** in the **Active Rows**. The assistant will calculate the required number of columns. The canvas and assistant pattern are updated. The array now will be **4 x 16**.
3. In the **Aspect Ratio**, just type **4** in the **Cols**. The array now will be **16 x 4**.
4. You can also type both rows and columns.

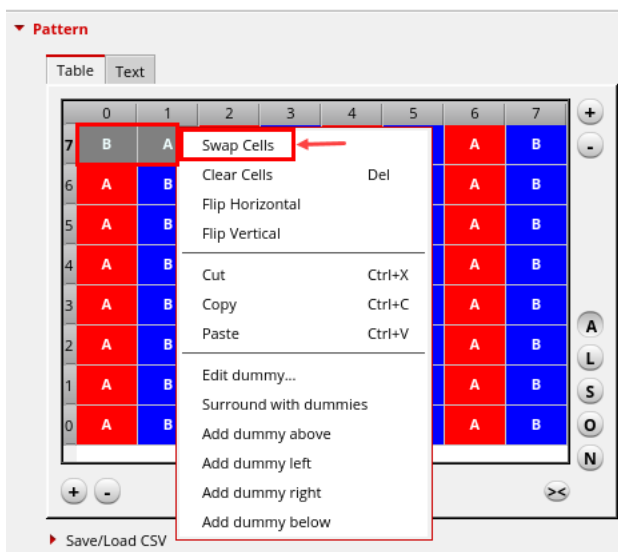
- You can add/delete **Rows** or **Columns** using the dedicated buttons on the right and bottom of the assistant.



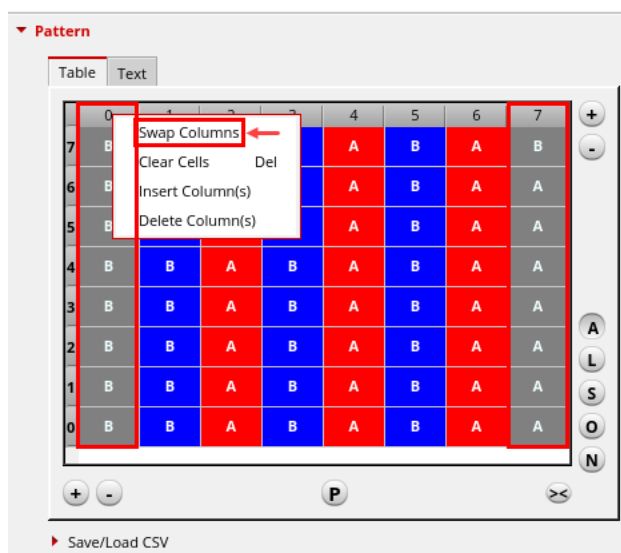
Interactive Modification Using the Placement Tab

The user can modify the devices' placement directly on the **Pattern**.

- If needed, stretch the **Pattern** to allow you to see the complete **8x8** array.
- Place your cursor on one of the cells in the array, **LMB**, to select it. Place the cursor on a different cell, **<Ctrl>LMB**, to select it. **RMB** will display a pop-up menu. Choose **Swap Cells** to swap between the two selected cells.



3. **LMB** click on a column number to select a complete column. **<Ctrl>LMB** on a different column number to select another column. **RMB** will display a pop-up menu. Choose **Swap Columns** to swap between two selected columns. You can swap between **Rows** in the same manner.

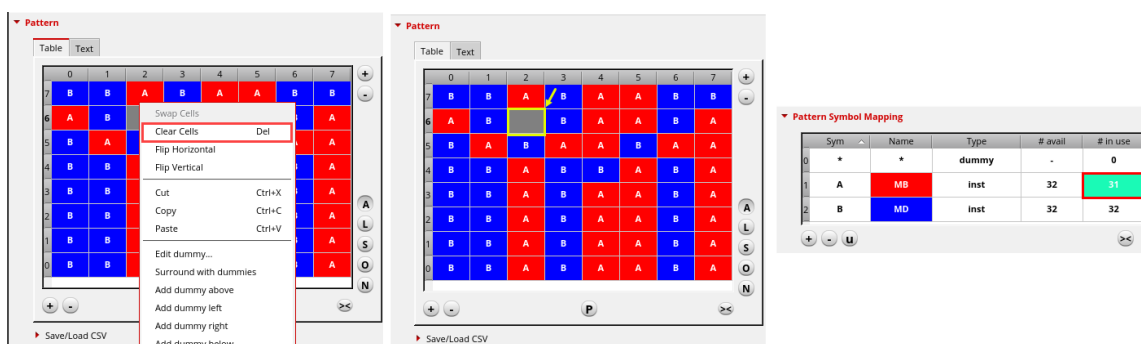


4. Place the cursor on one of the cells in column 0. **LMB** down and drag the cell to a different place with a different letter, release the **LMB**. You will see that you just have swapped between the cell you selected and the one that was in the slot.
5. **LMB** on a cell and **<Shift>LMB** on another cell. This will select all the cells between those two cells. **LMB** down on one of the selected cells and drag the selected set to a different place on the array, release the **LMB**. You just have swapped a group of cells.
6. **LMB** on a column/row number; this will select all the cells in the column/row. **LMB** on one of the selected cells and drag the column/row to overlap another column/row and drop.

As you just experience, select drag and drop will swap between elements.

7. Place the cursor on one of the cells in column 0. **LMB** down and drag the cell to a different place with a different letter and release the **LMB**. You will see that you just have placed the dragged cell in the slot. Cells around moved left and up to release the room for the dropped cell and fill up the slot where the dragged cell was initially. To really see the chain reaction, click first on the **L** button to display the unique layout name of each cell. The change is also reflected on the layout canvas.
8. **LMB** on a cell and **<Shift>LMB** on another cell. This will select all the cells between those two cells. **LMB** down on one of the selected cells and drag the selected set to a different place on the array and release the **LMB**. You just have inserted a group of cells.

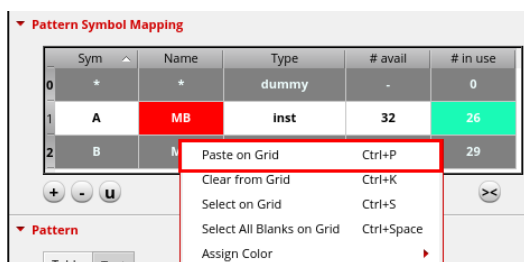
9. **LMB** on a column/row number to select the complete column/row. You can drag and insert the complete column/row.
10. Select a cell in the array. Choose the option **Clear Cells** in the pop-up menu. The cell in the array will be empty. Look in the **Pattern** assistant in the row that corresponds to the letter you selected. The **# in Use** has decreased by **one**, as shown here.



11. Select a row/column. Using the pop-up menu, clear all the cells in the row/column.

Assigning Device(s) to Empty Slot(s) Using the Pattern Symbol Mapping

1. **LMB** to select an empty slot in the Array Assistant. In the **Pattern Symbol Mapping**, **RMB** on a letter in **Sym** column for unplaced device and choose **Paste on Grid** to place that device in the selected slot in the Array Assistant.



2. You can select several slots in the Array Assistant, and using the same technique as in **Step 1**, fill them in one operation.
3. Keep all the cellviews open and the Virtuoso session running and continue to explore the use of the Guard Ring tab.

Using the Guard Ring Tab

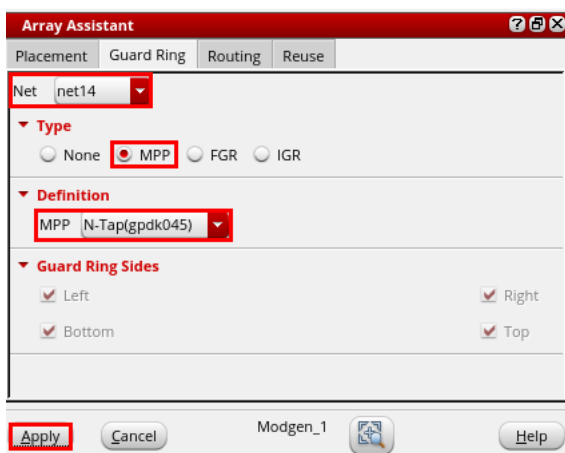
Guard rings enclose devices or device chains. Modgen tool creates MPP, fluid, and identical guard rings.



Creating the Multipart Path (MPP) Guard Ring

A multipart path is made up of a main path and one or more sub-parts. The main path is the primary object, and all sub-parts are related to and based on it. You have the option to define multiple sub-parts for a multipart path, which may include offset sub-paths, enclosure sub-paths, and sets of sub-rectangles. Additionally, you can specify connectivity for any or all the parts of a multipart path.

The tool supports only surround MPP guard rings. Although the sides are listed, you cannot control the sides along which the guard ring is added.



1. In **Type** field, select *MPP*.
2. Select **Net** as *net14*.
3. In the **Definition** field, choose an MPP for the guard ring, say *N-Tap(gpd045)*.
4. Click **Apply**.

MPP guard ring is created, as shown here.



Creating the Fluid Guard Ring (FGR)

With the Array Assistant, you have the option to create fluid guard rings (FGRs) around objects in addition to multipart path (MPP) guard rings. FGRs are a type of fluid Pcells, which can enclose one or more objects such as devices or device chains.



1. Select Create/Replace FGR.

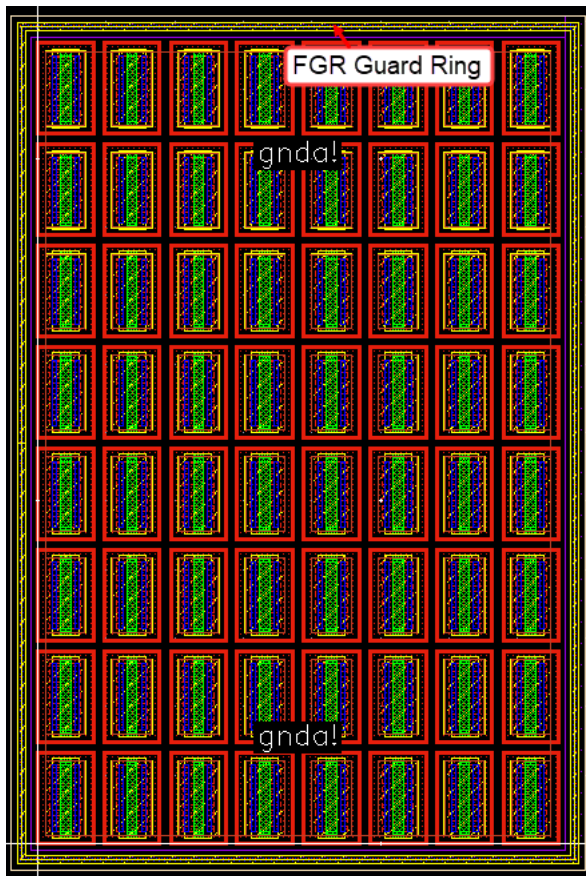
The Create Guard Ring form is displayed.

The screenshot shows the 'Create Guard Ring' dialog box with the 'Wrap' tab selected. The 'Device' dropdown is set to 'nring'. The 'Contact Rows' is 1, 'Path Width' is 0.12, and 'Net Name' is 'gndal'. The 'Main Layer' is 'Diffusion'. The 'Rectilinear' radio button is selected, and 'Wrap common' is checked. 'Place at Minimum Distance' is also checked, and 'Enclose by' is set to 0. The 'Contact Settings' sub-tab is active, showing 'Path Width' as the calculated parameter, 'Contact Spacing' of 0.08, and 'Contact Dimension' of 0.06. 'Match Contact Enclosures' is unchecked. The 'Minimum Diffusion Over Contact' and 'Minimum Metal Over Contact' are both set to W: 0.03 and L: 0.03. The 'Close' and 'Help' buttons are at the bottom right.

Note: FGRs can be created by drawing a path, a rectangle, or a polygon, or by using the wrap mode.

2. In the Wrap mode, an FGR is created around the objects you select. The FGR parameters are stored in the Modgen constraint.
 - a. Select nring in the device.
 - b. Remaining with the default setting, click on the Close button.
3. Click Apply in the Guard Ring tab.

FGR guard ring is created as shown here.



4. Close all the cellviews without saving the changes and the Virtuoso session running and continue with the next lab.

Summary

In this lab, you learned:

- ◆ How to use the Placement and Guard Ring tabs in the Array Assistant.



Lab 3-2 Working with Pattern Preset

Objective: To explore the Pattern Preset.

The **Pattern Preset** specifies the arrangement of devices within a Modgen. You can choose a pattern preset from a drop-down list, specify the base symbol pattern, or specify the base orientation pattern.

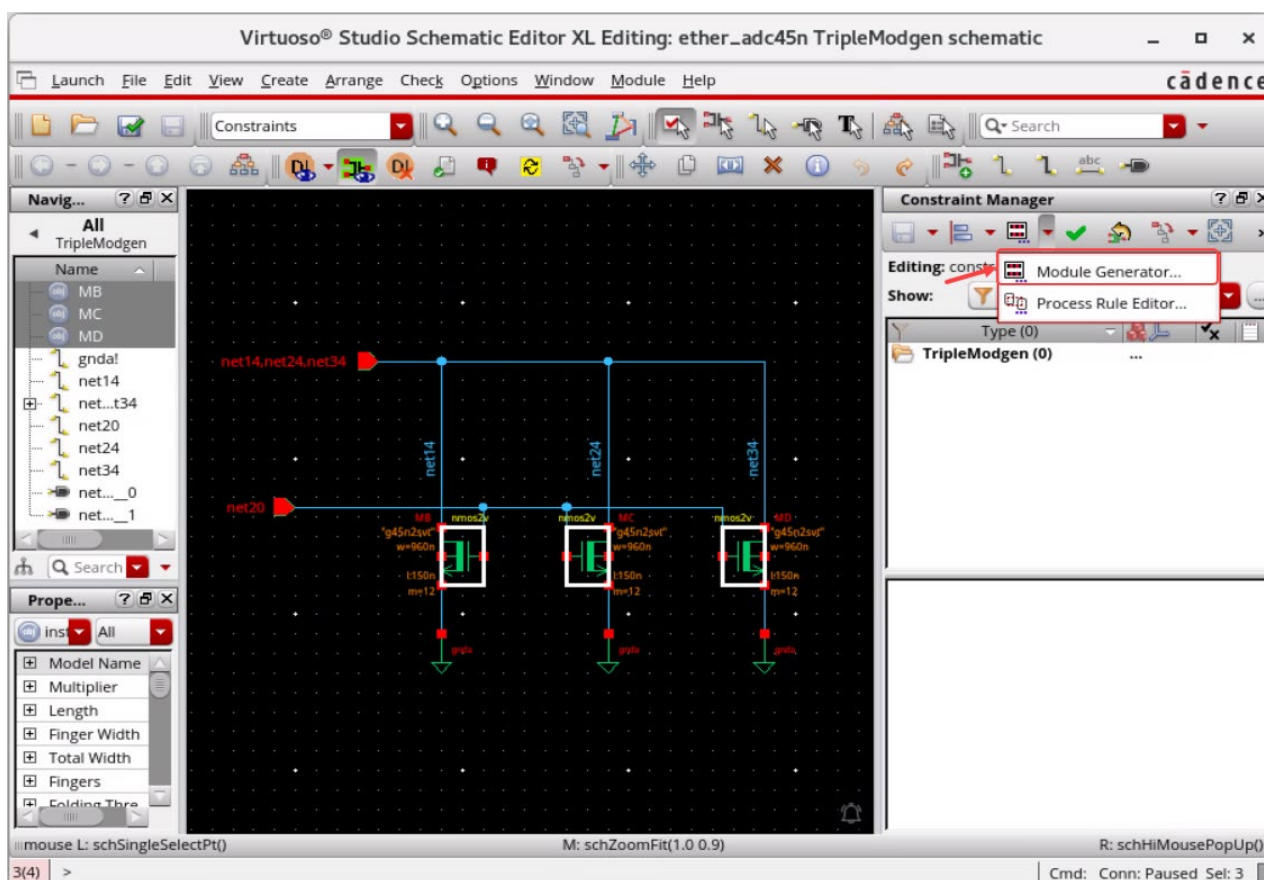
In this lab, you:

- ◆ Explore and use the Pattern Preset and use the Input Base Symbol Pattern and Input Base Orientation Pattern.

1. Use the Library Manager window to select and open the following in XL mode:

Library:ether_adc45n ; Cell Name:TripleModgen ; View Name:schematic

2. Select the devices **MB**, **MC**, and **MD** and invoke the Modgen Editor, as shown here.



The Modgen window appears.

3. Set the *Smart Display Configuration* to **Abstracted** in the Modgen window.

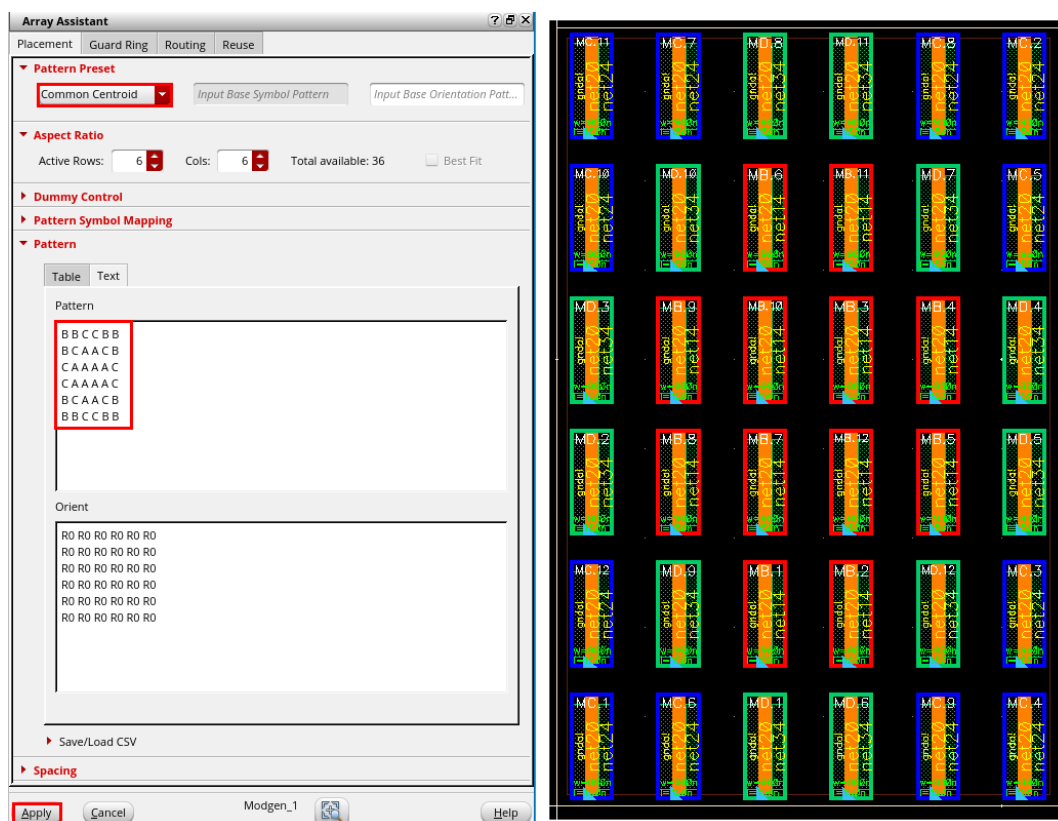
- In the **Pattern** section, in the **Table** tab, observe the Grid result. Similarly, observe the placement result in the **Text** tab, as shown here.



5. In the **Text** tab, enter the required *Pattern* and *Orient*. Click **Apply** to generate the pattern in the layout canvas, as shown in the graphic here.



6. In the **Placement** tab, in the Pattern Preset field, choose **Common Centroid**. It automatically updates in Pattern and click **Apply** to generate the pattern in the layout canvas, as shown in the graphic here.



7. Close all the cellviews without saving the changes and the Virtuoso session running and continue with the next lab.

Summary

In this lab, you learned:

- ◆ How to use the Pattern Preset, and use the Base Pattern and Base Orientation.

End of Lab

Lab 3-3 Using the Modgen On-Canvas Commands

Objective: To explore the Modgen on-canvas editing features.

In this lab, you:

- ◆ Use the Modgen on-canvas editing features.

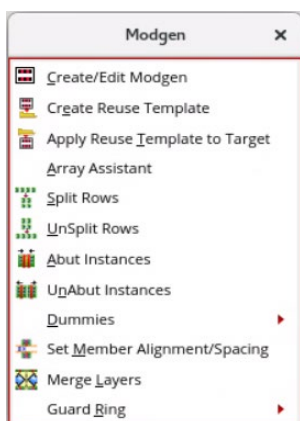
On-Canvas MODGEN commands aim to allow the users to do MODGEN operations (such as split Rows/Columns, abutment, and spacing) directly on the canvas without having to invoke the MODGEN Editor.

1. Use the Library Manager window to select and open the following in EXL mode:

Library:ether_adc45n ; Cell Name:adc_cascode_opamp ; View Name:layout

It already contains a Modgen array. You can iconize the schematic.

2. In the layout canvas, choose **Place – Modgen**. Underneath, you will find all the operations that can be made on a Modgen directly from the canvas, as shown here.

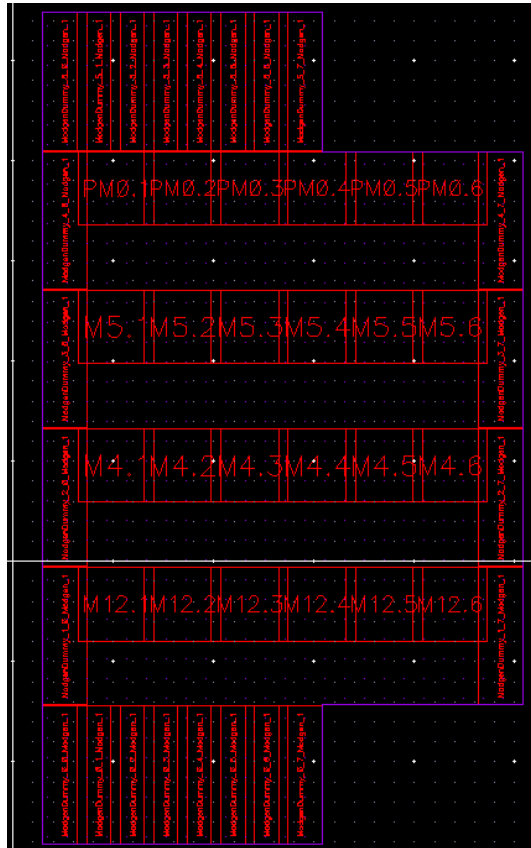


Note: The same set of operations appear in the **Placement** toolbar as well.

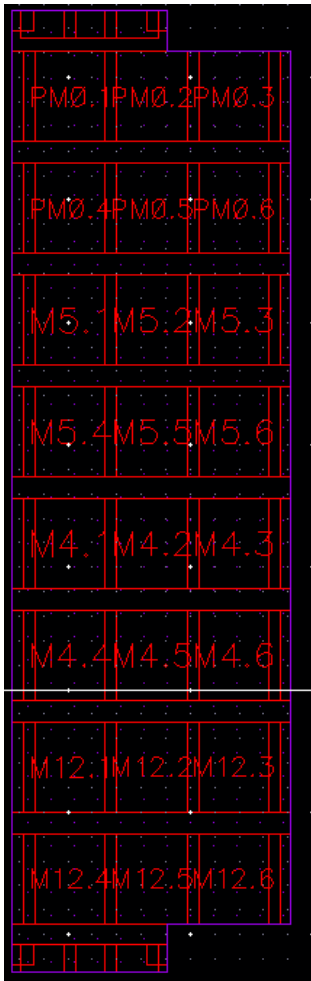
3. In the layout window, select the Modgen and choose the **Abut Instances** option.
4. Select the Modgen and choose the option **Set Member Alignment/Spacing**. Set *Vertical Alignment* as **custom** and *Spacing to Bottom* as **1**. Click **OK** in the form.
5. Choose **Options – Display** or press the bindkey **E** and in the Display Options form, set *Show Name Of* as **instance** and *Stop Display Levels* to **0** and click **OK**.
6. Select the Modgen and choose the option **Dummies – Add Surround Dummies** to open the Surround Modgen form.

- In the *Surround Modgen* form, in the **Connectivity** tab, in the *Connectivity* section, keep **Default** as option **VDD**.
- Keep the other values as defaults and click **OK** in the form.

An example result is shown here.



7. Select the Modgen and choose the option **Split Rows**. The new array has also **Surround Dummies**, as shown here.



Note: Explore the other Modgen operations as time permits.

8. Close all the cellviews without saving the changes and the Virtuoso session running and continue with the next lab.

Summary

In this lab, you learned:

- ◆ How to use the Modgen on-canvas editing.



Lab 3-4 Using the Topology Editor

Objective: To explore the Topology Editor.

In this lab, you:

- ◆ Route a Modgen array using the Topology Editor environment.

Using the set of available commands, you can control the routing that will be created by the PinToTrunk (P2T) router. In addition, you will learn how to modify the existing routing by changing the constraint and re-run the router on specific nets.

What Are Topologies?

Topologies are new OA objects which can be created automatically and edited before running the pinToTrunk router. The topologies have constraints associated with them that define the object's layer and width, but also can have constraints that control the router behavior, such as number and placement control of Vias, Pin cover, and matched net requirements. You will see the creation, editing, and constraints used by topologies and the PinToTrunk router in this section.

Setting Up a Default Workspace

For this lab, you will use the “workspace” that opens when the Modgen preview editor is opened. This is controlled by an environment variable that defines what workspace should be used when the Modgen preview editor is opened. The variable must be defined before opening the Modgen. `envSetVal("layoutXL" "modgenWindowConfigFile" 'string "modgen")`. The Modgen workspace is defined and saved already. It can be found at the following path: `./cadence/dfII/workspaces/Virtuoso_XL/`.

Creating a Modgen from Selection

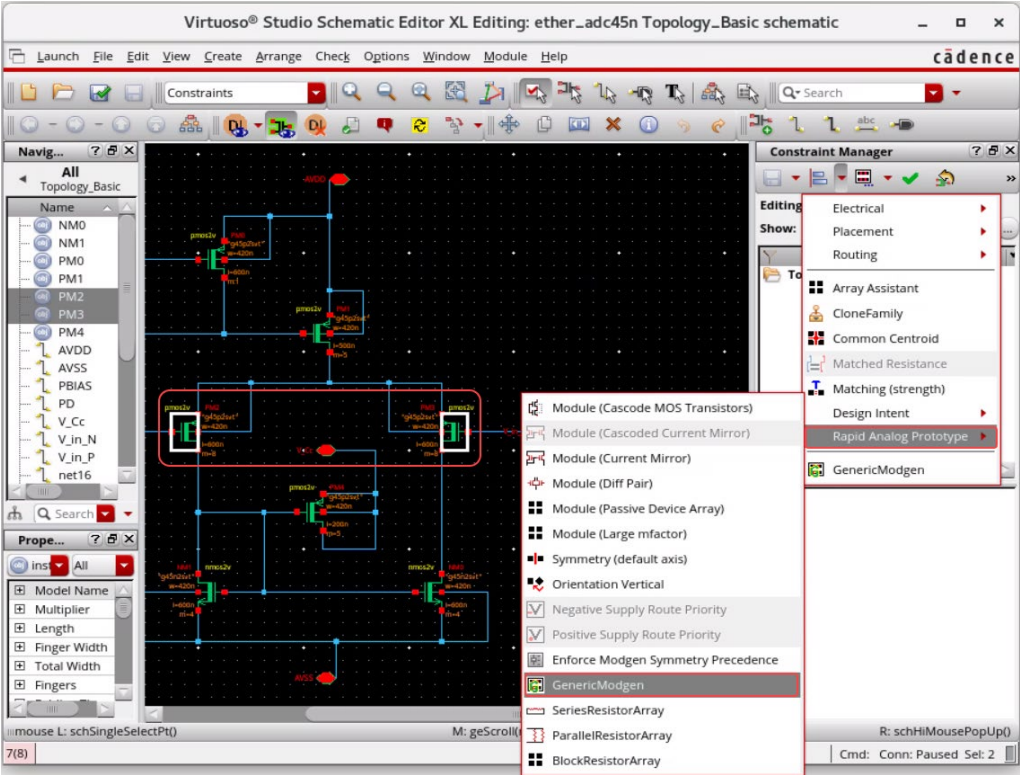
You will use the GenericModgen generator to create a basic Modgen from the schematic for use with the Topology Editor.

1. Use the Library Manager window to select and open the following in XL mode:

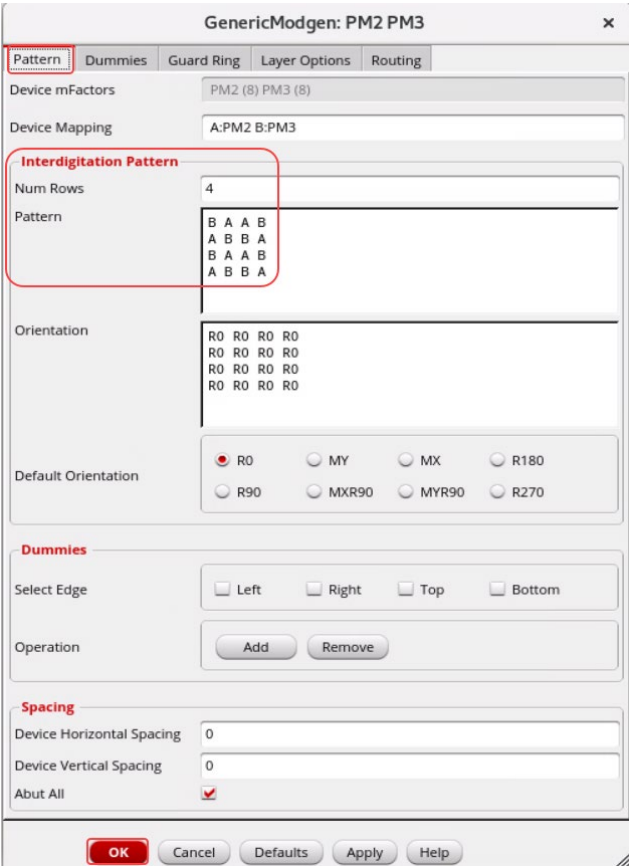
```
Library:ether_adc45n ; Cell Name:Topology_Basic ; View Name:schematic
```

2. Select the instances **PM2** and **PM3**.
3. Click the **Constraint Manager, Constraint menu – Rapid Analog Prototype – GenericModgen** icon to invoke the GenericModgen form.

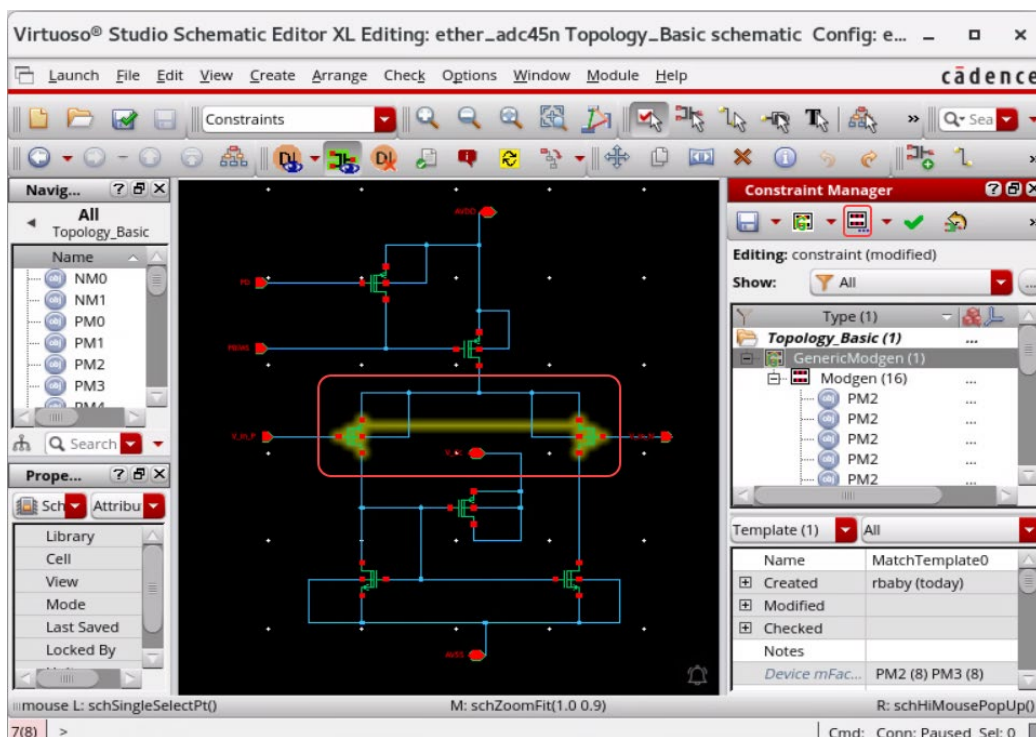




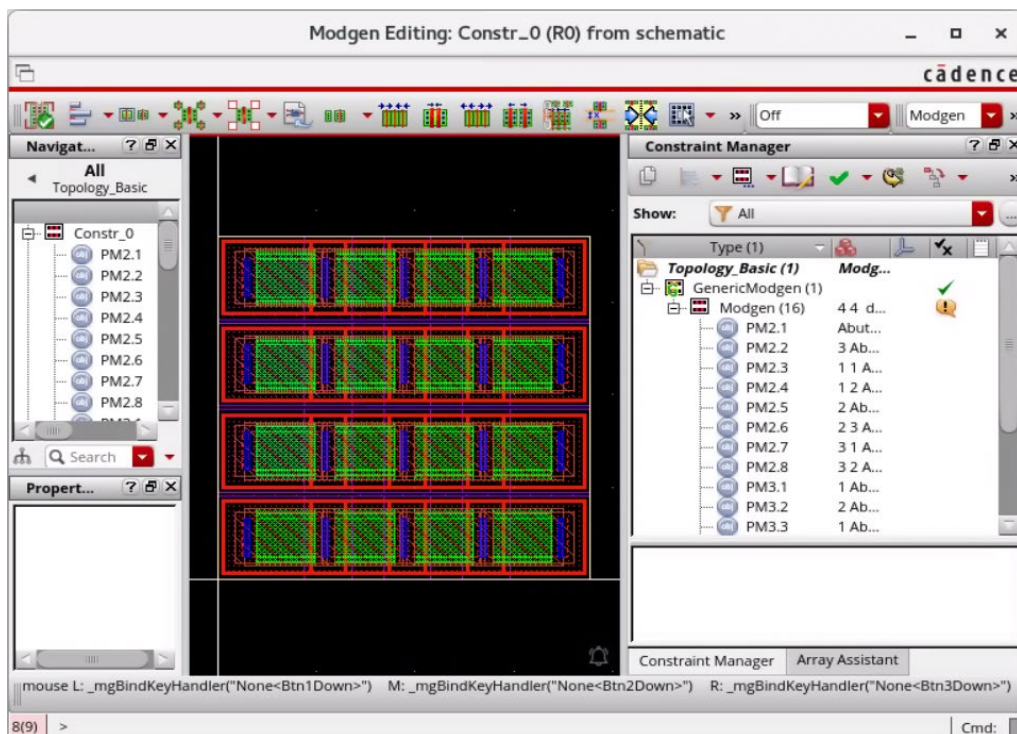
4. In the GenericModgen form, enter the pattern, as shown here, and click **OK**.



5. Select the **GenericModgen** constraint from the **Constraint Manager**.
6. Click on the **Module Generator** icon in the **Constraint Manager**, as shown here.



The Modgen window appears. It contains **16** members since each instance has a Multiplier factor of 8. There are **4** rows each based on the pattern given in the generator.

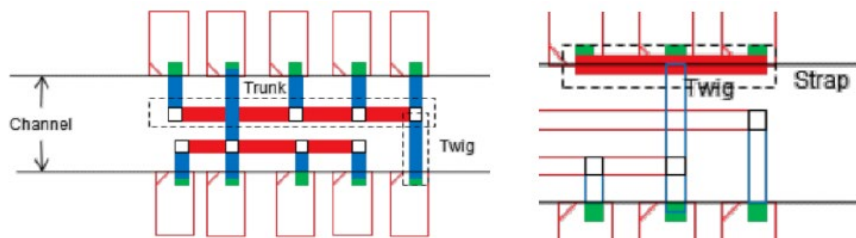


Creating Trunk Topologies

Instance terminals are connected to trunks through twigs, or straps and twigs, and as such, the Trunk Topologies must first exist before assigning twig or strap connections.

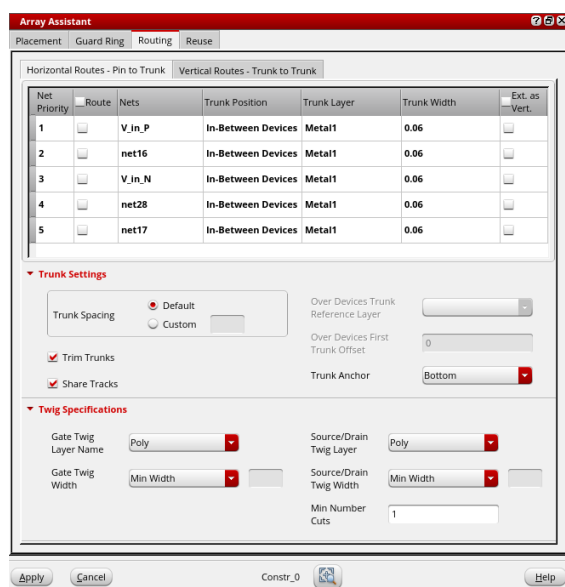
Modgen Topology Patterns contain the set of rules for the Pin-to-Trunk router. The different topology patterns specify specific routing types:

- ◆ Trunk – Straight line interconnect used to connect instTerms.
- ◆ Twig – Straight line interconnect between an instTerm and a Trunk.
- ◆ Strap – The connection between a number of instTerms. A strap is connected to a trunk using a twig.
- ◆ Constraints are added to the topology objects giving greater control of the routing results.



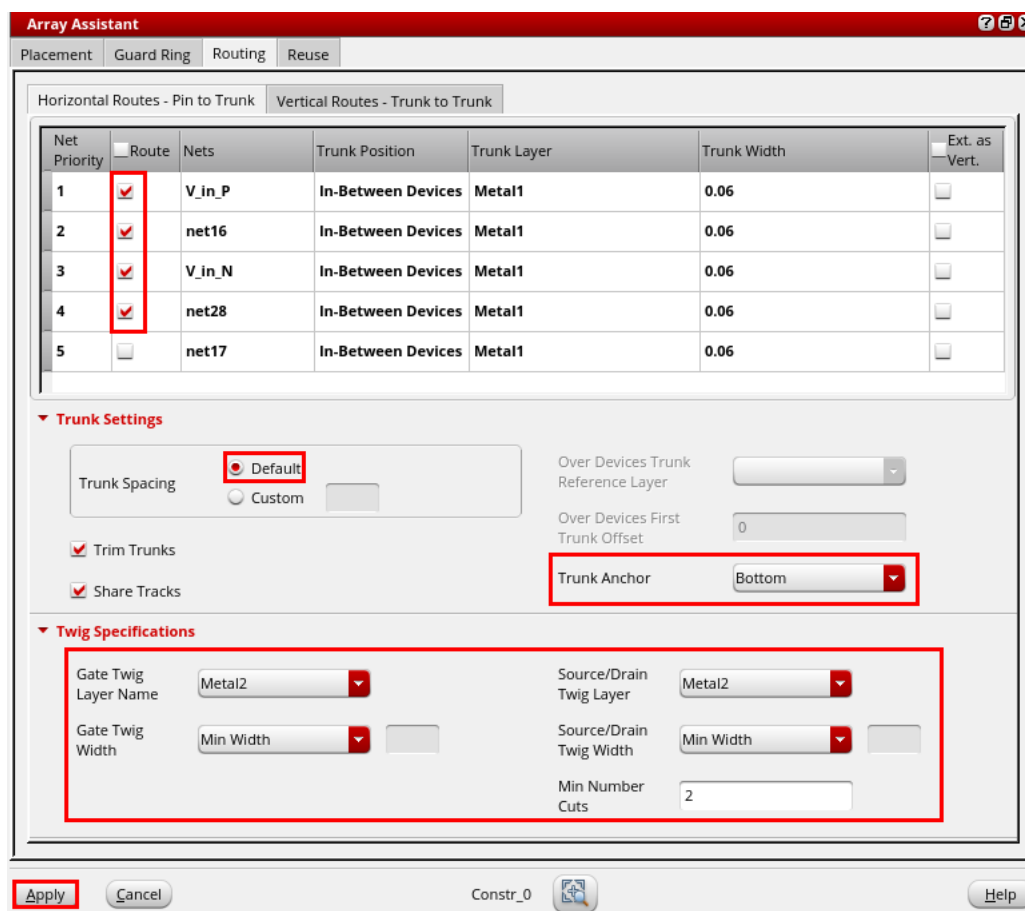
Creating trunks and trunk topologies in the modgen can be done using the **Routing** tab in the Array Assistant.

In the Array Assistant, you can find the Routing tab which offers various options to define Modgen topology patterns using the pin to trunk router. This process includes specifying the routing preferences for both the horizontal and vertical nets. Modgen topology patterns are useful aids as they allow you to easily visualize, configure, and store routing information.



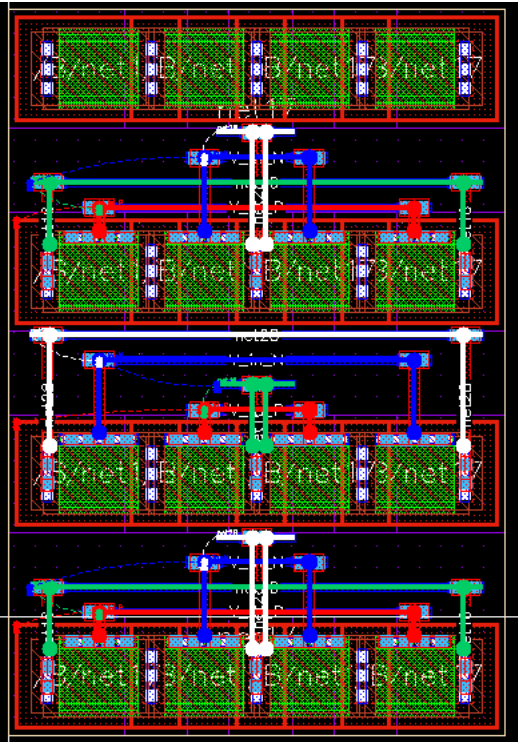
Creating Horizontal Routes - Pin to Trunk

1. In the **Routing** tab, in the Horizontal Routes - Pin to Trunk tab:
 - a. Select individual rows in the Route column to specify the nets for routing.
 - b. In the trunk Setting,
 - Keep Trunk Spacing as a Default.
 - Set Trunk Anchor to Bottom (bottom-left vertex of the device row).
 - c. In the Twig Specifications,
 - Select a Metal 2 layer from the Gate Twig Layer Name list.
 - Specify the width of the gate twigs to Min Width in the Gate Twig Width field.
 - Select a Metal 2 layer from the Source/Drain Twig Layer list.
 - Specify the source and drain twig width TO Min Width in the Source/Drain Twig Width field.
 - Min Numbers Cuts – 2



- d. Click **Apply**.

Pin to Trunk routing is created as shown here.

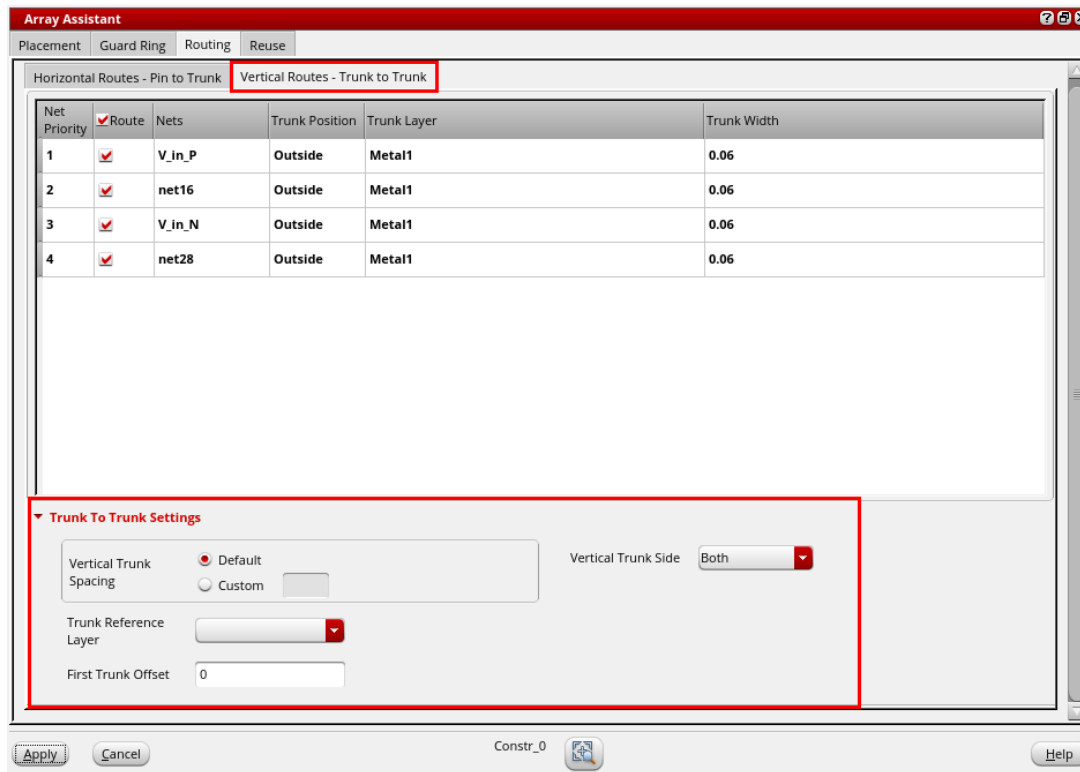


Creating Vertical Routes - Trunk to Trunk

Select **Ext. as Vert.** in the Horizontal Routes - Pin to Trunk tab to specify the net is to be listed on the Vertical Routes - Trunk to Trunk tab.

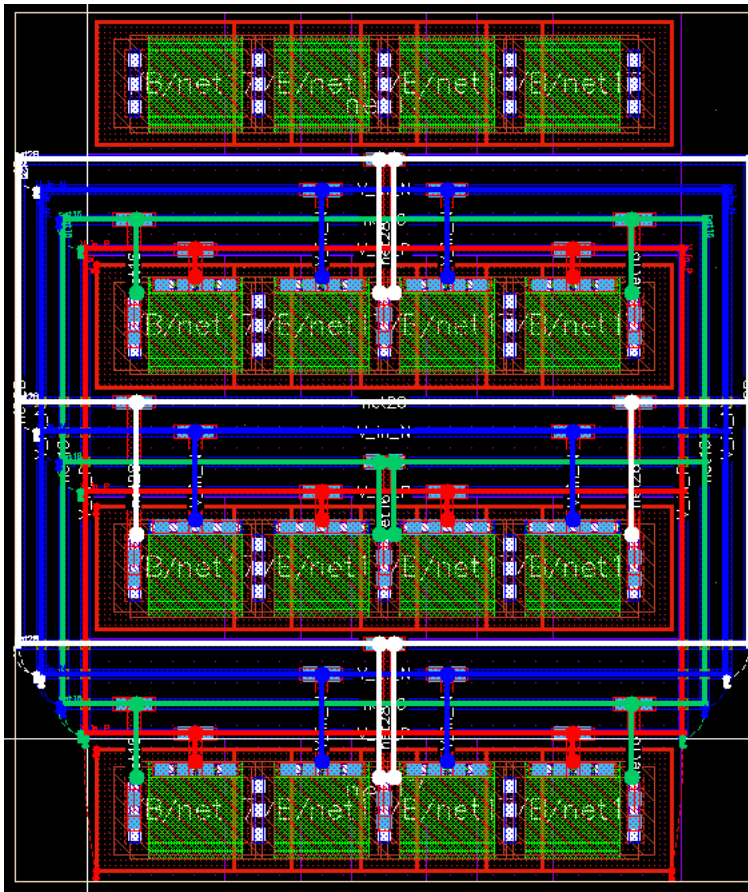
Horizontal Routes - Pin to Trunk							Vertical Routes - Trunk to Trunk	
Net Priority	Route	Nets	Trunk Position	Trunk Layer	Trunk Width	Ext. as Vert.		
1	☒	V_in_P	In-Between Devices	Metal1	0.06	☒		
2	☒	net16	In-Between Devices	Metal1	0.06	☒		
3	☒	V_in_N	In-Between Devices	Metal1	0.06	☒		
4	☒	net28	In-Between Devices	Metal1	0.06	☒		
5	☐	net17	In-Between Devices	Metal1	0.06	☐		

Switch to the Vertical Routes - Trunk to Trunk tab to specify routing preferences for vertical nets.



1. In the **Routing** tab, in the Vertical Routes - Trunk to Trunk tab:
 - a. In Trunk to Trunk Settings,
 - Set Vertical Trunk Spacing to Default.
 - Select a Vertical Trunk Side to both the side along which vertical trunks are to be generated.
 - First Trunk Offset – 0
2. Click Apply to apply the routing settings.

Trunk to Trunk routing is created, as shown here.



3. Close all the cellviews without saving the changes and exit the Virtuoso session.

Summary

In this lab, you learned:

- ♦ How to use the Topology Editor.



Module 4: Floorplanner

Lab 4-1 Defining the Cell Type Attribute

Objective: To define the cell type attribute.

Introduction

This course steps through the Virtuoso® Floorplanner Flow. The objective of this flow is to increase the layout productivity through the improved Floorplanner functionality aimed at block-level interconnect.

Basics

The Virtuoso Floorplanner is an IC physical block floorplanner from the Virtuoso Layout Suite EXL and MXL. The Virtuoso Floorplanner contains automatic and interactive floorplanning tools and technologies that will help custom design and mixed-signal (non timing driven) engineers develop a layout floorplan from the schematic in a more methodical manner.

The Virtuoso Floorplanner can be used in multiple flows. For example, it can be used to create a schematic-driven top-level floorplan. This floorplan is then pushed down to the block designers for further implementation, during which the floorplanner may again be used to create a floorplan for the blocks this time. The floorplans could be frozen before push-down, or it could be a rough topology that's finalized iteratively using floorplanning utilities. The floorplan menu contains commands that manipulate hierarchy, help define floor plans, tools to edit the floorplan efficiently and offers automated pin-and-block placement engines.

In this module, you will learn how to create a top-level floorplan.

In this lab, you:

- ◆ Define cell type attribute.

You will use the Virtuoso Floorplanner to create a top-level floorplan by:

- ◆ Using a top-level schematic for connectivity.
- ◆ Using exhausting layout and abstract cells for area and connectivity.
- ◆ Calculating or assigning area and aspect ratio for abstracts.
- ◆ Configuring the physical hierarchy to specify different cells for placement.
- ◆ Placing the blocks of the design.
- ◆ Editing objects on Level 1 to reshape and optimize pins.
- ◆ Optimizing and aligning block pins.

Note: Floorplanner does not require an abstract or autoAbstract view for the top-level blocks.

Floorplanning Design Flow (*commonly used*)



Note: Pin optimization also does pin alignment. However, the designer can also opt to do pin alignment post pin optimization step if required.

Floorplanning Strategy

The strategy involves:

- ◆ Using existing block sizes
- ◆ Estimating block sizes for unfinished layout
- ◆ Establishing placement preferences:
 - Blocks that have key importance
 - Prioritize key blocks that need to have short nets, such as RAM, ROM, and custom analog.

- ◆ Top-down and bottom-up floorplanning:
 - Freedom to start with abstracts and completed cells
 - Create soft blocks using area estimation
 - Calculated from schematics
 - Calculated from script

Floorplanning Design Flow Lab

In this course, you will learn the Floorplanning flow, which involves using various features:

- ◆ Defining Cell Type
- ◆ Setting up Physical Configuration View (Defining Soft Block parameters)
- ◆ Generating Physical Hierarchy
- ◆ Running Block Placer
- ◆ Snap Pins, Editing Soft Blocks, and Pin Optimization
- ◆ Pin Alignment
- ◆ Creating Feed Through Terminal Pins

Design Information

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in.

```
cd ../VLPT7_IC_25_1/mod04_VFP
```

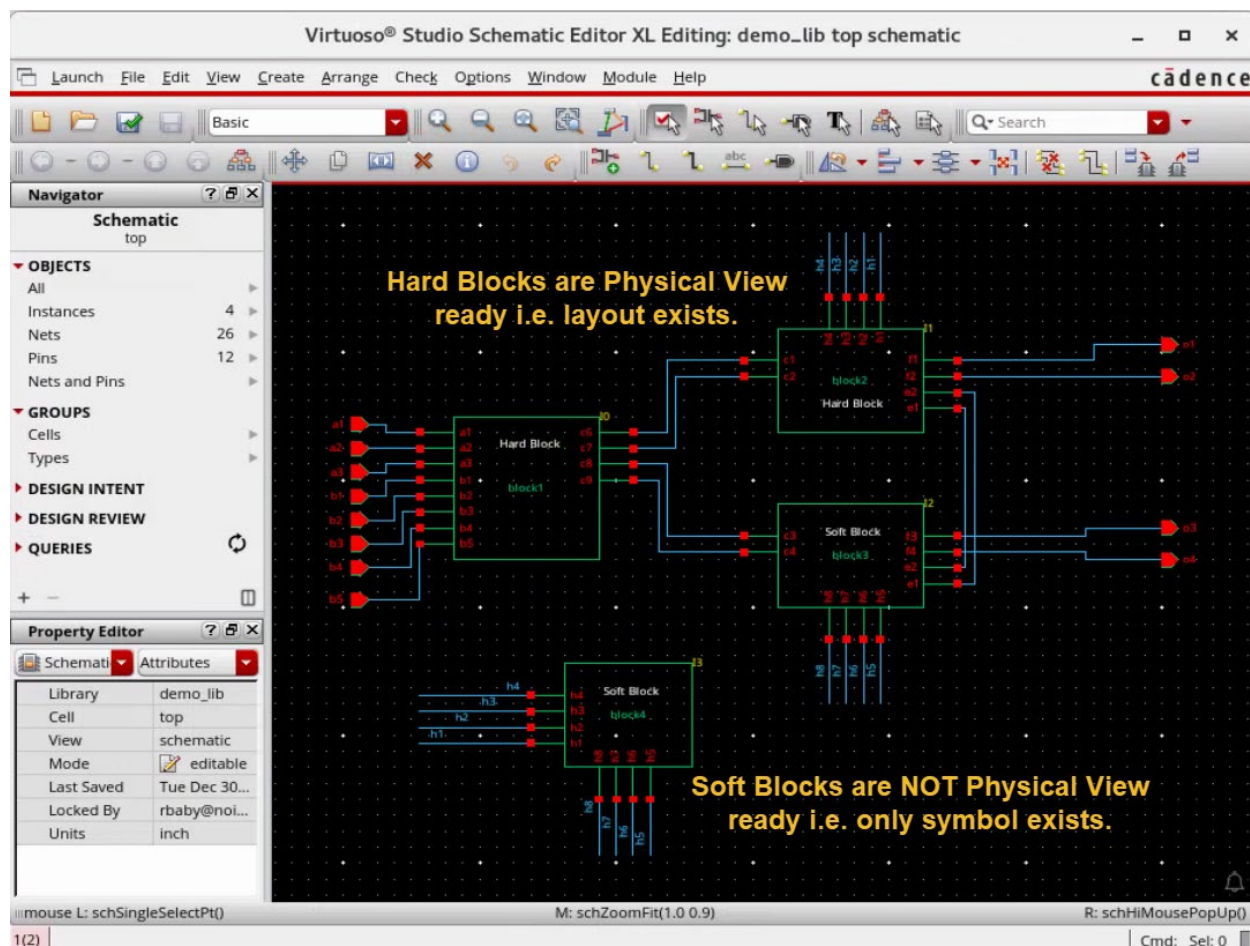
2. To start Cadence® Virtuoso software, enter this command in the same terminal window.

```
virtuoso &
```

The Command Interpreter Window (CIW) appears.

In this design, two blocks are already implemented (*hard blocks*), i.e., **block1** and **block2**, and two blocks are yet to be implemented (*soft blocks*), i.e., **block3** and **block4**.

For both the *Soft Blocks*, **block3** and **block4**, you will estimate the area and try to build a top-level layout for the top cellview.



Setting the Cell Type from the CIW

In this section, you will define the *Cell Type* attribute.

On the basis of Cell Type, the Virtuoso Floorplanner (VFP) decides how the given block should be treated.

The *Cell Type* attribute is required in all the physical views in the design.

- ◆ *Hard Blocks* – No changes are made by VFP to these blocks.
- ◆ *Soft Blocks* – These can be modified by VFP. You can make changes to Boundary, I/O Pins (Soft Pins), and Obstruction.

Hence, if you have already implemented some blocks or, for some reason, you don't want to modify a block during floorplanning, set it to a hard block.

However, if you want to modify a block during floorplanning, such as modify its boundary/shape to create a more compact floorplan, want to add some blockages, and want to optimally place the pins or optimize its pins, then it should be set to soft block so that Floorplanner can work on it.

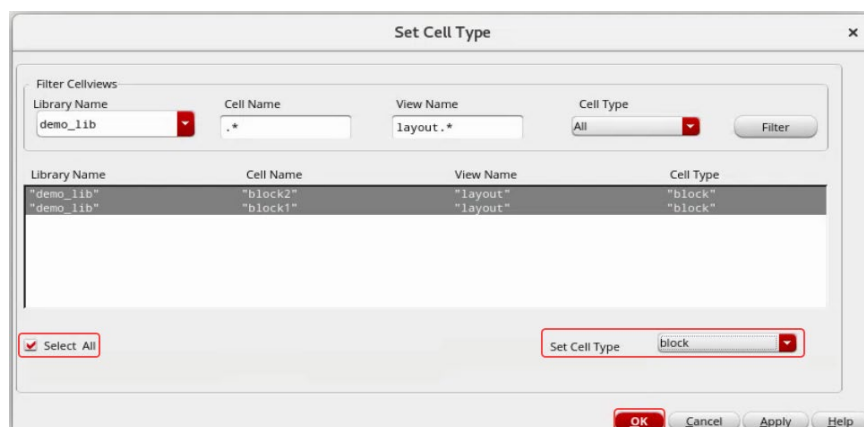
You will set *Cell Type* for **block1** and **block2** layout views.

1. In the CIW window, choose **Tools – Set Cell Type**.

The Set Cell Type form opens. Here **layout** views are listed for both **block1** and **block2**.

Note: Click **Filter** if the cell names are listed. By clicking **Filter**, they will be listed.

2. Select the options, as shown here.



- Click **Select All**.
- Change *Set Cell Type* to **block** and click **Apply**.
- Click **OK** in the Set Cell Type form.

This will enable the Virtuoso Floorplanner to treat both blocks **block1** and **block2** as *hard blocks*.

3. Leave the Virtuoso session running and continue with the next lab.

Summary

In this lab, you learned:

- ♦ How to define the cell type attribute.



Lab 4-2 Using the Configure Physical Hierarchy Utility

Objective: To use the CPH to create and edit the soft blocks.

This *CPH* mode has been specifically designed for use in the Virtuoso Floorplanner. You can create *stop points* (soft blocks) and specify the generation parameters for each of them. You can also configure how to generate the physical hierarchy of your design.

In this lab, you:

- ◆ Use the CPH to create and edit the soft blocks.

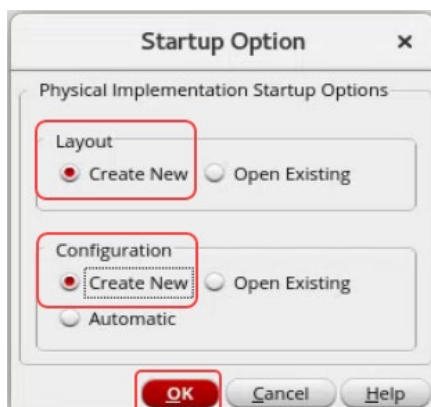
Opening the Configure Physical Hierarchy Utility

Verify that your session is still open from the previous lab. If not, open now.

1. In the **CIW**, choose **Tools – Library Manager** and open the following cellview:

Library	demo_lib
Cell	top
View	schematic

2. From the schematic window, choose **Launch – Layout EXL**.
3. When the Startup Option form appears, select **Create New** for the Layout and **Create New** again for the Configuration. Click **OK**.



The New File form appears.

- Set the following in the New File form.

Library	demo_lib
Cell	top
View	layout
Type	config

- Click **OK** on the New File form.

If prompted that the cell already exists, choose **Yes** to overwrite.

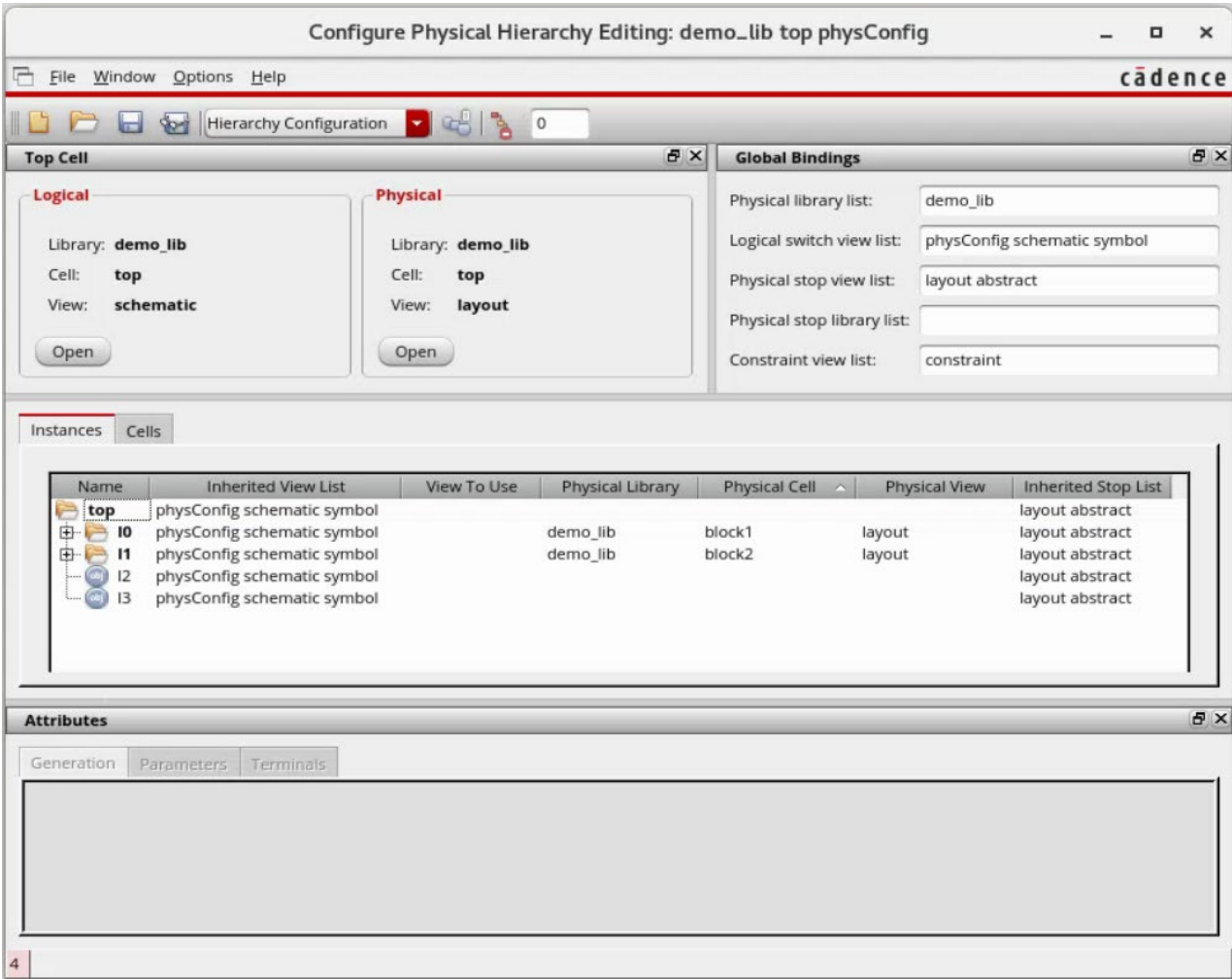
The layout EXL window for *demo_lib top layout* opens.

The **Create Physical Configuration View** form opens with the default view name of *physConfig*.

- Click to enable **Open CPH**.

- Click **OK** in the Create Physical Configuration View form.

8. Click **OK** to overwrite the existing view.
- The Configure Physical Hierarchy (CPH) form opens, as shown here.



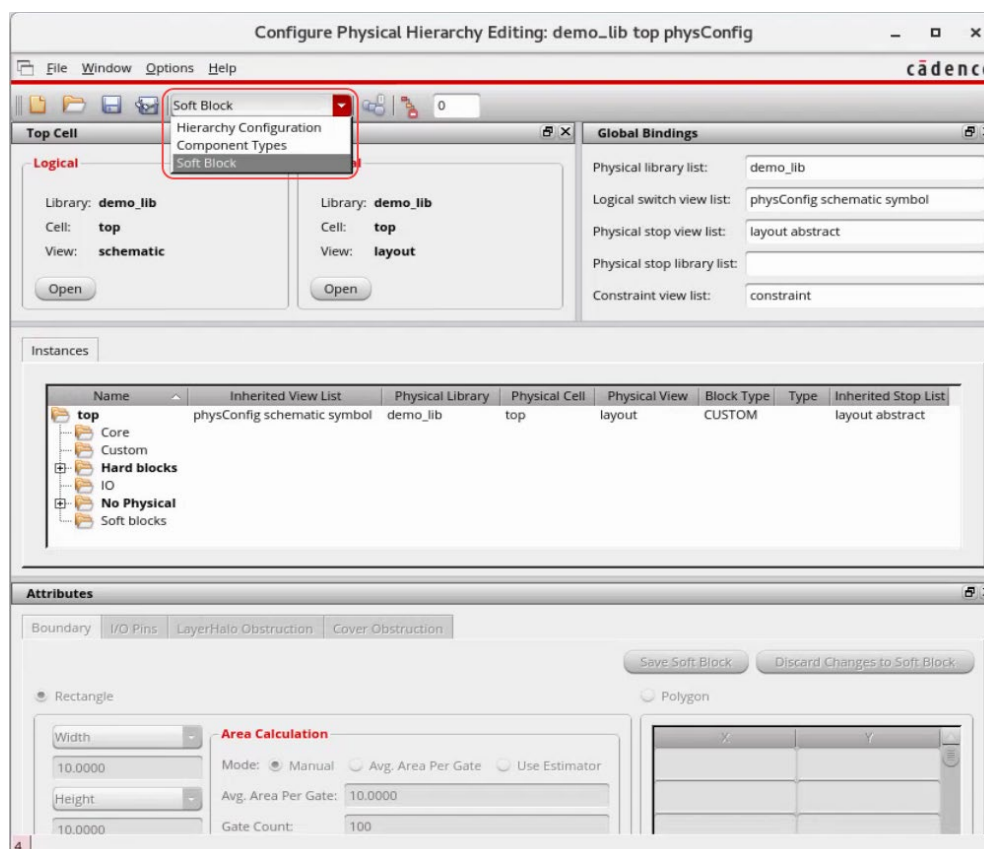
Using the Soft Block Mode

The Virtuoso Floorplanner utilizes only the *Soft Block* mode of CPH.

From the CPH Soft Block mode, you can define which blocks are soft and hard, distribute them into different bins, define the correspondence between the schematic and layout views of blocks, and most importantly, define the soft block parameters, such as boundary definition, I/O pins, and blockages.

After this, the Generate Physical Hierarchy (GPH) is used to generate Soft Blocks based on the parameters defined in CPH.

1. In the CPH form, click on the cyclic field to change from **Hierarchy Configuration** mode to **Soft Block** mode.



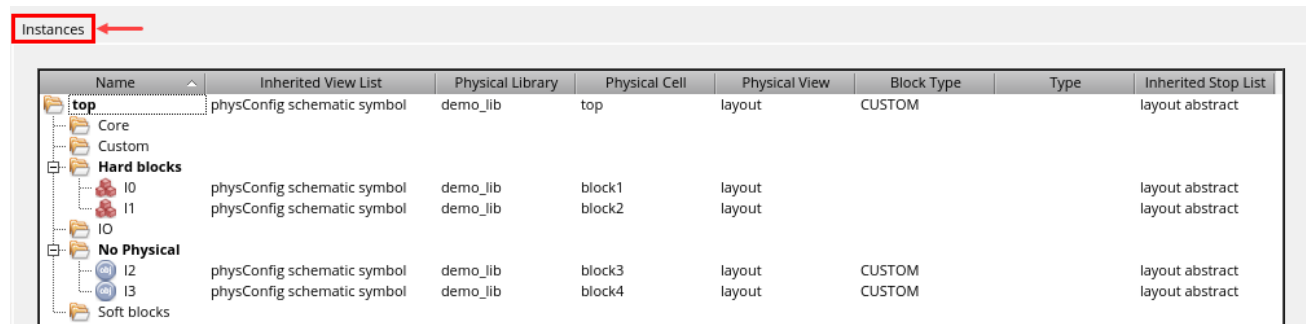
You can use this mode to define the soft block parameters for layout generation: *Boundary*, *I/O Pins*, *LayerHalo Obstruction*, and *Cover Obstruction*.

The **Top Cell** and **Global Bindings** fields at the top of the CPH form are common to the three modes.

The three modes in the CPH form are *Hierarchy Configuration*, *Component Types*, and *Soft Block*.

The display in the **Instances** tab lists the categories.

Refer to the **Instances** section.



Name	Inherited View List	Physical Library	Physical Cell	Physical View	Block Type	Type	Inherited Stop List
top	physConfig schematic symbol	demo_lib	top	layout	CUSTOM		layout abstract
Core							
Custom							
Hard blocks							
I0	physConfig schematic symbol	demo_lib	block1	layout			layout abstract
I1	physConfig schematic symbol	demo_lib	block2	layout			layout abstract
No Physical							
I2	physConfig schematic symbol	demo_lib	block3	layout	CUSTOM		layout abstract
I3	physConfig schematic symbol	demo_lib	block4	layout	CUSTOM		layout abstract
Soft blocks							

- **block1 (I0)** and **block2 (I1)** will appear as *Hard blocks*.

The hard blocks are bound to a physical view (layout or abstract), which is defined in the **Physical stop view list** in the **Global Bindings** section at the top right of the CPH form.

- **block3 (I2)** and **block4 (I3)** will appear as *No Physical* since no physical view, and no cell types are defined yet.

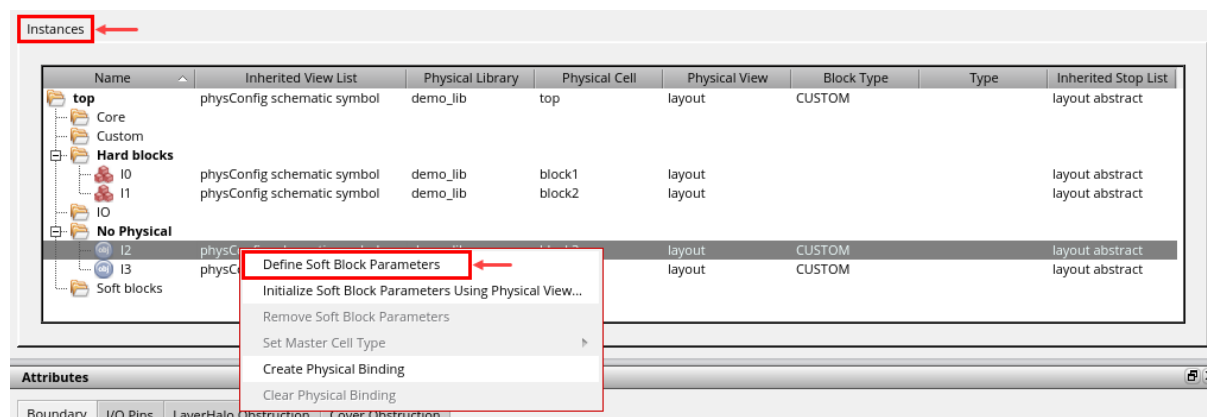
Take a case where you have a block with an existing layout view. Now, you need to create a new layout. How do you handle this?

Answer: In such a case, specify the new physical view name for the block in the CPH. As soon as a new view name is specified, the cell will move from *Hard blocks* to *No Physical* view. Now the same set of steps can be followed on this block as mentioned in this section.

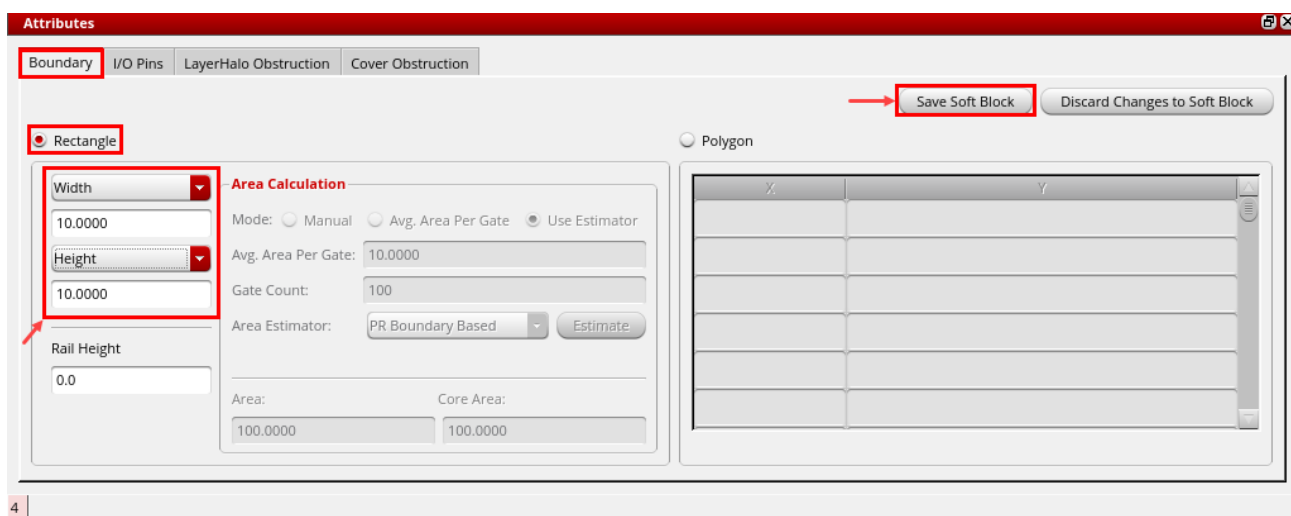
It is always a good practice to clearly specify all the blocks as *hard* or *soft*. For the block with no physical view appearing under *No Physical*, you can define *Cell Type* as **softMacro**. This will move the block from *No Physical* to *Soft blocks*.

Defining the Soft Block Parameters for I2 (block3)

1. Select **I2** with a **left-click** in CPH form.
2. **Right-click** and select **Define Soft Block Parameters**.



- On the **Boundary** tab, enable **Rectangle**.
Under the **Rectangle** radio button are cyclic fields to set parameters for the soft block.
- Specify the *Boundary* parameters: **Width** as **10** and **Height** as **10**.
- You won't be making any updates to *I/O Pins*, *LayerHalo Obstruction*, or *Cover Obstruction*. Leave these fields as default values.
- Click **Save Soft Block** to save the Soft Block Attributes for **I2 (block3)**.



Defining the Soft Block Parameters for I3 (block4)

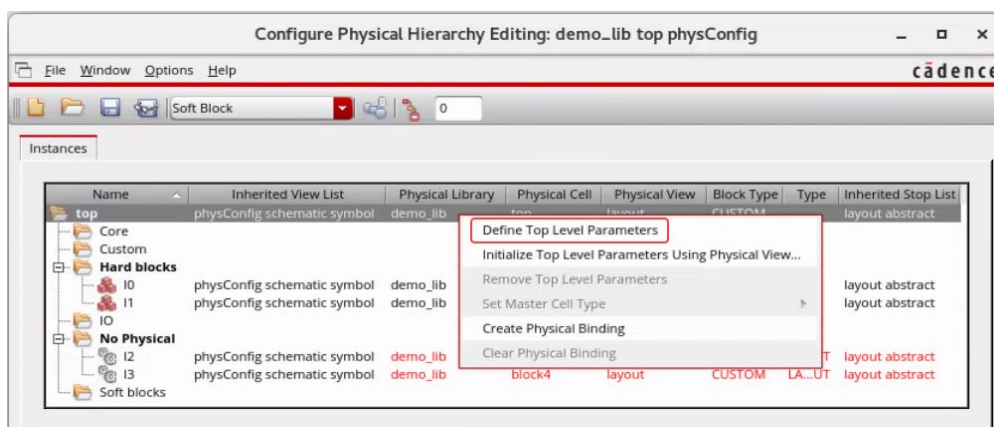
- Select **I3** with a **left-click** in CPH form.
- Right-click** and select **Define Soft Block Parameters**.
- On the **Boundary** tab, enable **Rectangle**.
- Specify **Aspect Ratio** as **0.5** and **Height** as **10**.
- You won't be making any updates to *I/O Pins*, *LayerHalo Obstruction*, or *Cover Obstruction*. Leave these fields as default values.
- Click **Save Soft Block** to save the Soft Block Attributes for **I3 (block4)**.

Defining the Soft Block Parameters for the Top-Level Cell (top)

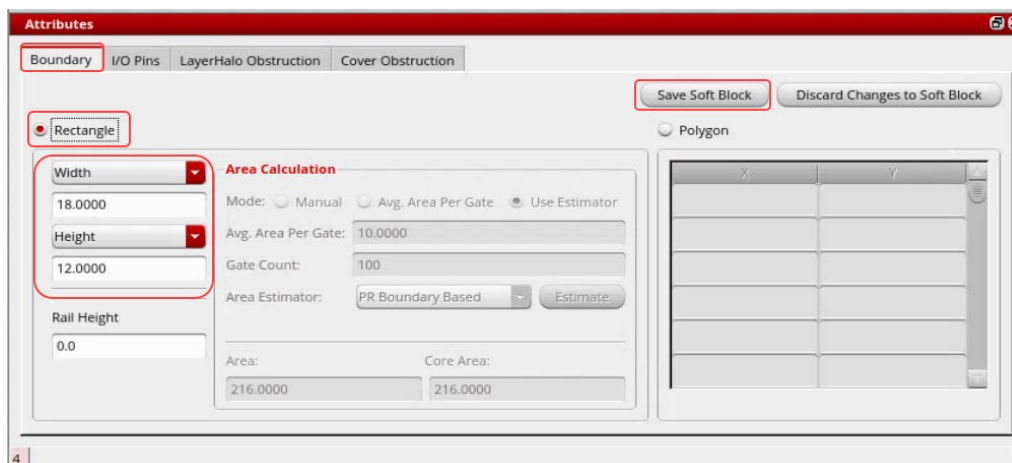
Just like the other soft blocks, the top-level needs parameters to generate the boundary and pins.

The top cell is also a soft block because you are modifying its contents, such as creating soft blocks inside it and optimizing the top-level pins.

1. Select **top** with a **left-click** in CPH form.
2. **Right-click** and select **Define Top Level Parameters**.



3. On the **Boundary** tab, enable **Rectangle**.
4. Specify **Width** as **18** and **Height** as **12**.
5. You won't be making any updates to *I/O Pins*, *LayerHalo Obstruction*, or *Cover Obstruction*. Leave these fields as default values.
6. Click **Save Soft Block** to save the Soft Block Attributes for the top level.



Saving the Floorplan Properties File

The floorplan properties file contains the soft blocks' parameters.

1. Choose the **File – Save** menu command or click the **Save** icon in the CPH window.
2. Choose **File – Save Floorplan File** in the CPH form.
The Save Floorplan Property File form appears.
3. In the Save Floorplan Property File form, enter the filename name as *floorplanPropertyFile.txt* and click **Save**.
4. Choose to overwrite if there is an existing file of that name.
5. From the Configure Physical Hierarchy form, choose **File – Exit**.
6. Click **No** in the dialog box that asks whether you want to save the floorplan file.
7. Keep the cellviews open, and the Virtuoso session running and continue with the next lab.

Summary

In this lab, you learned:

- ◆ How to use the CPH to create and edit the soft blocks.



Lab 4-3 Generating and Placing the Blocks with the Floorplanner

Objective: To generate the physical hierarchy and place the blocks of the design.

Generate Physical Hierarchy (GPH) generates the physical hierarchy based on the configuration defined in Configure Physical Hierarchy (CPH).

GPH allows the user to generate a physical hierarchy from the logical hierarchy in a schematic. It also creates soft layouts and soft abstracts in the layout canvas corresponding to the configuration view generated by CPH.

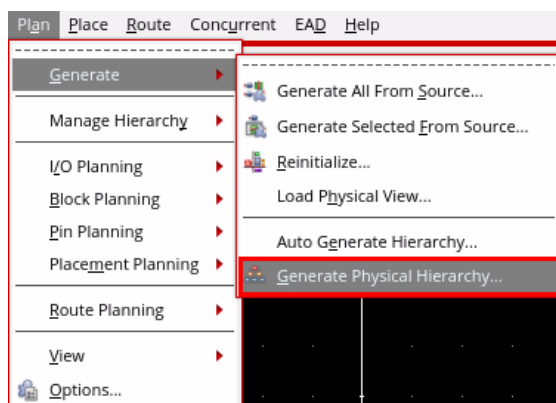
In this lab, you:

- ◆ Generate the physical hierarchy and place the blocks of the design.

Generating the Physical Hierarchy

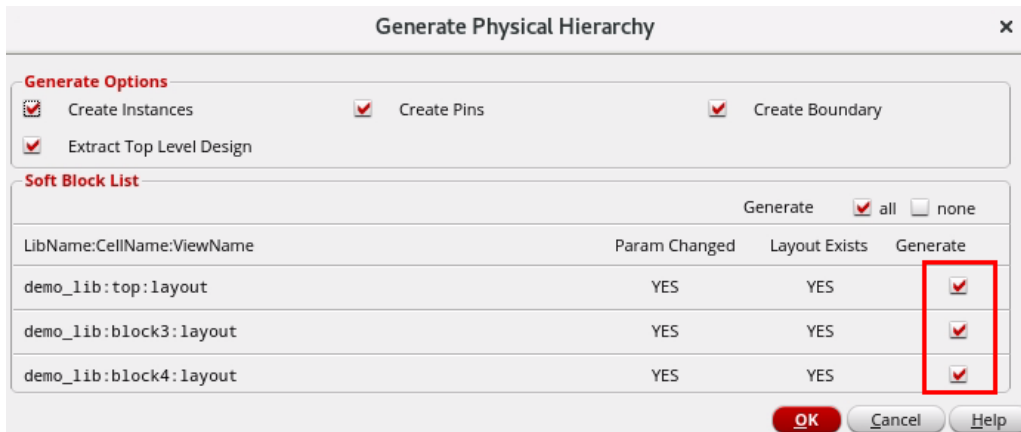
Verify that your session is still open from the previous lab. If not, open them now.

1. Choose **Plan - Generate – Generate Physical Hierarchy** or click the **Generate Physical Hierarchy** icon  in the layout window.

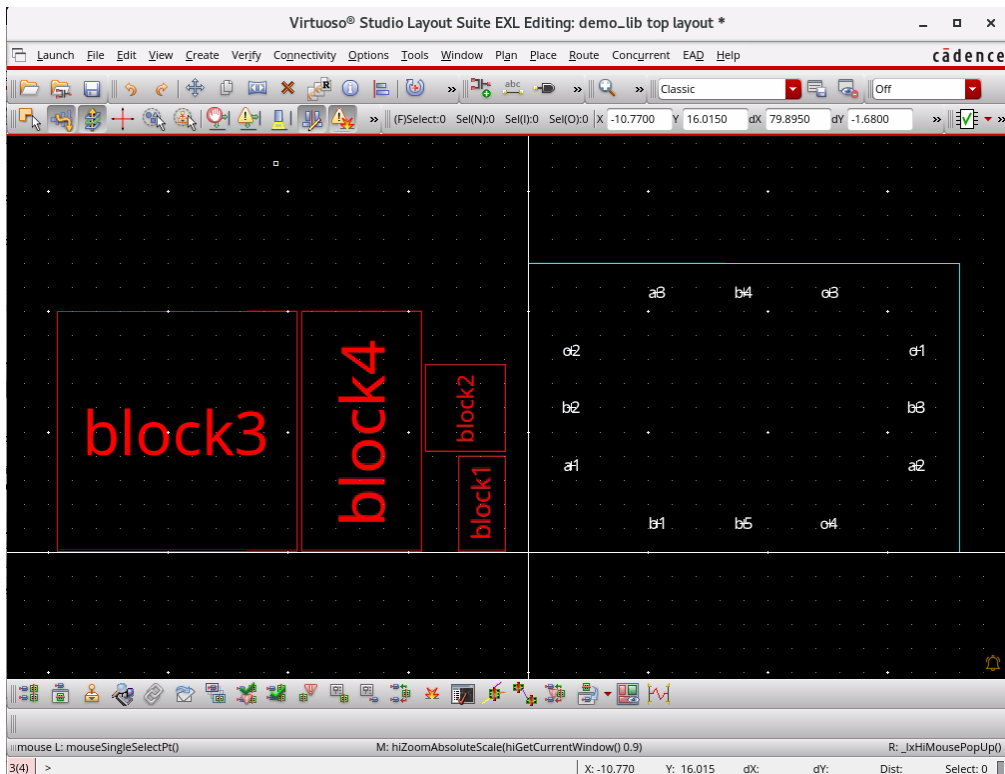


The Generate Physical Hierarchy form opens.

2. Enable the **Generate** button for each of the cellviews in the list and click **OK**.



The blocks are generated and placed to the left of the *prBoundary* in the cellview.
The layout hierarchy has been created successfully.



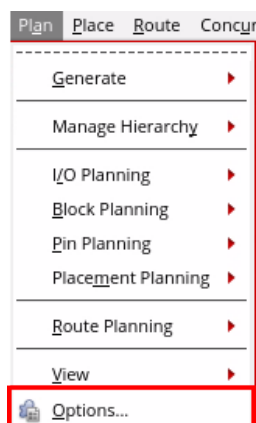
Note: Enable *Pin Names* in the **Display Options** form to display the pin names in the layout.

3. Observe the CIW window for the Generate Physical Hierarchy (GPH) results.

Color Coding the Hard and Soft Blocks

Use the Design Planning and Analysis Options form to control the display of virtual hierarchies and blocks in a design.

1. Choose **Plan – Options** or click the **Options** icon  in the layout window.



The **Design Planning and Analysis Options** form is displayed.

2. Select the following in the **Display** section on the **Design Planning and Analysis Options** form.
 - a. Select the **Macro coloring** option.
 - b. Choose the desired **hilite** color for the soft and hard blocks and click **OK**.

The hard and soft blocks are highlighted in different colors.

Design Planning and Analysis Options

▼ Generate

Virtual hierarchyall

Positiondefault

▼ Auto Adjustment

Auto place on editarea boundaries

Auto place virtual hierarchyoutside instances

☒ Fix placement status on move

Auto adjust☒ area boundary

☐ preserve enclosure

Area boundary snapAll Points

Snap unknown block type asCustom

▼ Display

Name ofVirtual hierarchy

☐ Symbol overlay

☒ Use bind keys

☒ Dashed line for placement statusNone

Clonehilite drawing9

Generatedhilite drawing5

Createdhilite drawing6

☒ Macro coloring

Soft blockhilite drawing1

Hard blockhilite drawing5

► Pins

OK

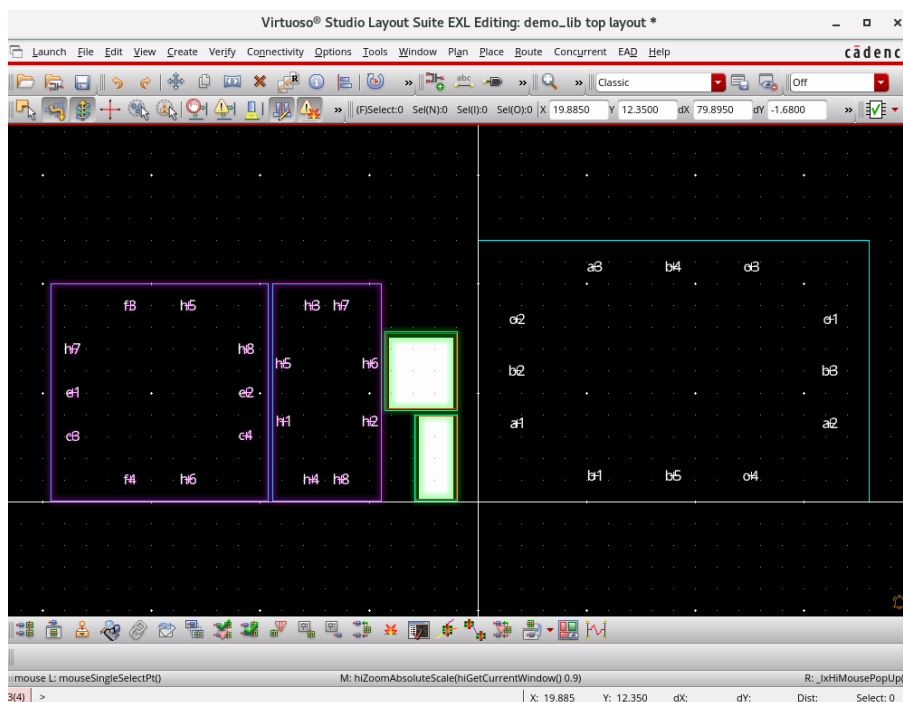
Cancel

Apply

Defaults

Help

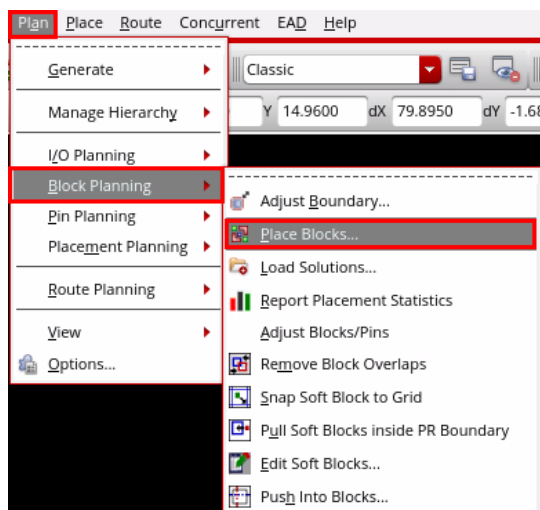
3. Press **Shift+F** to view all levels of the hierarchy, as shown here.



Using the Block Placer

The Block Placer command enables you to automatically place all the hard and soft blocks in a design and minimize the wire length and chip area, i.e., places all blocks in *prBoundary*.

1. Choose **Plan – Block Planning – Place Blocks** in the layout window.



The Block Placer form opens.

2. Set the following in the **Options** section on the Block Placer form.
 - a. *Effort* as **5**.
 - b. *Number of Solutions* as **5** and click **OK**.

The screenshot shows the 'Block Placer' dialog box with the 'Constraints' tab selected. The 'Options' section is highlighted with a red box. The 'Options' section contains the following settings:

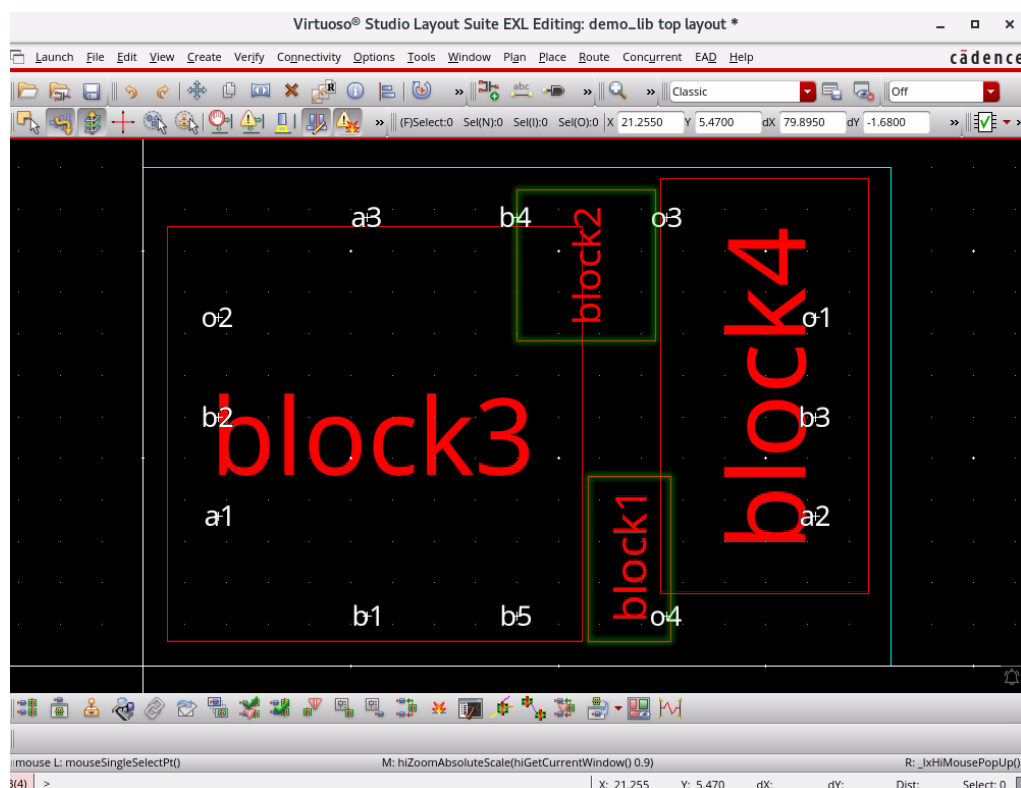
- Default Minimum Distance between Instances: ☐ Pins per Side, ☒ User Defined, 1.24
- Optimize Top-Level Pins: ☐
- Effort: 1 to 5 (set to 5)
- Number of Solutions: 1 to 5 (set to 5)
- Enable Multi Threading: ☐
- Enable Congestion Based Refinement: ☐ Congestion Value: 150
- Number of Iterations: 1 to 5 (set to 5)

The 'Chip Dimensions' section shows 'Fixed Die' checked, 'Minimum Boundary to Instance Distance' set to 0.52, and 'Specify Chip Dimension Range' set to 'Width and Height' with 'PRBoundary Width Range' and 'PRBoundary Height Range' both set to 200 to 200.

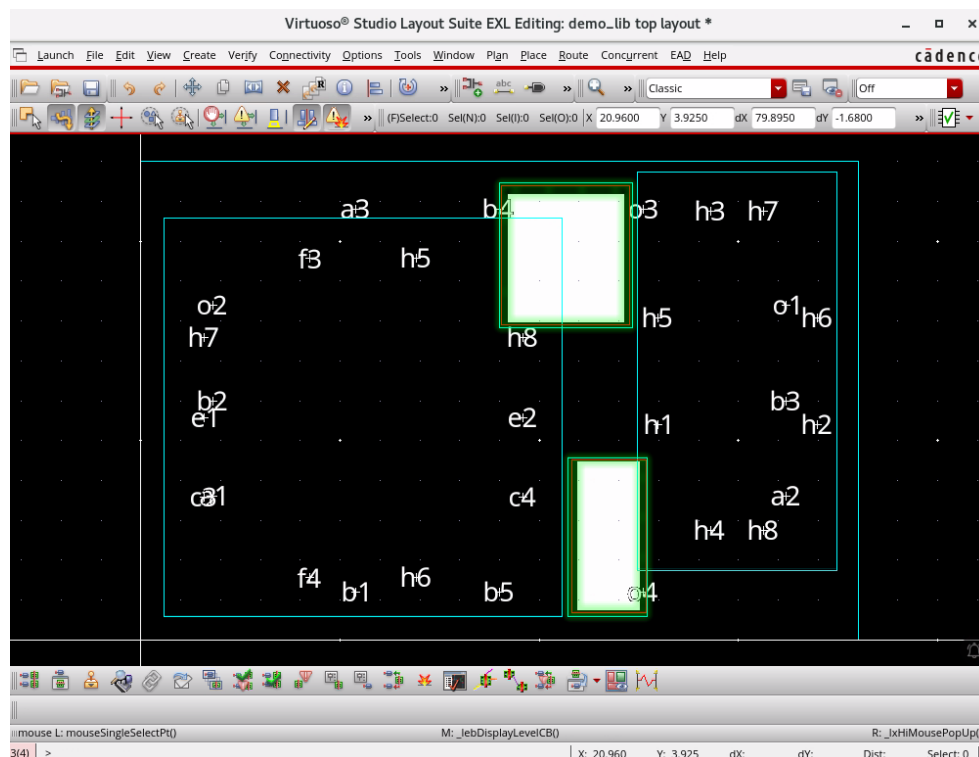
The 'Cost Order' section shows a list of cost factors: 1. Overlap, 2. Boundary, 3. Order/Alignment, 4. Distance, 5. Net Length, 6. Cluster, 7. Cluster Boundary, 8. Net Criticality, 9. Halo. A 'Default Cost Order' button is at the bottom.

At the bottom of the dialog are buttons for 'OK', 'Cancel', 'Apply', 'Reload All Constraints from Constraint Manager', and 'Help'.

All the four blocks are placed inside the *prBoundary*, as shown here.



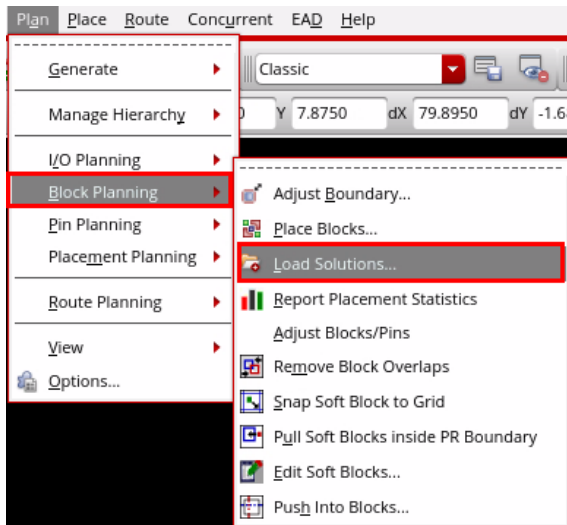
If they appear to be overlapped, press **Shift+F** to view the contents of the blocks.



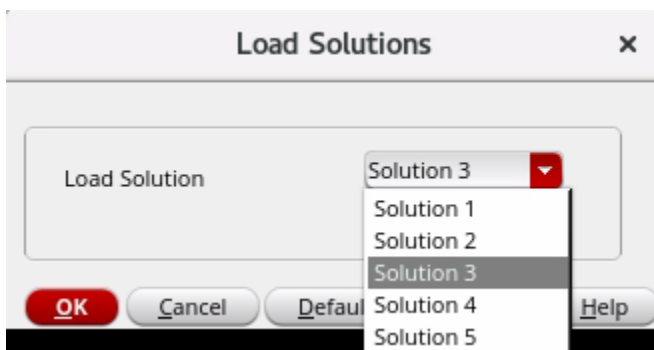
Refer to the CIW window for the Block Placer run details.

Loading the Solutions

1. Choose **Plan – Block Planning – Load Solutions** in the layout window.

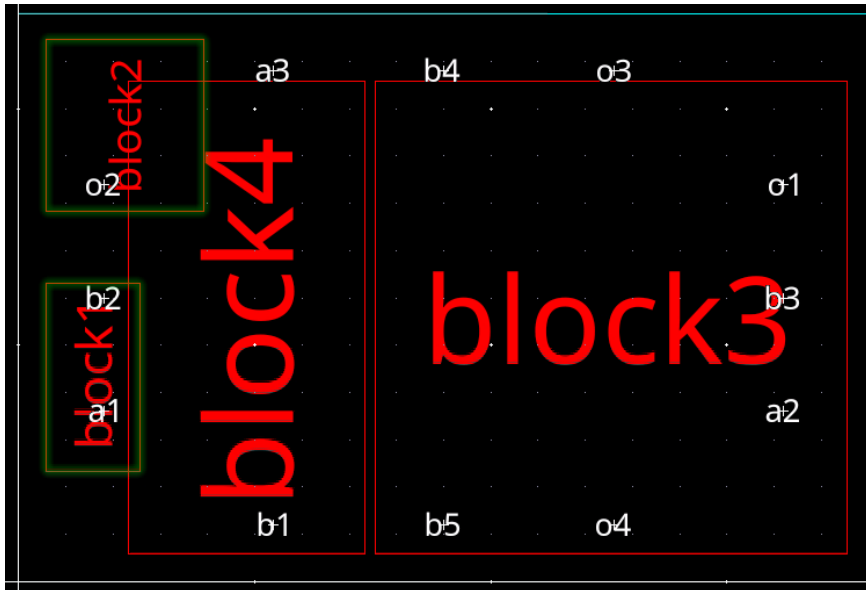


The Load Solutions form appears.

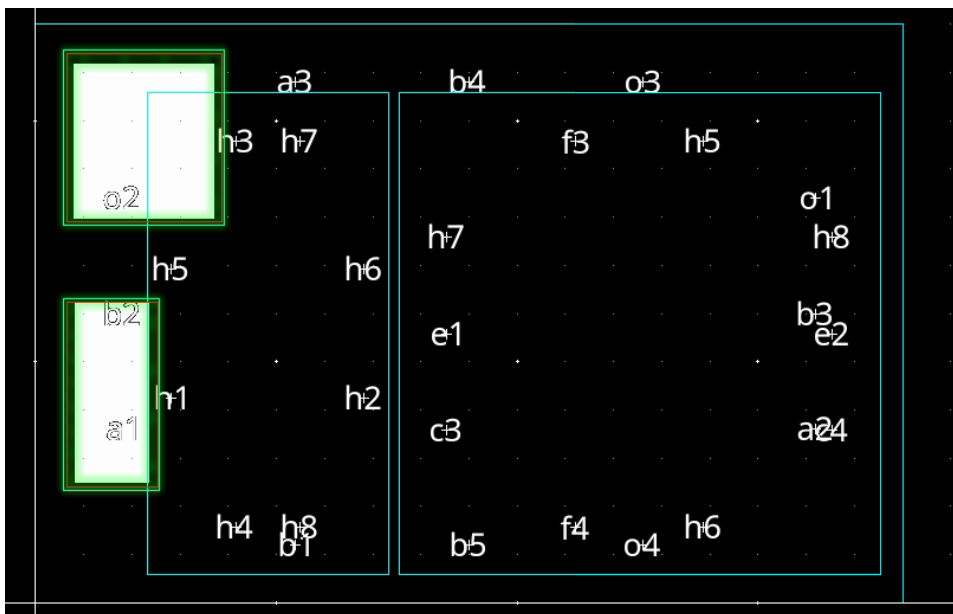


2. Cycle through the **five** solutions.

- One example placement is shown here, but you will probably find a different placement that will work well.

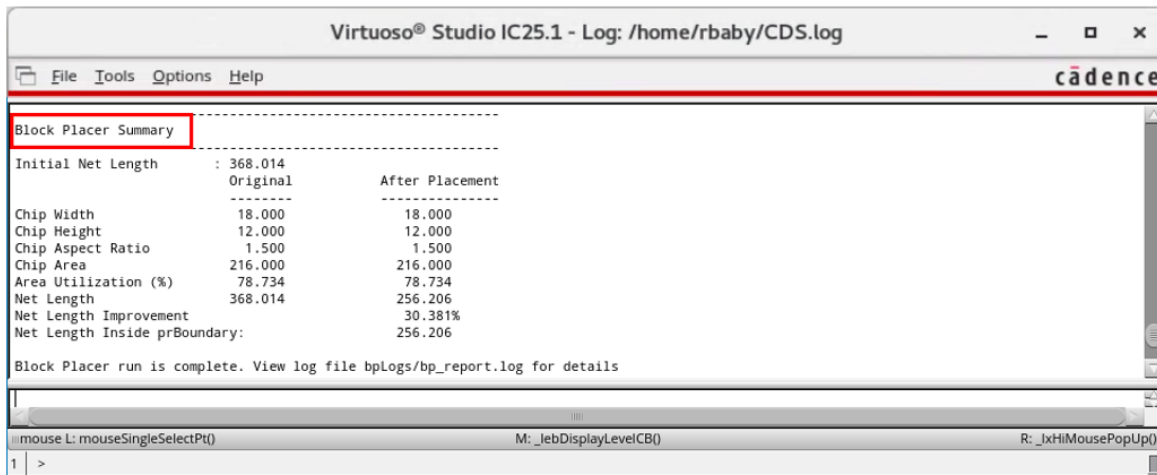


- There are overlapping blocks. Use **Shift+F** to see the actual amount of area used in each block.



- Press **Ctrl+F** to set the **Display Stop Level** field to **0** to see the top level of the hierarchy.

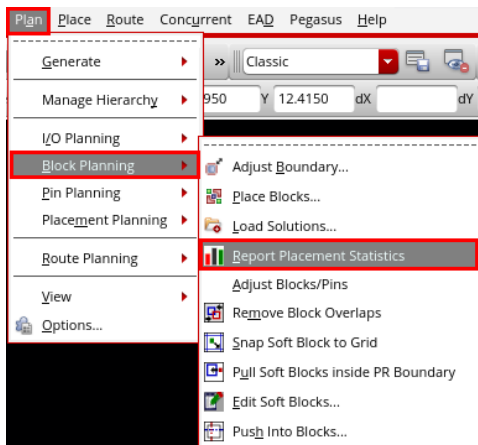
Block Placer Summary in the CIW



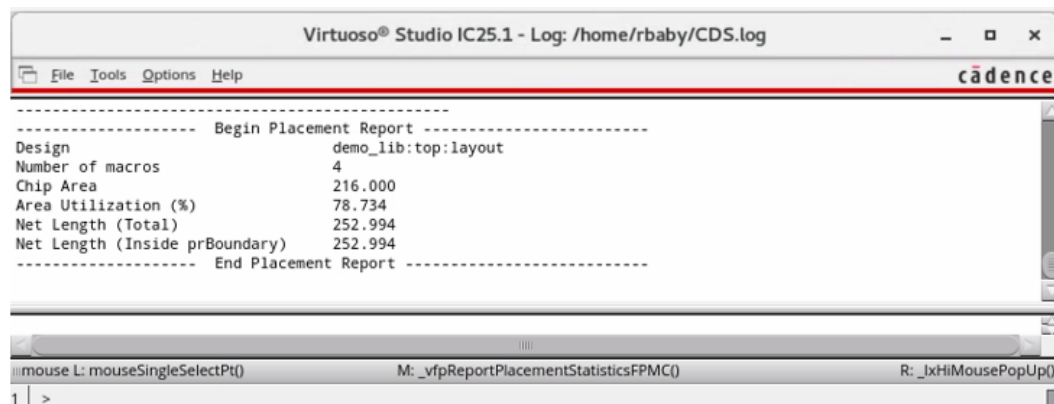
Reporting the Placement Statistics

You can view the placement statistics report at any time after block placement.

1. Choose **Plan – Block Planning – Report Placement Statistics** in the layout window.



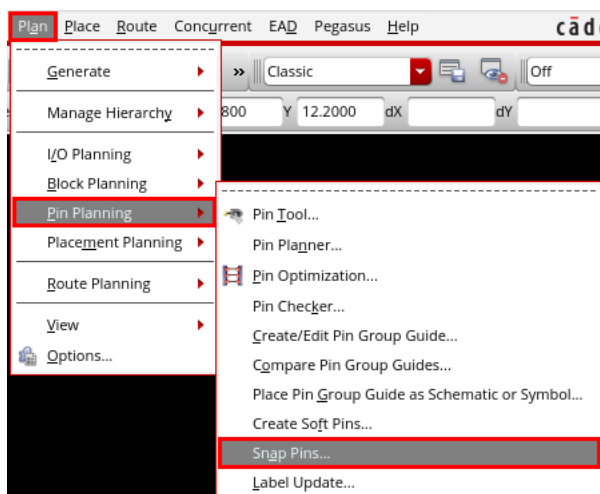
This command displays the placement report in the CIW window, as shown here.



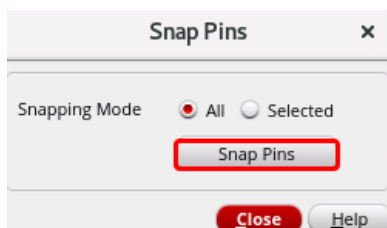
Snapping the Soft Pins

The soft pins (pins of the soft block are called soft pins) in the generated blocks, and the top-level pins are yet to be snapped to the *prBoundary*.

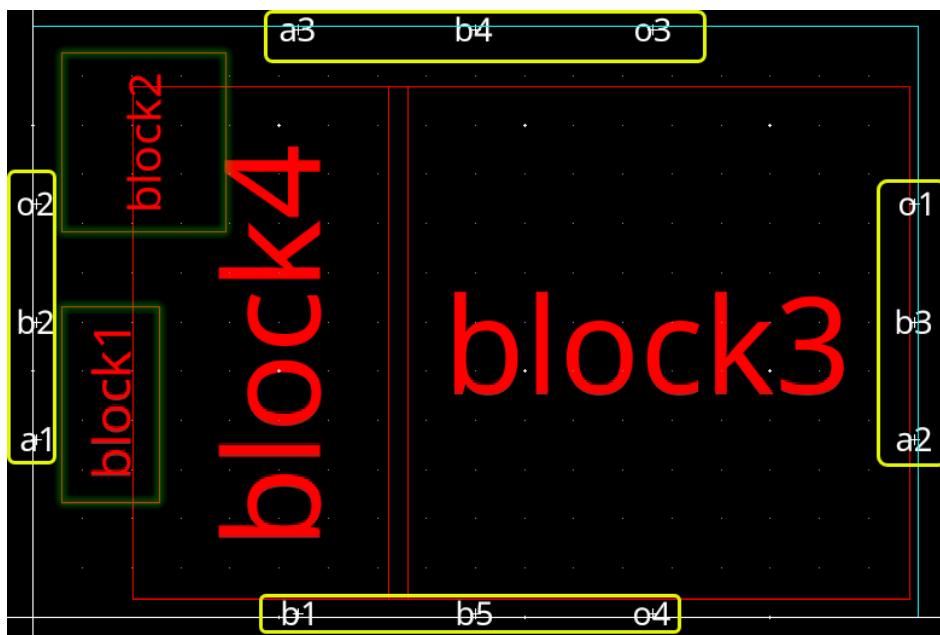
1. Choose Plan – Pin Planning – Snap Pins.



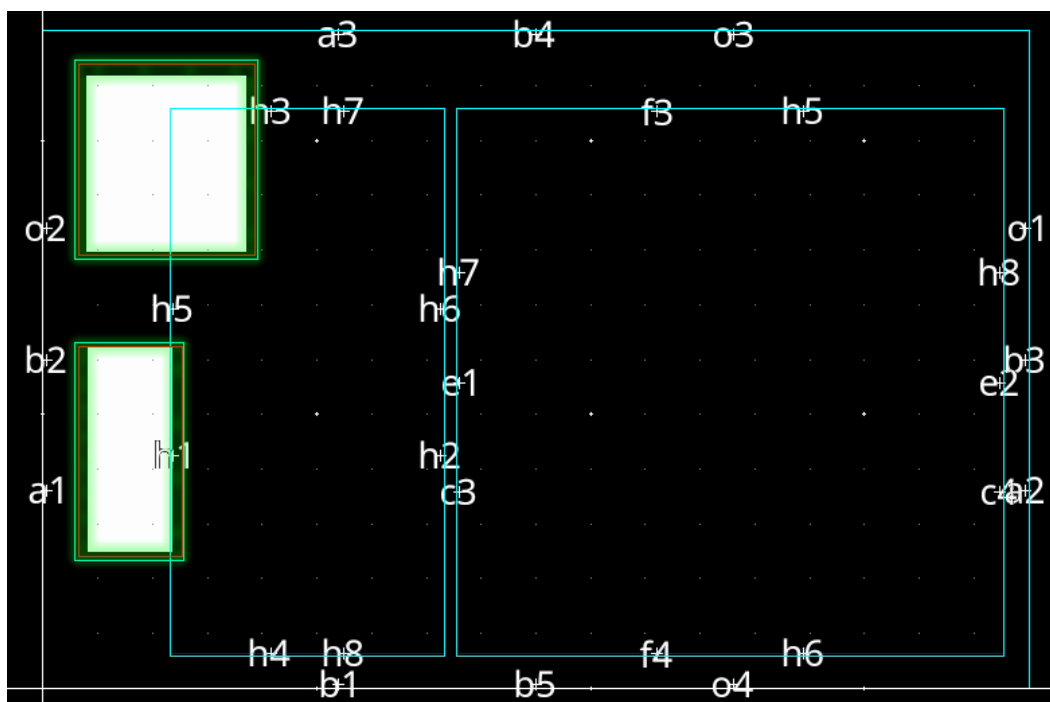
2. Choose Snapping Mode – All and click Snap Pins.



Soft pins get snapped to the *prBoundary*. Custom block soft pins are snapped to the mfg grid, while digital ones are snapped to the routing grid.



Notice how all the top-level and level-1 pins are snapped to the appropriate grid.

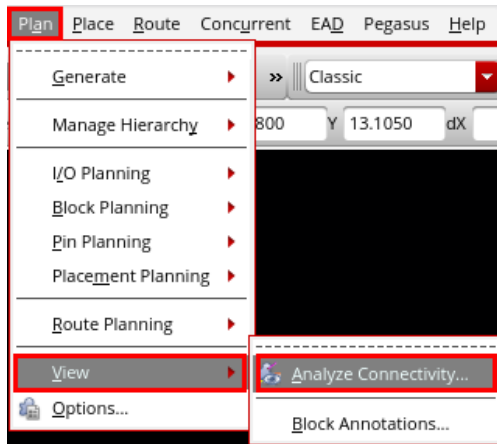


3. Click **Close** on the Snap Pins form to close it.

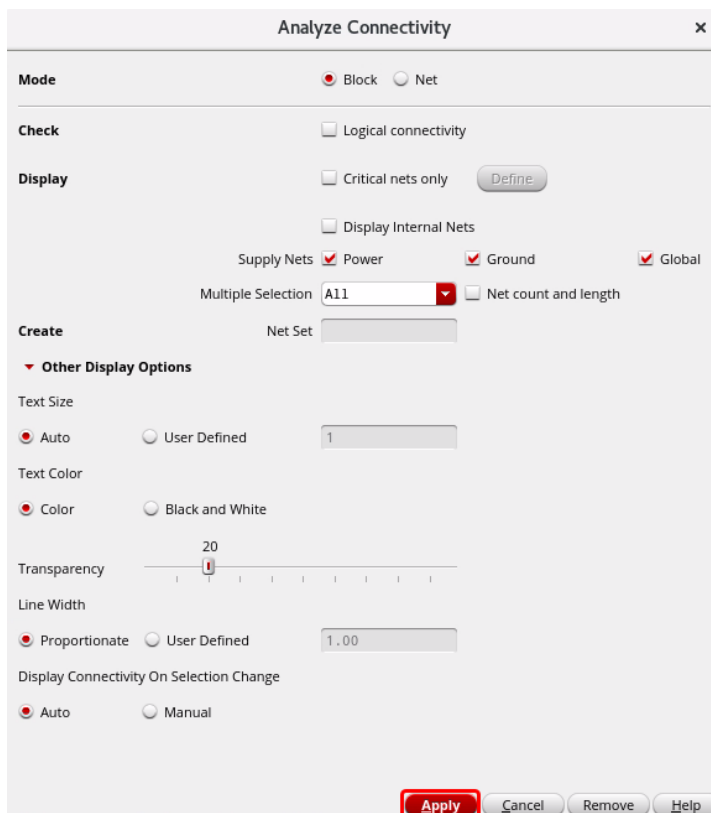
Analyzing the Connectivity

You can analyze the connectivity between blocks to check the number of connections between preselected blocks.

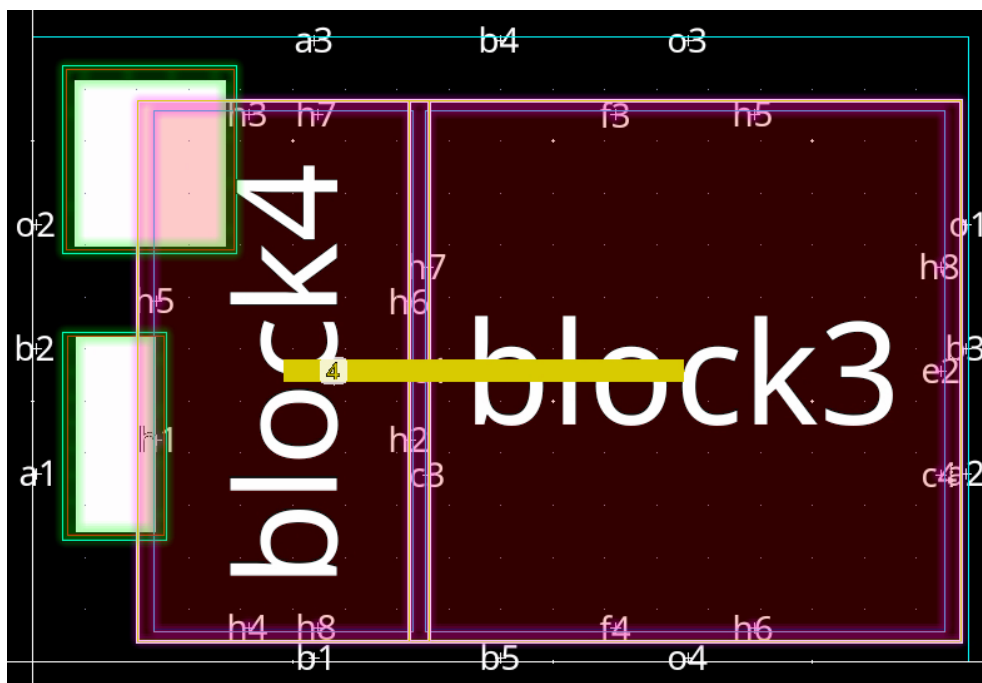
1. Select **block3** and **block4**.
2. In the layout canvas, choose **Plan – View – Analyze Connectivity** or click the **Analyze Connectivity** icon to open the Analyze Connectivity form.



3. Keep the values as shown here, and click **Apply** in the Analyze Connectivity form.



The **numeric 4** represents the number of connections between the selected blocks.



4. Click **Cancel** in the Analyze Connectivity form.
5. Press **Ctrl+D** to deselect all the objects in the layout window.
6. Keep the cellviews open and the Virtuoso session running and continue with the next lab.

Summary

In this lab, you learned:

- ♦ How to generate the physical hierarchy and place the blocks of the design.



Lab 4-4 Editing the Level-1 Objects

Objective: To use Level-1 editing to modify the soft blocks in the hierarchy.

In this lab, you:

- ♦ Edit the soft blocks at **Level-1** without having to descend into the block itself.

Level-1 Editing

Level-1 Editing is a way to edit the soft block from the top level without using the **Edit In Place** or **Descend** commands. You can access only the *prBoundary* and the pins.

Verify that your session is still open from the previous lab. If not, open them now.

1. In the layout window, choose **Options – Display** or press the bindkey **E** and make sure that the **Stop Level** field is set to **1**. You can set it deeper, but it must be at least set to **1**.

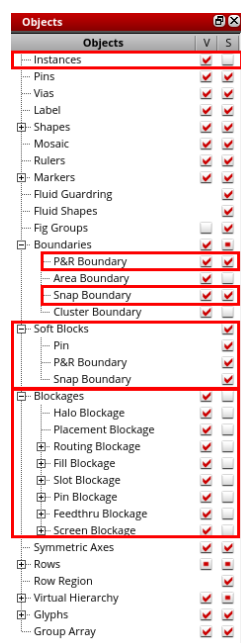
Note: Setting it to 1 gives you the cleanest view of Level-1 shapes.

2. In the **Objects** assistant, turn **ON** the Selectability for the following:

- Boundaries: P&R Boundary and Snap Boundary
- Soft Blocks: Pin, P&R Boundary, and Snap Boundary

3. In the **Objects** assistant, turn **OFF** the Selectability for the following:

- Instances and Blockages

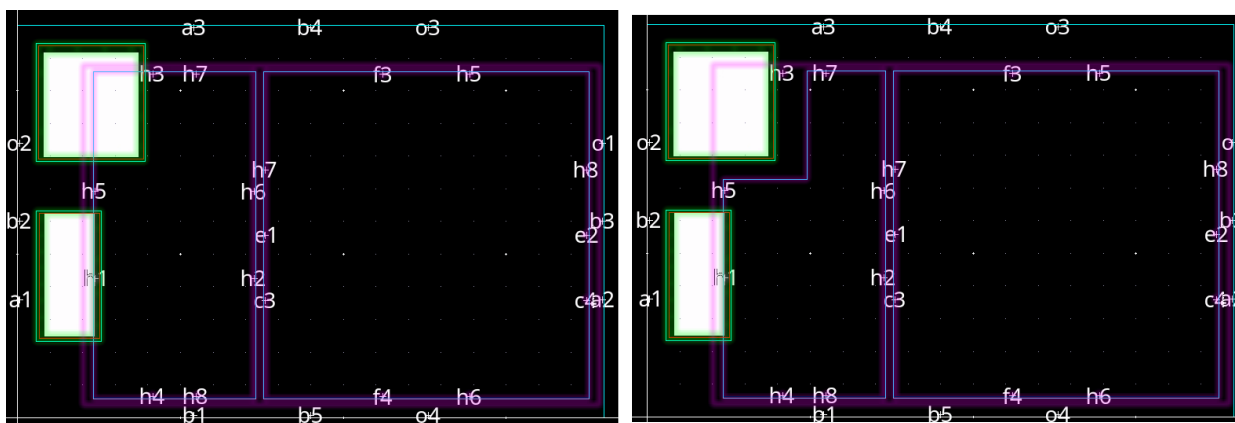


Using the Chop Command to Resize the Soft Block Boundary

You can use the **Chop** command to cut away part of an object or cut an object into pieces.

Note: Partial chopping of instances and vias is not supported.

1. In the layout canvas, select the boundary of the desired soft block to chop (**block4**).
2. Choose **Edit – Basic – Chop** or press the bindkey **Shift+C** and chop, as shown here.



3. Press the **Esc** key to stop the **Chop** command.

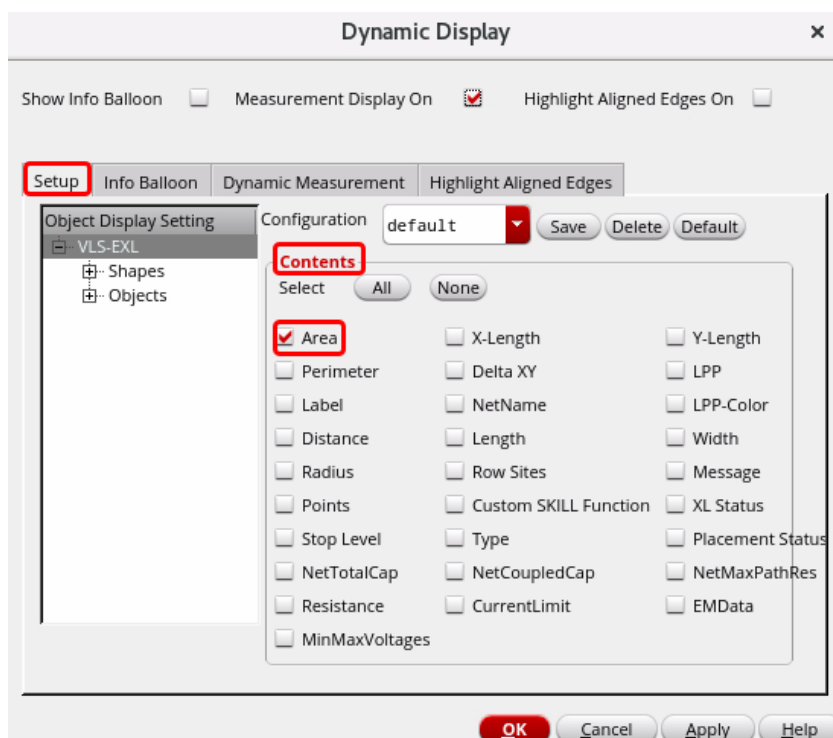
Using the Constant Area Stretch Option in Stretch Command

You can stretch the boundary keeping the area constant. It helps to resize the aspect ratio of the block with a fixed area.

Note: **Constant area stretch** supports both rectilinear and rectangular blocks.

1. Press **Ctrl+D** to deselect all the objects in the layout window.
2. Choose **Options – Dynamic Display** in the layout window.
The Dynamic Display form appears.
3. In the Dynamic Display form, enable the **Measurement Display On** option.

- Click the **Setup** tab, and in the **Contents** field, click **None**.



- Enable **Area** and **OK** the form.
- In the layout window, choose **Edit – Stretch** or click the **Stretch** icon or press the bindkey **S**.
- Press **F3** to display the Stretch form.
- Enable the **Constant area stretch** option in the **Boundary** section and expand the Progressive Disclosure Widget.

As the **Boundary** section gets expanded, the form gets updated with additional options to adjust the desired edge.

9. Enable the **Left edge** and **Right edge** options and disable the **Top edge** and **Bottom edge** options, as shown here.

Stretch [X]

☒ Lock angles

Snap Mode: orthogonal [v]

Delta X,Y: 0 0 [Apply]

☒ Display draglines

▼ Connections

☐ Move Connected Instances And Wires

Keep Wires Connected to: Spine wires only [v]

☐ Keep wires connected to shapes

☐ Lock Unselected Vias

☐ Keep guard ring ends connected

Chain Mode: Selected [v]

▼ Special Objects

☐ Mosaic partial selection

☐ Via partial selection

Via Mode: Stretch Metal [v]

☐ Allow opposite metal extend

▼ Snap

☐ Pins and boundary to grid

☐ Pins to boundary

Soft Block Pins to: ☒ Modified Edge ☐ Closest Edge

▼ Boundary

☒ Constant area stretch

☒ Left edge

☒ Right edge

☐ Top edge

☐ Bottom edge

☐ Stretch with P&R boundary

[Hide] [Cancel] [Help]

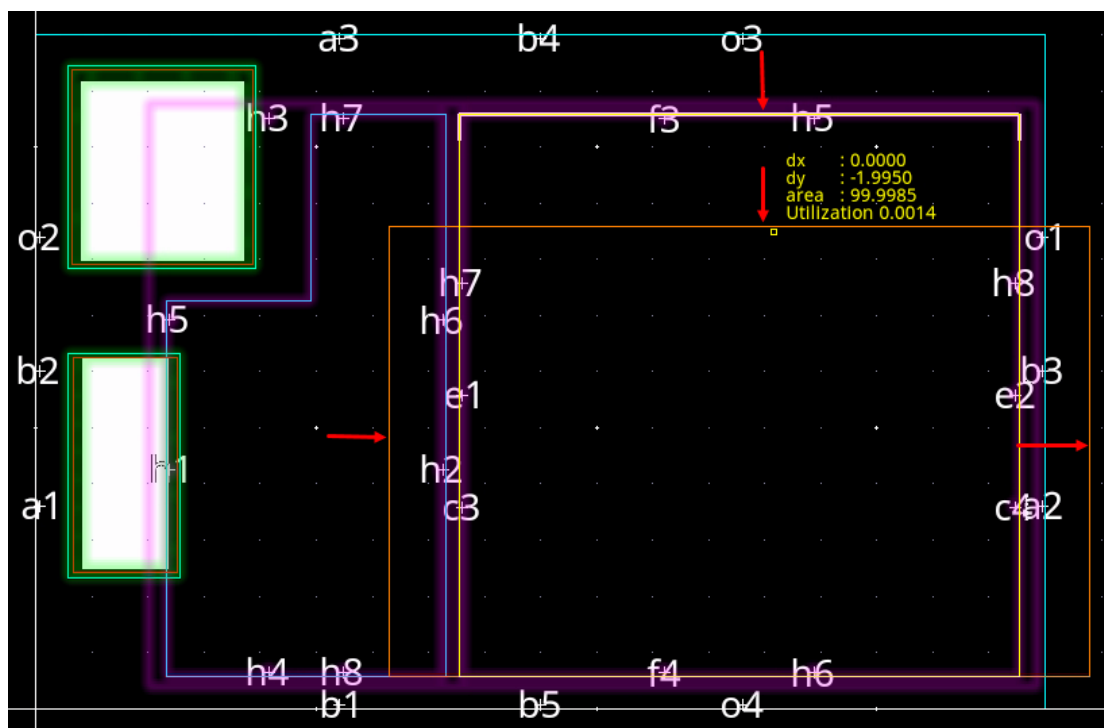
This setting allows the movement of the left and right edges to adjust the area.

10. Click **Hide** on the Stretch form.

11. Click the top edge of **block3** *prBoundary*.

12. Stretch the top edge and notice that the left and right edges move to keep the area constant.

Notice that the area has remained the same after stretching, as shown here.



Note: You may choose not to make the above changes. After post-editing, the *prBoundary* of the soft block **Snap Pins** command needs to be re-run.

13. Press the **Esc** key to stop the **Stretch** command.

14. Leave the Cadence windows and session open for the next lab.

Summary

In this lab, you learned:

- ♦ How to use Level-1 editing to modify the soft blocks in the hierarchy.



Lab 4-5 Using the Pin Optimizer

Objective: To use the pin optimizer to optimize the level-1 and top-level pins.

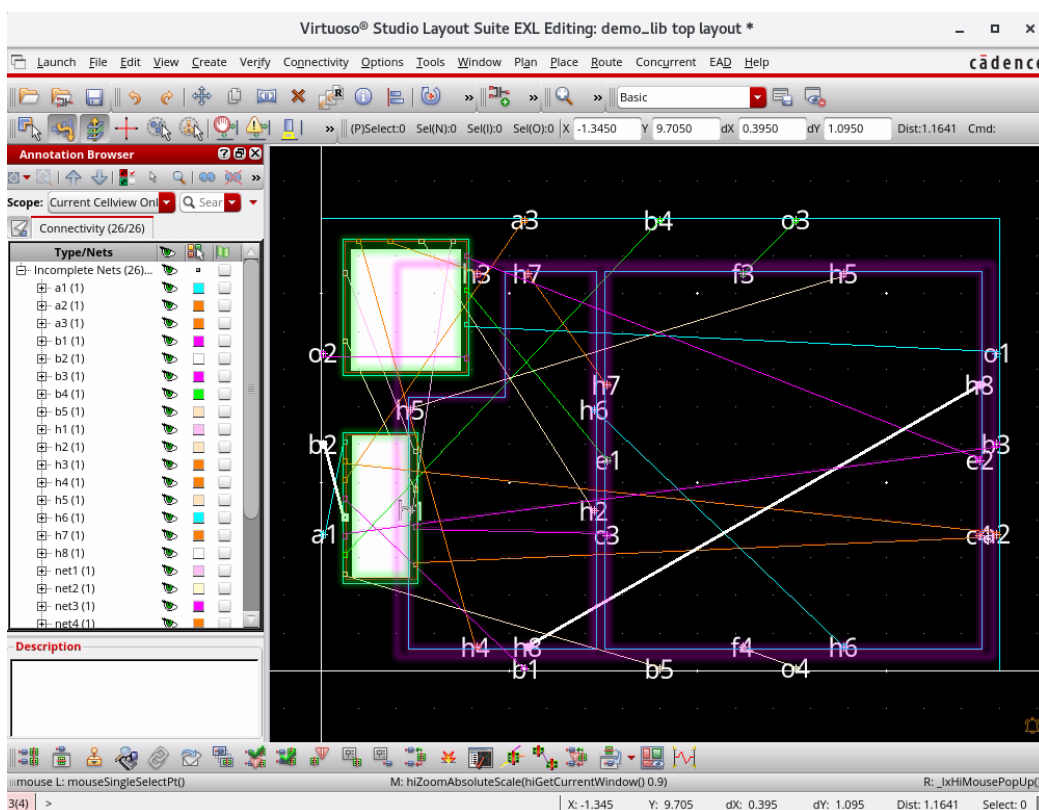
In this lab, you:

- ◆ Use pin optimization on the soft block pins and top-level pins.

Your cellviews should still be open from the previous lab. If not, re-open the cellviews.

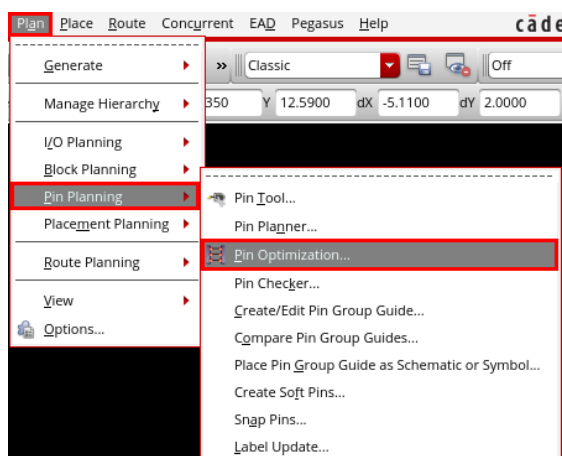
Optimizing the Soft Block Pins and Top-Level Pins

1. Press **Shift+F** to view all levels of the hierarchy in the layout window.
2. In the layout canvas, choose **Window – Assistants – Annotation Browser**.
3. Click the **eye** icon in the Annotation Browser to turn on the flight lines to review the connectivity across the blocks.

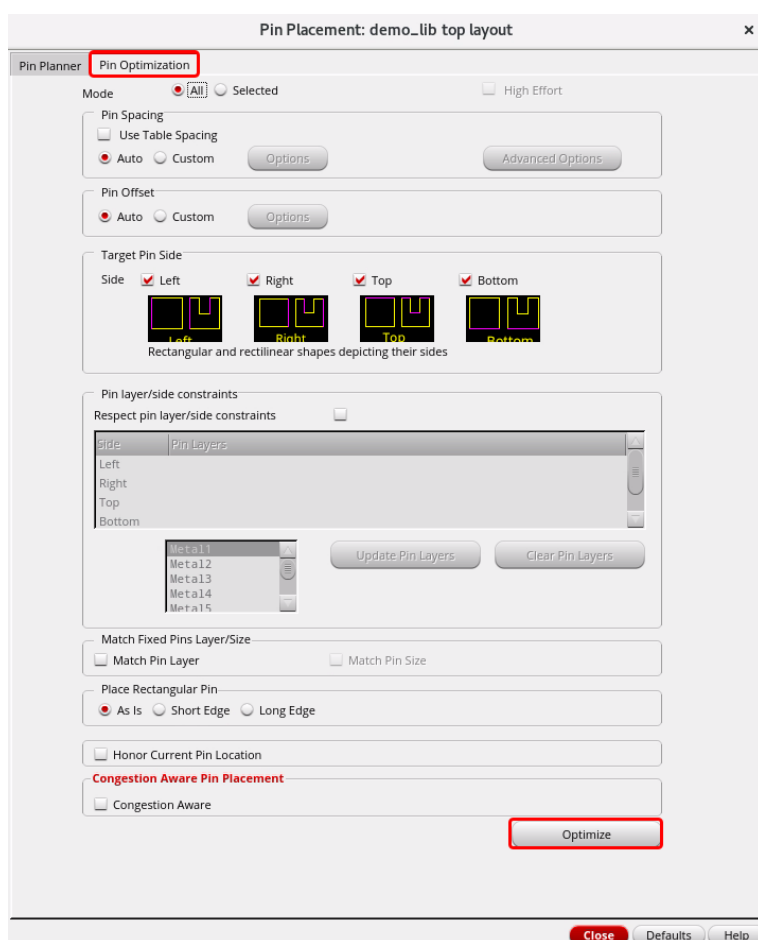


Currently, the connectivity looks messed up. Note that the pins are yet to be optimized.

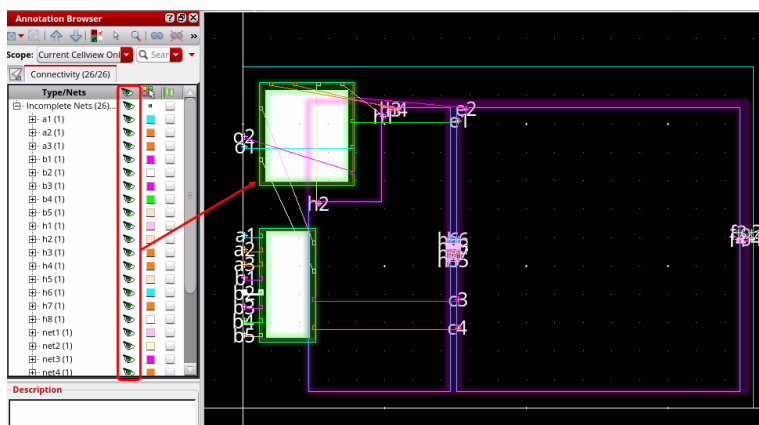
4. In the layout window, choose **Plan – Pin Planning – Pin Optimization** or click the **Pin Optimizer for All Pins** icon or choose **Plan – Pin Planning – Pin Planner** to optimize the pins.



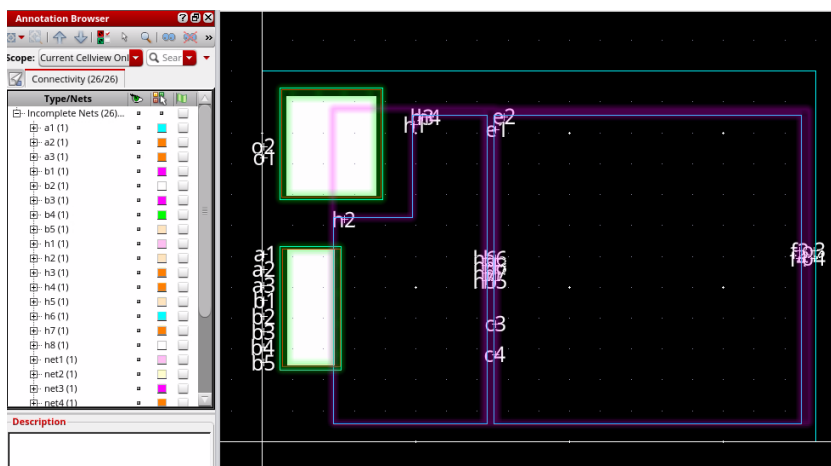
5. In the Pin Placement form, in the **Pin Optimization** tab, select the following options:
 - a. Select *Mode* as **All**.
 - b. With other options as default values, click **Optimize**.



All the pins are optimized. The connectivity flightlines are cleaned up now across blocks. You can see that the pins are perfectly aligned.



6. Turn off the flight lines by clicking the eye icon.



This being a simple design, notice that the pins are also aligned between the blocks. However, in aggressive designs, you might not achieve pin alignment during the pin optimization step. In such cases, you will choose to run the Pin Alignment step, which you will see in the next lab.

7. Close all the cellviews without saving the changes and keep the Virtuoso session running.

Summary

In this lab, you learned:

- ◆ How to use the pin optimizer to optimize the soft block pins and top-level pins.

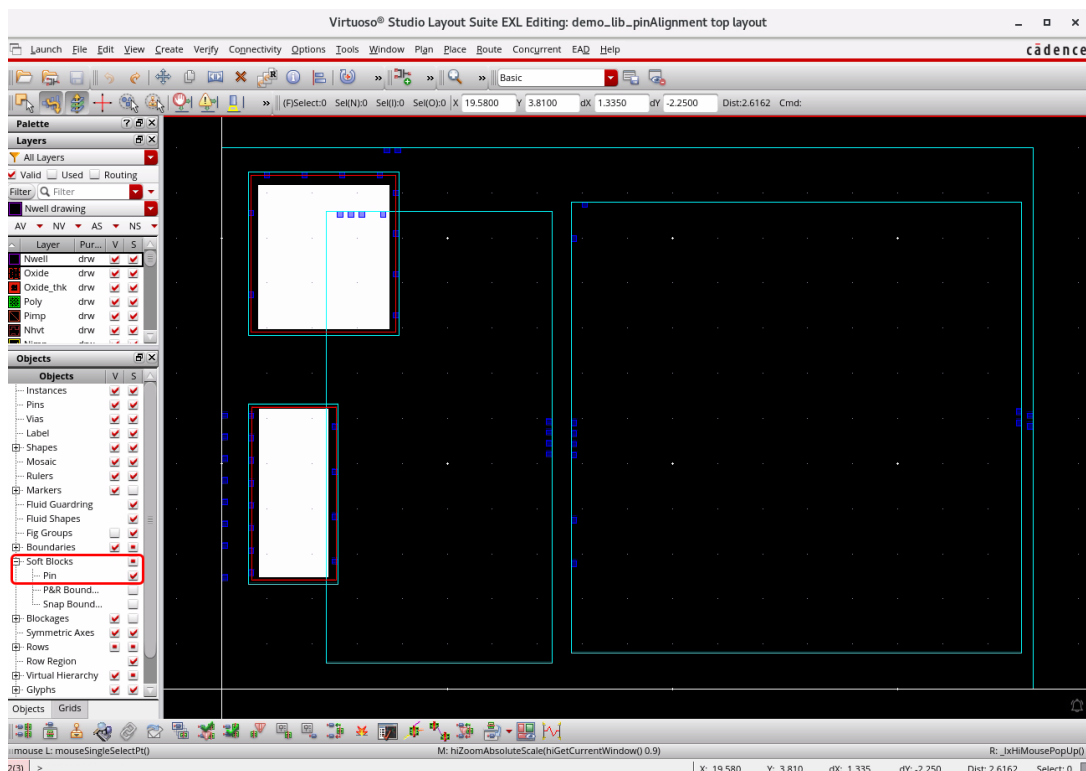


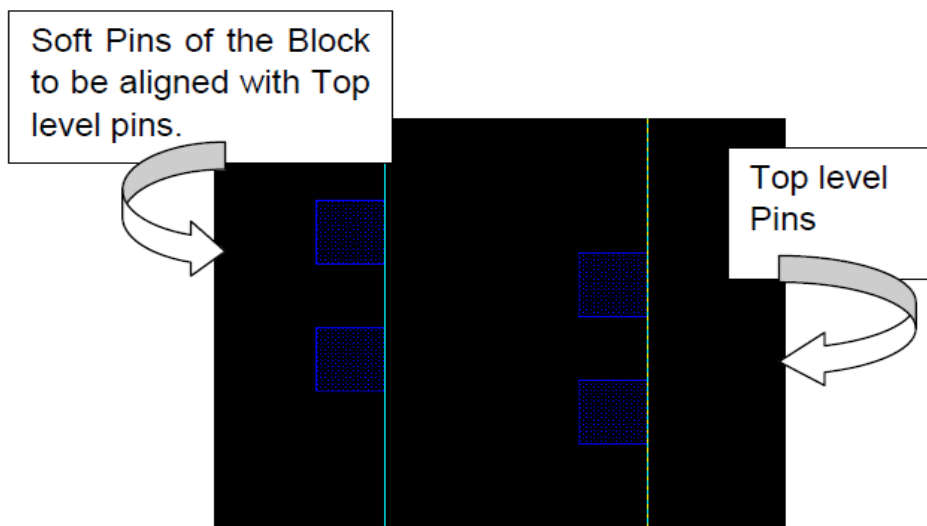
Lab 4-6 Aligning the Soft Block Pins with the Top-Level Pins

Objective: To align the soft block pins with the top-level pins using the pin alignment command.

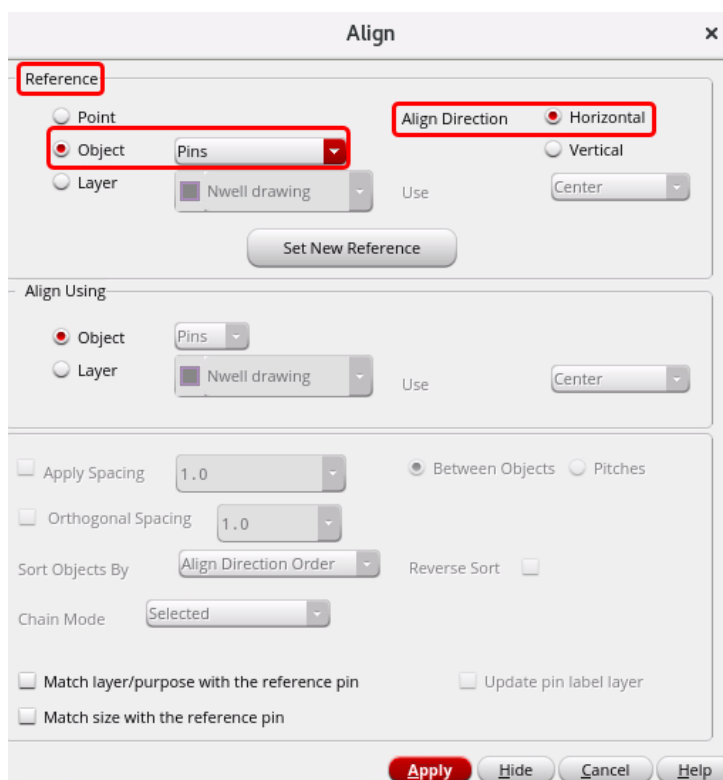
In this lab, you:

- ◆ Align the soft block pins with the top-level pins using the pin alignment command.
1. In the **CIW** window, open the **demo_lib_pinAlignment/top/schematic** cellview.
 2. From the schematic, choose **Launch – Layout EXL** and open the existing layout view **demo_lib_pinAlignment/top/layout**.
You will align the pins enclosed in the **red box** shown here.
 3. Press **Shift+F** to view all levels of the hierarchy.
 4. Turn on Selectability of **Soft Blocks Pin** in the **Objects** assistant, as shown here.

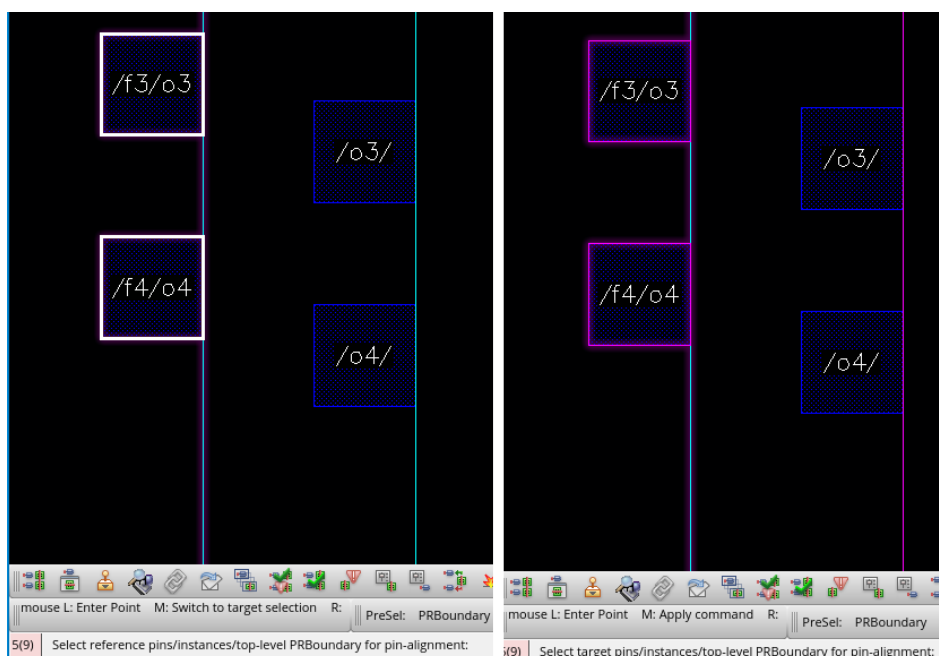




5. In the layout canvas, choose **Edit – Advanced – Align** or use the **Align** toolbar icon to invoke the Align command.
6. In the Align form, in the **Reference** section, select the options, as shown here.
 - a. *Object: Pins*
 - b. *Align Direction: Horizontal*

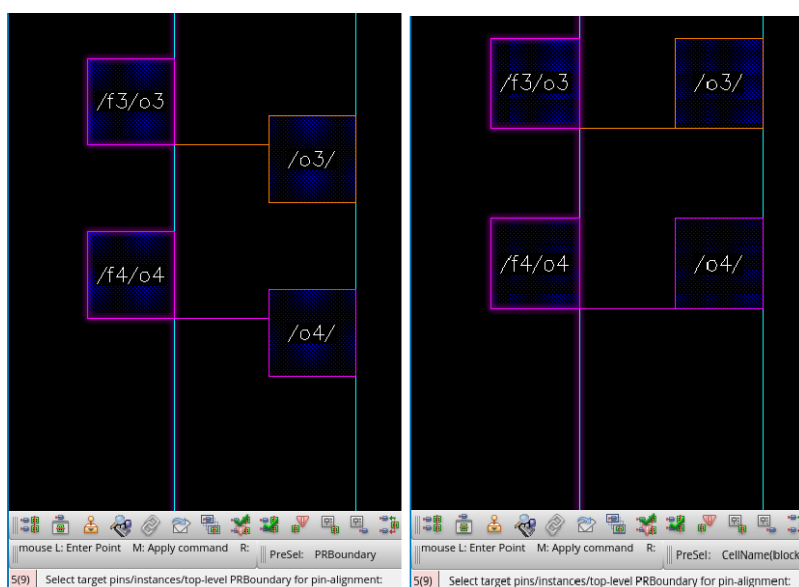


7. After selecting the reference soft pins in the block, click the middle mouse button.



Notice the soft pins in the reference block turns pink in color, and command mode changes to select target pins.

8. Turn on the **Annotation Browser** assistant to turn on the connectivity between reference and target pins.
9. After selecting the target pins of the top level on the RHS, click the **MMB**.



The soft block pins are aligned with the top level pins.

10. Press the **Esc** key to cancel the **Align** command.

Pin Alignment Report in the CIW

Refer to the *Pin Alignment Report* in the CIW window.



In case you do not get the pin alignment report, you will get information stating the reason for the same in the CIW window.

1. Close all the cellviews without saving the changes and keep the Virtuoso session running.

Summary

In this lab, you learned:

- ◆ How to align the soft block pins with the top-level pins using the pin alignment command.

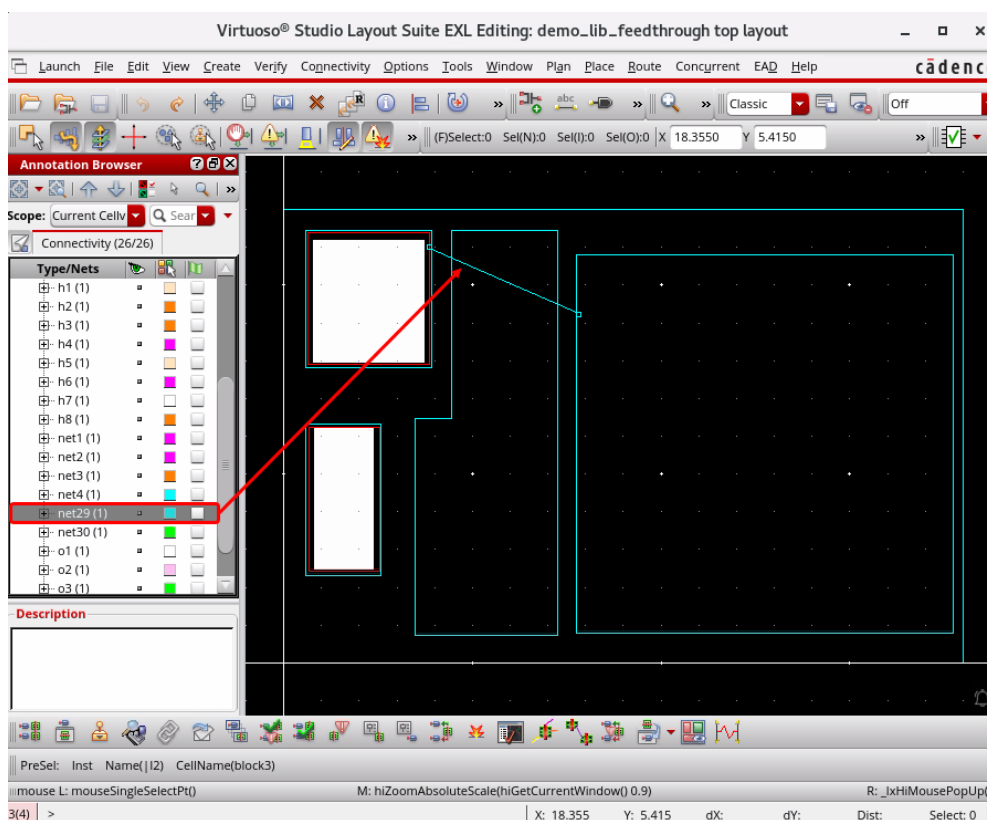


Lab 4-7 Creating Feed Through Terminal Pins

Objective: To create feed through terminal pins.

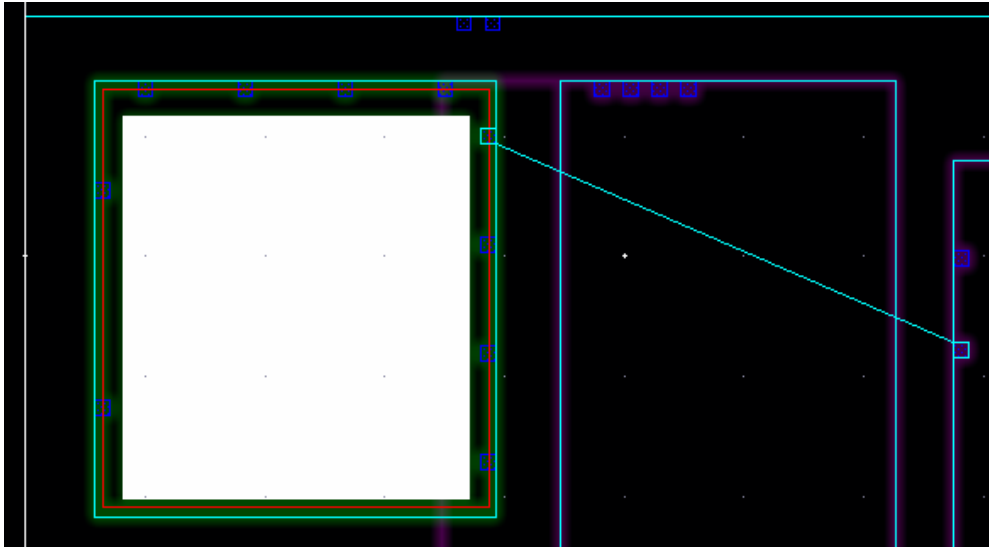
In this lab, you:

- ◆ Create Feed Through Terminal Pins. You can use this feature to improve the connectivity and reduce the area resource during the routing phase in your design.
1. In the **CIW**, open the **demo_lib_feedthrough/top/schematic** cellview.
 2. From the schematic, choose **Launch – Layout EXL** and open the existing layout view **demo_lib_feedthrough/top/layout**.
 3. In the canvas, from the **Objects** assistant, turn **ON** the Selectability for **Instances**.
 4. In the layout window, turn on the **Annotation Browser** assistant.

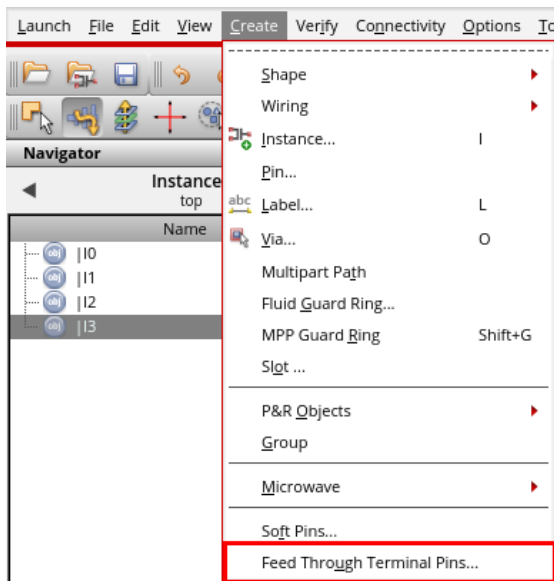


5. Zoom in around the pins connecting **net29**, which appear between Instances **I1** and **I2**.

Soft block **I3** is placed between the two blocks (Instances **I1** and **I2**) and will be used as a feedthrough block for **net29**, as shown here.



6. Select Instance **I3** (**block4**) to be used as a feedthrough block.
7. In the layout canvas, choose **Create – Feed Through Terminal Pins**.



This opens the Create Feed Through Terminal Block Pins form.

You can notice that Soft Block **block4** appears under the *Soft block list*. **block4** will be used as a feedthrough.

8. Specify the layer, pin size, and the block side on which feedthrough pins need to be created in the pin parameter section using the **Add** button.
 - a. Select *Net/Bus/NetClass* **net29**.

- b. Select **Layer Metal1 drawing**; **Width 0.12**; **Height 0.12**; **Num 1**; **Side Left** – Click **Add**.
- c. Select **Layer Metal1 drawing**; **Width 0.12**; **Height 0.12**; **Num 1**; **Side Right** – Click **Add**.
- d. Select **Align Pins** to enable automatic pin alignment.
- e. Click **OK** in the Create Feed Through Terminal Block Pins form.

Create Feed Through Terminal Block Pins

Soft block list: demo_11b_feedthrough.block4.layout

Lib: demo_11b_feedthrough
Cell: block4
View: layout

Select Net/Bundle

Search for: Nets Name: .* Filter

Net/Bus/NetClass

- h8
- net29
- net30
- o1
- o2

Define TTB pin parameters

User defined terminal name: ftTerm_net29

Layer	Width	Height	Num	Side
Metal1 drawing	0.12	0.12	1	Left
Metal1 drawing	0.12	0.12	1	Right

Layer: Metal1 Width: 0.12 Height: 0.12 Num: 1 Side: Right

Add Update Delete

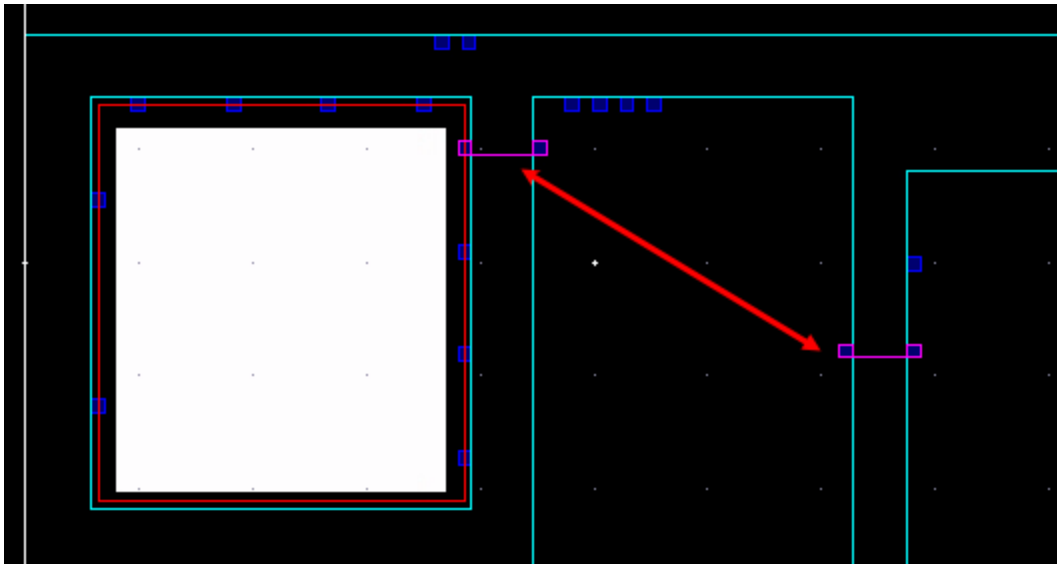
Align Pins ☒

Left Right Top Bottom

Rectangular and rectilinear shapes depicting their side constraints

OK Cancel Apply Help

The Feed Through Pins will be created on instance **I3 (block4)**, as shown here.



This will help with less area resources during the routing phase.

9. Close all the cellviews without saving the changes and exit the Virtuoso session.

Summary

In this lab, you learned:

- ♦ How to create Feed Through Terminal Pins.



Module 5: Top-Down Floorplanning

Lab 5-1 Creating a Top-Down Floorplan

Objective: To floorplan to a specific die size, assuming you have no physical design available and complete a top-level floorplan using all soft blocks.

In this lab, you:

- ◆ Use the abstract view in the **Configure Physical Hierarchy** tool to create the new soft blocks for your floorplanning exercise.

Using the Soft Block Features for Top-Down Planning

1. Start a new Virtuoso® session in the `./VLPT7_IC_25_1/mod05_VFP_pads/design/custom` directory.
2. Use the Library Manager window to select and open (in read-only mode) the following schematic view in XL mode:

Library	ether
Cell Name	top
View Name	schematicWithPads

3. From the schematic window, choose **Launch – Layout EXL**.
The Startup Option form appears.
4. Select **Create New** in *Layout* and **Create New in Configuration** fields.
5. Click **OK** in the Startup Option form.
6. In the New File form, set *View* as **abstract** and *Type* as **config**.
7. Click **OK** in the New File form.
8. In the Create Physical Configuration View form, select the **Use Template** and **Open CPH options**.
9. Click **OK** in the Create Physical Configuration View form and then **OK** in the dialog box to overwrite the existing file.

Setting the Cell Type from the CPH

1. In the Configure Physical Hierarchy form, select the **Soft Block** mode.

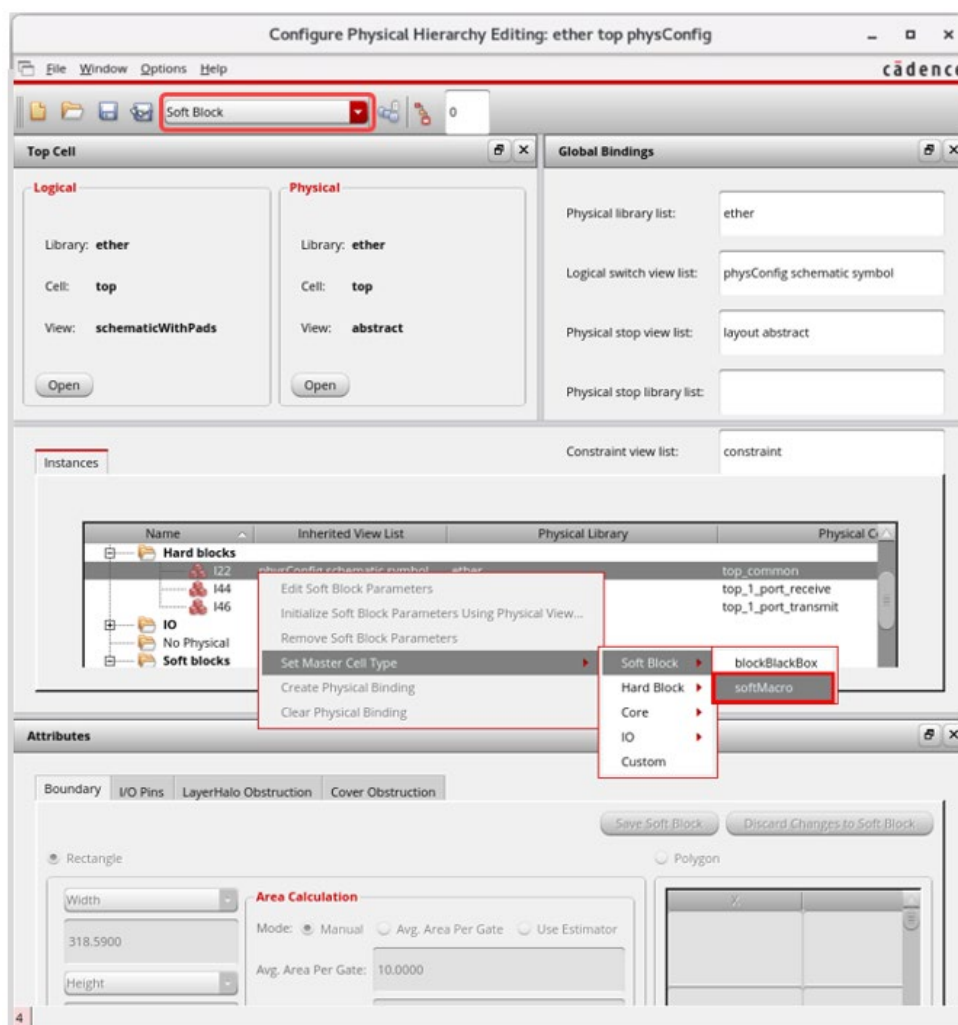
2. Expand the **Hard blocks** and **Soft blocks** folders.

Note: If blocks are defined and available in **Hard blocks**, follow the procedure given here and make sure the block types are set to **softMacro**. If all the instances are already in the **Soft blocks** section, proceed to the next part of the lab.

Do **NOT** change the **IO blocks**.

3. Select **I22**.

4. **Right-click** and select the *Set Master Cell Type* field to **Soft Block-softMacro**.



5. Repeat this for the remaining hard blocks **I44** and **I46**.
You are declaring the blocks to be adjustable.
6. Change the *Physical stop view list* to the **abstract layout**.
This loads the abstracts in priority over layouts.

Specifying the Top-Level Fixed Area Boundary

1. In the *Instances* field, select **top** with a left-click.
2. **Right-click** and select **Define Top Level Parameters**.
The Attributes section of the form becomes editable.
3. In the **Attributes** section, in the **Boundary** tab, enable the **Rectangle** radio button.
4. In the **Attributes** section, in the **Boundary** tab:
 - a. Set *Width* to **2100**.
 - b. Set *Height* to **1800**.
5. Click **Save Soft Block** in the Attributes section.

Setting the Soft Blocks to Calculate Area and Shape

Three-block areas are given. Use the known area values for the *top_common*, *top_1_port_transmit*, and *top_1_port_receive* soft blocks.

1. Select *top_common* in the **Soft blocks** listing.
2. **Right-click** the selection and select **Edit Soft Block Parameters**.
The Attributes section of the form becomes editable.
3. Start with **0.5 Aspect Ratio** and **90% Utilization** for *top_common* soft block.
 - a. Set the **Area Calculation Mode** to **Manual** and enter the *Core Area* value as **497700**.
 - b. Set the Default pin layer to **Metal3 drw**.
 - c. Set the *Power* and *Ground I/O* pin sizes to **11 x 11**.

- d. Set the *LayerHalo Obstruction* for the *pin* from **Metal1** through **Metal4**.

Attributes

Boundary I/O Pins LayerHalo Obstruction Cover Obstruction

Save Soft Block Discard Changes to Soft Block

Rectangle

Aspect Ratio (W/H) 0.5000

Utilization (%) 90.0000

Rail Height 0.0

Area Calculation

Mode: Manual Avg. Area Per Gate Use Estimator

Avg. Area Per Gate: 10.0000

Gate Count: 100

Area Estimator: PR Boundary Based Estimate

Area: 497700.0000 Core Area: 497700.0000

Polygon

Attributes

Boundary I/O Pins LayerHalo Obstruction Cover Obstruction

Save Soft Block Discard Changes to Soft Block

Create Label Clear All Filters Add Pin Delete Pin

Net Name	Term Name	Term Type	Layer	Width	Height	Number	Criticality	Sig Type
23 PLL160	PLL160	output	Metal3 (drw)	0.1400	0.1400	1	0	signal
24 RBIAS	RBIAS	inputOutput	Metal3 (drw)	0.1400	0.1400	1	0	signal
25 RBIAS_SENSE	RBIAS_SENSE	input	Metal3 (drw)	0.1400	0.1400	1	0	signal
26 SIDDQ	SIDDQ	input	Metal3 (drw)	0.1400	0.1400	1	0	signal
27 VDDA_160	VDDA_160	input	Metal3 (drw)	11.0000	11.0000	1	0	supply
28 VDDA_250	VDDA_250	input	Metal3 (drw)	11.0000	11.0000	1	0	supply

Attributes

Boundary I/O Pins LayerHalo Obstruction Cover Obstruction

Save Soft Block Discard Changes to Soft Block

Add Blockages

Blockage Type

Material all

Layer Cont

Offsets

Top 0.0000 Bottom 0.0000

Right 0.0000 Left 0.0000

Type	Layer	Top	Bottom	Right	Left
1 routing	Metal1	0.0000	0.0000	0.0000	0.0000
2 routing	Metal2	0.0000	0.0000	0.0000	0.0000
3 routing	Metal3	0.0000	0.0000	0.0000	0.0000

Add Delete

4. Click **Save Soft Block** in the Attributes section and continue.

5. Set these values for the first pass for the rest of the soft block attributes.
 - a. Default pin layer to **Metal3 drw**.
 - b. *Power* and *Ground I/O* pin sizes to **11 x 11**.
 - c. *Utilization* to **80%**.
 - d. *LayerHalo Obstruction* for *pin* from **Metal1** through **Metal4**.
 - e. *Bias_port Aspect Ratio* to **2.0**; **Area Calculation Mode** to **Use Estimator**.
 - f. *anlg_cntl Aspect Ratio* to **1.0**; **Area Calculation Mode** to **Use Estimator**.
 - g. *top_l_port_receive Aspect Ratio* to **1.0**; *Core Area* to **1290700**.
 - h. *top_l_port_pnr* fixed dimension *Width* to **735**; *Height* to **380**.
 - i. *top_l_port_transmit Aspect Ratio* to **1.0**; *Core Area* to **621600**.

Saving the Floorplan Properties File

The floorplan properties file contains the soft block's parameters.

1. Choose **File – Save** or click the **Save** icon in the CPH form.
2. Choose **File – Save Floorplan File** to open the *Save Floorplan Property File* form.
3. In the Save Floorplan Property File form, enter the **File name** as the default *floorplanPropertyFile.txt* and click **Save**.
4. Leave the CPH window **active**. You might be using it to change soft block *Aspect Ratio* or cell sizes if you choose to do so.

Generating the Estimated Block Sizes

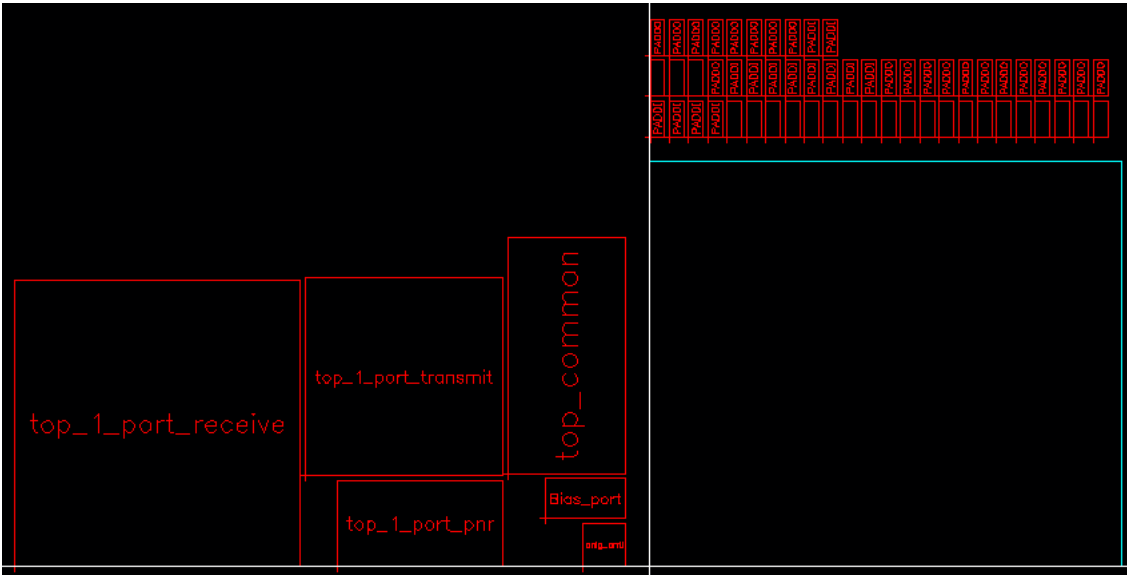
1. In the layout canvas, choose the **Plan – Generate – Generate Physical Hierarchy** menu command or click the **Generate Physical Hierarchy** icon in the abstract window.

The Generate Physical Hierarchy form opens.

Generate Physical Hierarchy			
Generate Options			
☒ Create Instances	☒ Create Pins	☒ Create Boundary	
☒ Extract Top Level Design			
Soft Block List			
Generate ☒ all ☐ none			
LibName:CellName:ViewName	Param Changed	Layout Exists	Generate
ether:top:abstract	YES	YES	☒
ether:top_common:abstract	YES	YES	☒
ether:Bias_port:abstract	YES	YES	☒
ether:anlg_cntl:abstract	YES	YES	☒
ether:top_1_port_receive:abstract	YES	YES	☒
ether:top_1_port_pnr:abstract	YES	YES	☒
ether:top_1_port_transmit:abstract	YES	YES	☒

2. Click **OK** to generate all the new abstract views using the parameters you specified.
3. Click **OK** to bypass any inherited connections issue.

Here is the generated view.



Using the I/O Placer to Create the Pad Rows

1. In the layout canvas, choose the **Plan – I/O Planning – I/O Row Create** menu command or click the **IO Row Create** icon to create the rows.

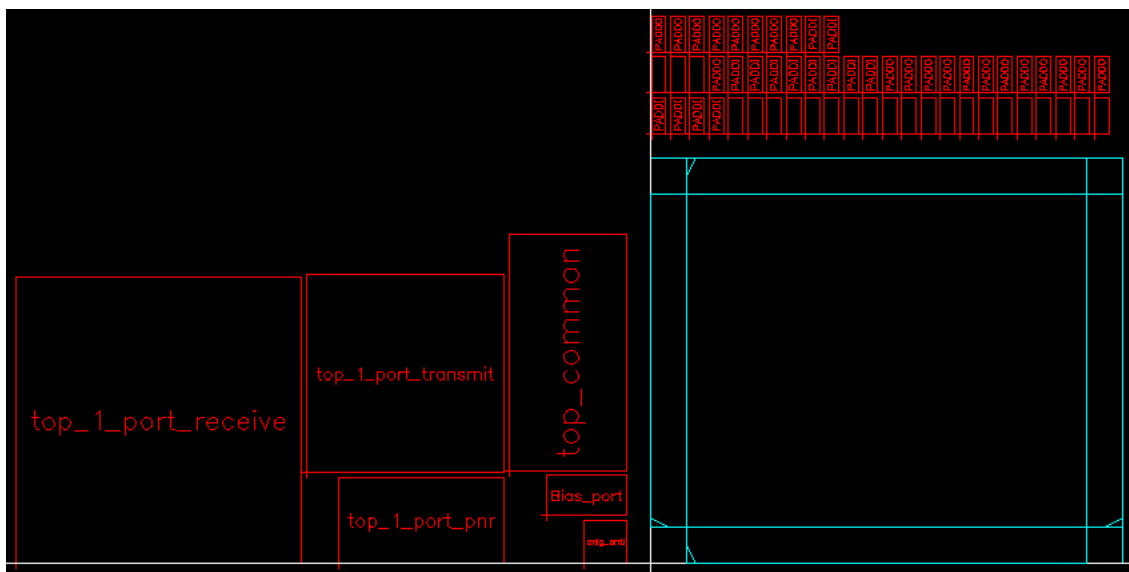
The Create IO Row form opens.

- a. Select the *Rows Per Side* to **One Row**.
- b. Select the *Site Pattern* to **pad** and *Site Orient* to **R0**.
- c. Click **OK** to create the rows.

Can you see the rows?

Answer: To make the rows visible, select the **Row** object in the **Objects** pane and then create the rows.

Answer: Alternatively, in the **Layers** assistant, set **AV** and **AS** to be able to see the rows when they are created.



Using the I/O Placer to Create the Pad Placement

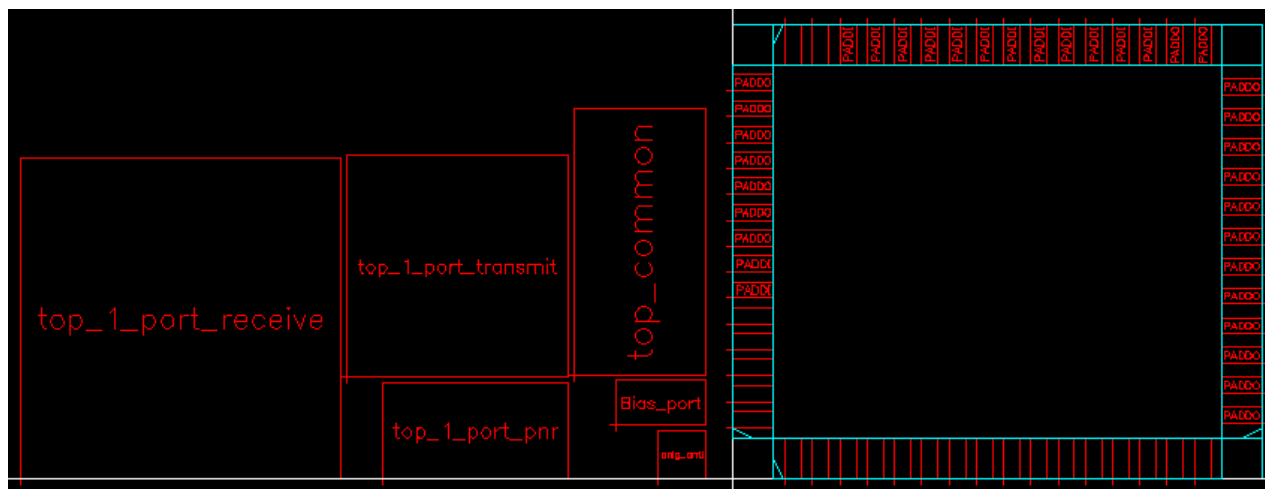
1. In the layout window, choose the **Plan – I/O Planning – I/O Pad Placer** menu command or click the **IO Placer** icon to place the pads.

The IO PAD Placement form opens.

The choices allow for equally spaced pads, abutting pads, or Custom spacing for the pads.

2. Click **OK** to accept the form defaults.

The pads are equally spaced around the perimeter, as shown here.



Using the Block Placer to Place the Blocks

1. In the layout canvas, choose the **Plan – Block Planning – Place Blocks** menu command to place the blocks.

The Block Placer form opens.

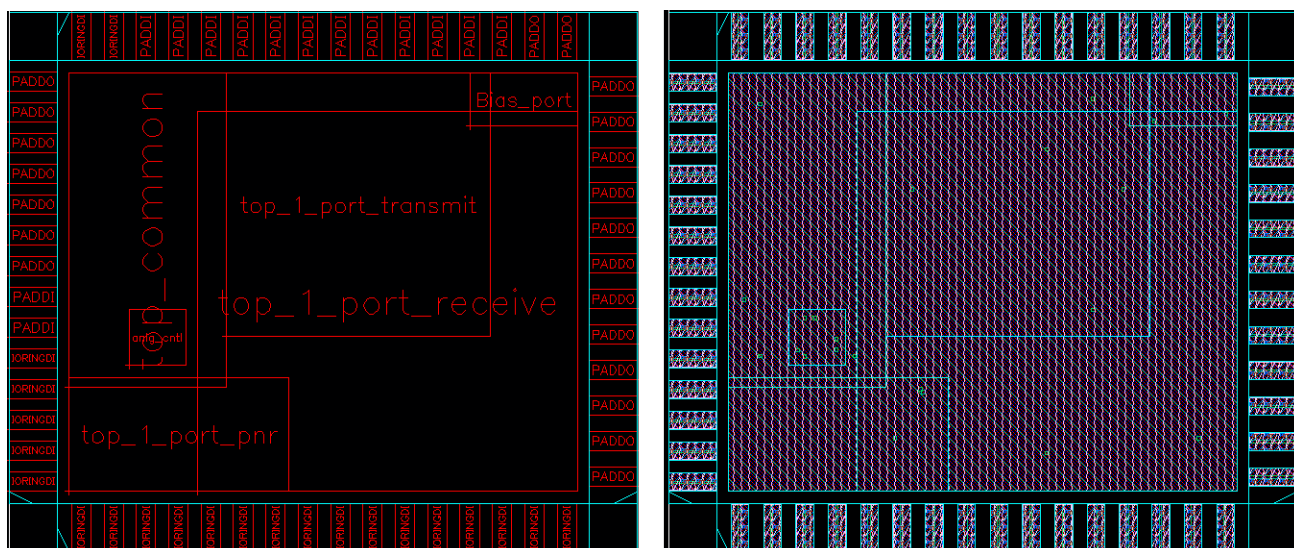
- a. Set *Minimum I/O to Core Distance* to **40**.
 - b. Set *Default Minimum Distance between Instances: User Defined* to **40**.
 - c. Enable **Optimize Top-Level Pins**.
 - d. Click **OK** to place the blocks.
2. In the layout canvas, choose the **Plan – Block Planning – Load Solutions** menu command to cycle through the three solutions.

What happened to the blocks?

Answer: There is not enough room for the blocks.

The result shown here is one possible result with *top level* and *full depth* views.

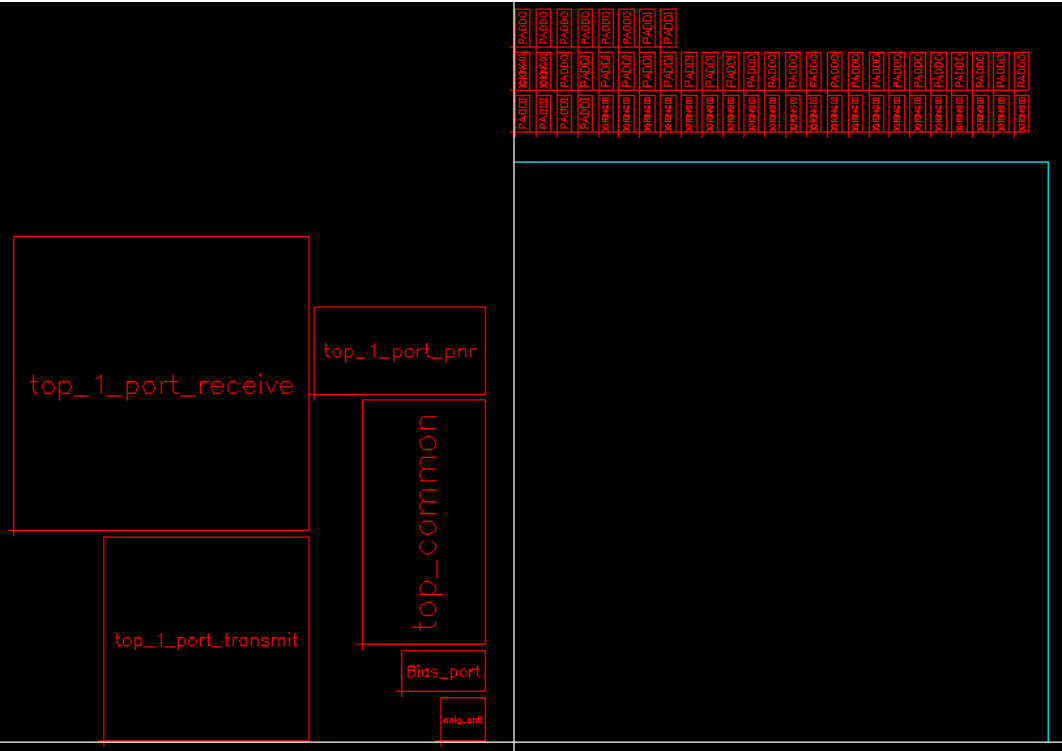
In the *full depth* view, you see the actual blockages in the abstracts.



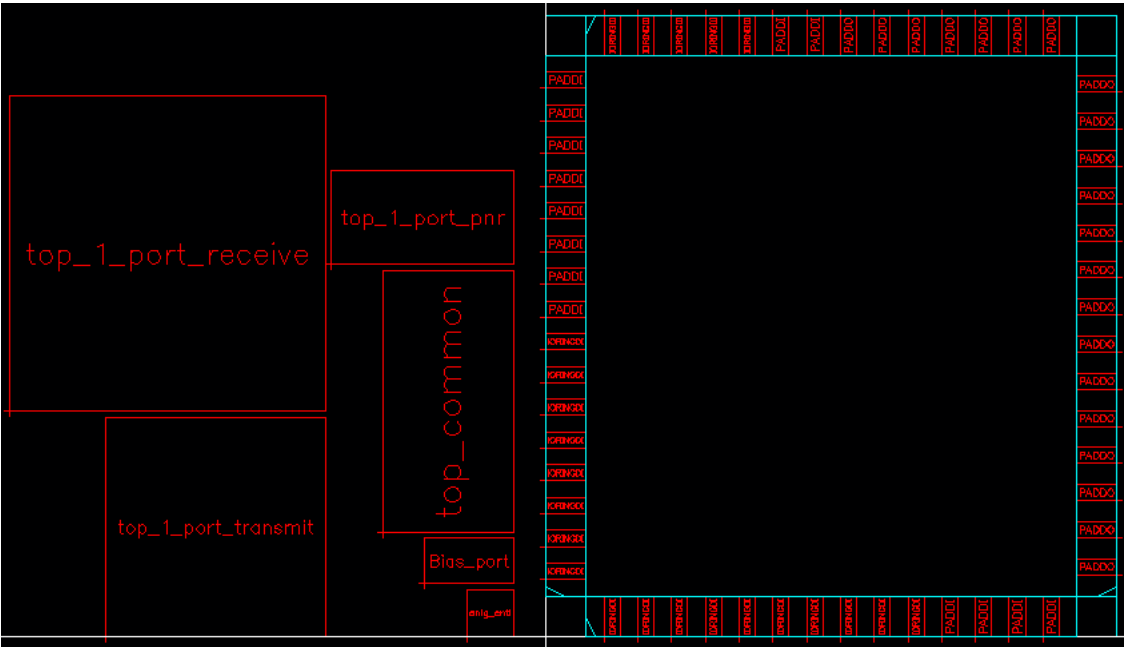
Recalculating the Die Size of the Design

1. Open the CPH form and change the top-level parameters to allow the die size to be calculated.
2. Change the top abstract parameters so the system can calculate the die size.
 - a. Set *Width* to **2300**.
 - b. Set *Height* to **2500**.
 - c. Select **Save Soft Block** in the Attributes section.
3. Choose **File – Save** or click the **Save** icon in the CPH form and save the floorplan file for later use.
4. In the layout canvas, choose the **Plan – Generate – Generate Physical Hierarchy** menu command or click the **Generate Physical Hierarchy** icon in the abstract window.
 - a. Select all the checkboxes to allow for **Regeneration**.
 - b. Click **OK** in the form to generate all the new abstract views.
 - c. Click **OK** to bypass any **inherited connections** issue.

You can notice that the *prBoundary* is larger this time.



- 5. Create the rows and place the pads, as shown here.



Using the I/O Placer to Insert the Filler Cells

1. In the layout window, choose the **Plan – I/O Planning – Insert I/O Filler** menu command to insert the filler cells.

The Insert Filler Cells form opens.

Insert Filler Cells

Region Selection Mode

☒ Entire Design
☐ Selected Region

Draw

ltx: 0 lty: 0 urx: 0 ury: 0 Zoom to

Select Filler Cells

Library Name: giolib090 Cell Name: IoringFEED60 View Name: abstract Filter

Site Name	Library Name	Cell Name	View Name
pad	giolib090	IoringFEED60	abstract

<< >>

"giolib090" : "IoringFEED01" : "abstract"
 "giolib090" : "IoringFEED001" : "abstract"
 "giolib090" : "IoringFEED1" : "abstract"
 "giolib090" : "IoringFEED3" : "abstract"
 "giolib090" : "IoringFEED3X" : "abstract"
 "giolib090" : "IoringFEED0005" : "abstract"
 "giolib090" : "IoringFEED5" : "abstract"
 "giolib090" : "IoringFEED10" : "abstract"
 "giolib090" : "IoringFEED60" : "abstract"

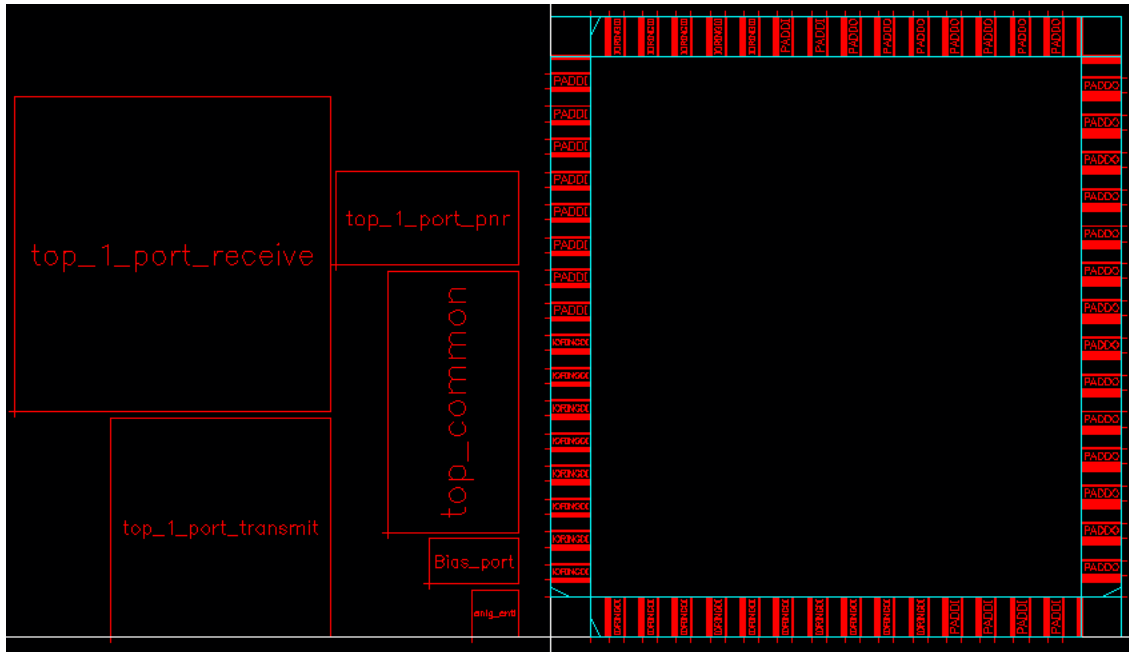
Filler Instance Attribute

Instance Name Prefix: Filler_ Mark Fillers As Fixed: ☐

OK Cancel Apply Help

2. In the Select Filler Cells section, select the Library Name, Cell Name, and View Name.
 - a. Use the *Filter* option to filter **giolib090/IoringFEED*/abstract** cells.
 - b. Add each of the **IoringFEED* abstract** cells to the Insert Filler Cells form.
 - c. Click  **Add Filler**.
The cell name is displayed in the Insert Filler Cells form.
 - d. Continue to select the individual **FEED** cells and add them.
3. To fill only a specific region, you can change the selection to *Selected Region*. However, for now, leave the selection to *Entire Design*.
4. Click **OK** in the Insert Filler Cells form.

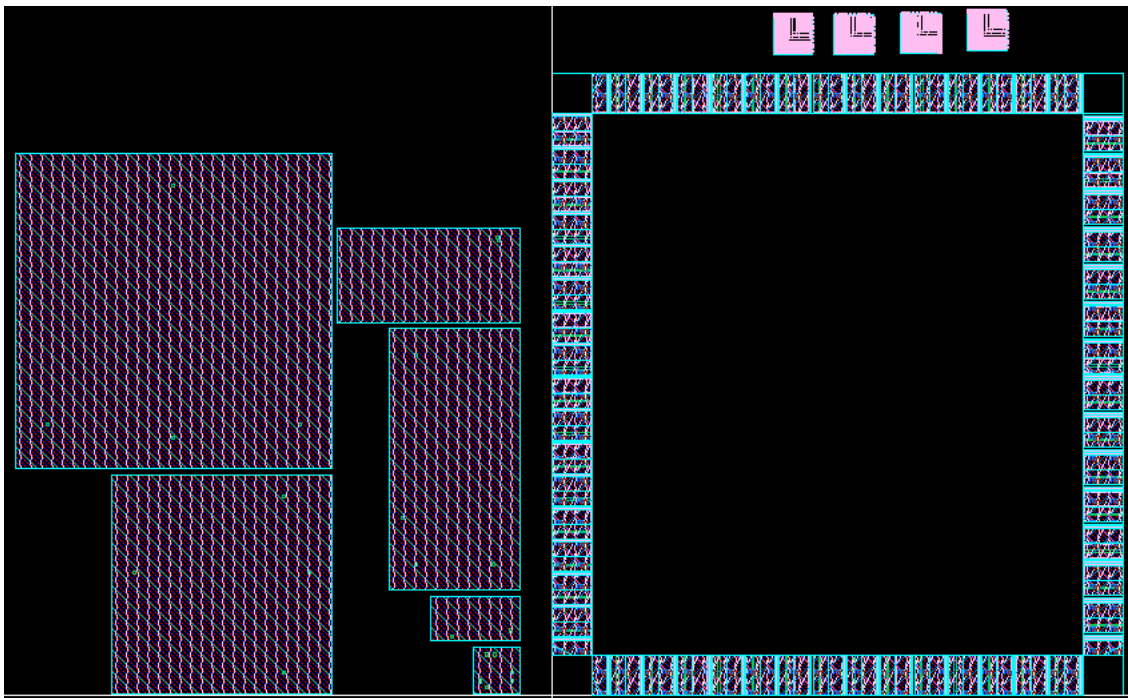
The fillers are inserted around the I/O ring, as shown here.



Using the I/O Placer to Place the Corner Cells

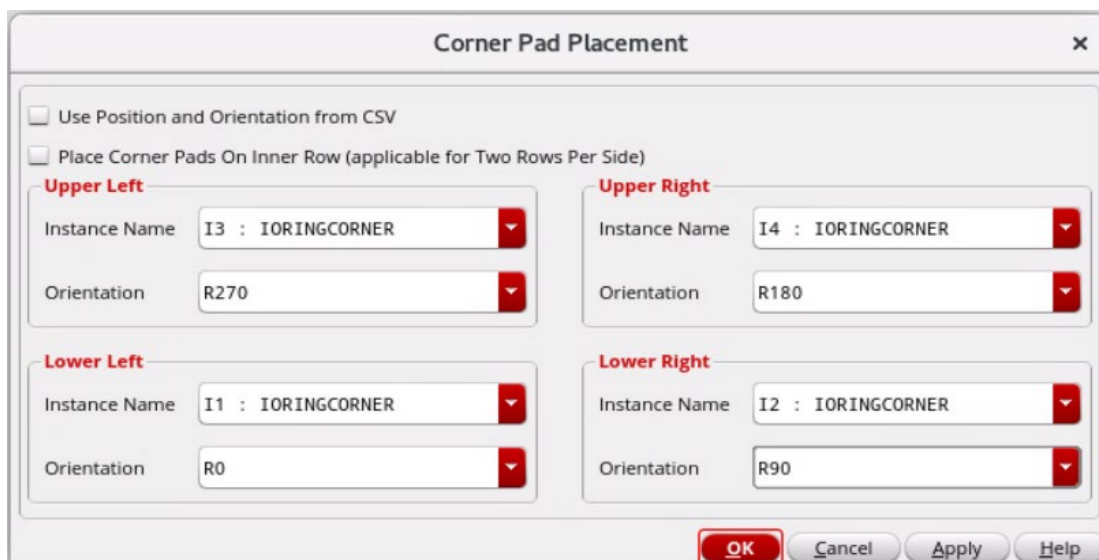
You need to place the corner cells in the design before applying the **Corner Pad Placer**.

1. Create four instances of **giolib090/IORINGCORNER/layout** above the *prBoundary*.



2. Select the four corner pads in the layout.
3. In the layout canvas, choose the **Plan – I/O Planning – Corner Pad Placer** menu command to place the corner pads.

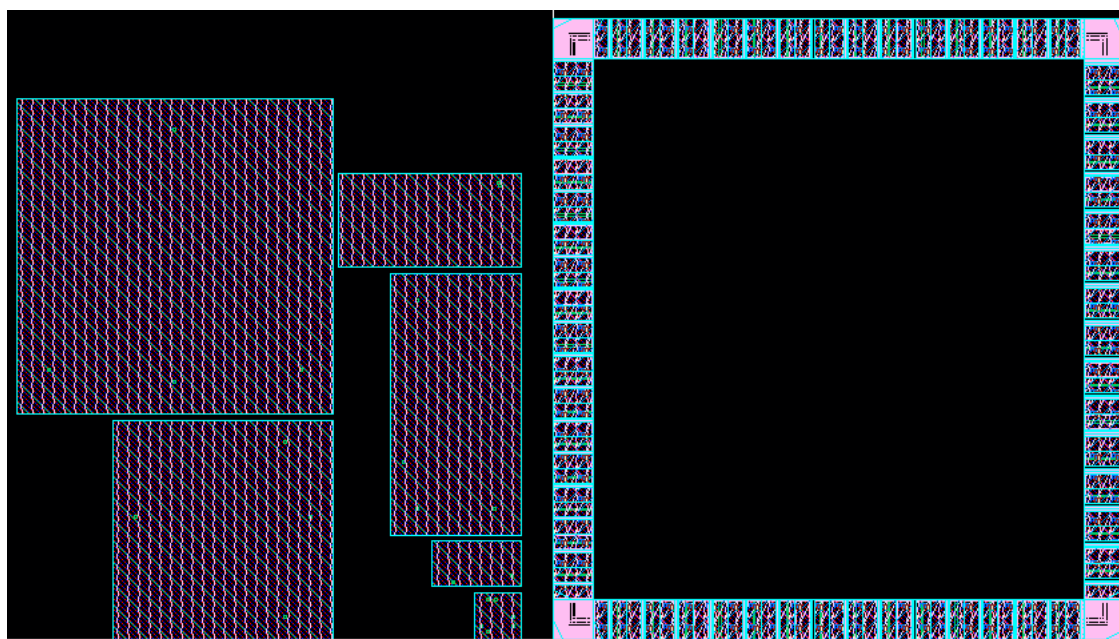
The Corner Pad Placement form opens.



The screenshot shows the 'Corner Pad Placement' dialog box. It has a title bar with a close button (X). Inside, there are two checkboxes: 'Use Position and Orientation from CSV' and 'Place Corner Pads On Inner Row (applicable for Two Rows Per Side)'. Below these are four sections for corner pads: 'Upper Left', 'Upper Right', 'Lower Left', and 'Lower Right'. Each section contains an 'Instance Name' dropdown (all set to 'I3 : IORINGCORNER', 'I4 : IORINGCORNER', 'I1 : IORINGCORNER', and 'I2 : IORINGCORNER' respectively) and an 'Orientation' dropdown (all set to 'R270', 'R180', 'R0', and 'R90' respectively). At the bottom right are 'OK', 'Cancel', 'Apply', and 'Help' buttons.

- a. Set **Orientations** as shown here to rotate the corner cells properly.
- b. Click **OK** in the Corner Pad Placement form.

The corner cells are placed around the I/O ring, as shown here.

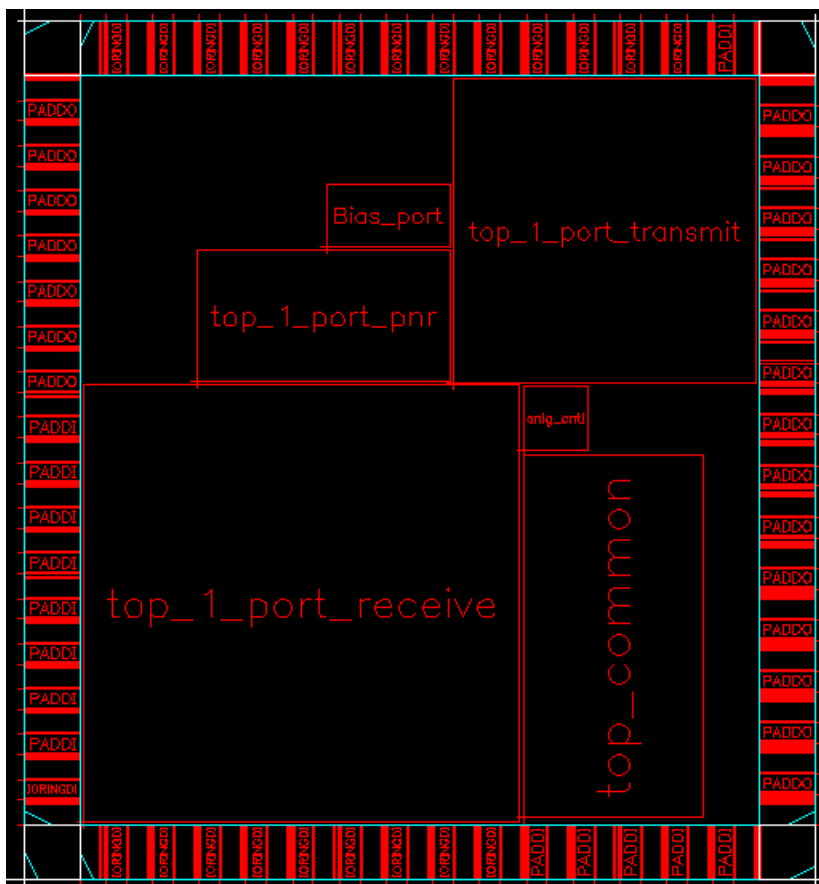


Placing the Blocks with Routable Solution

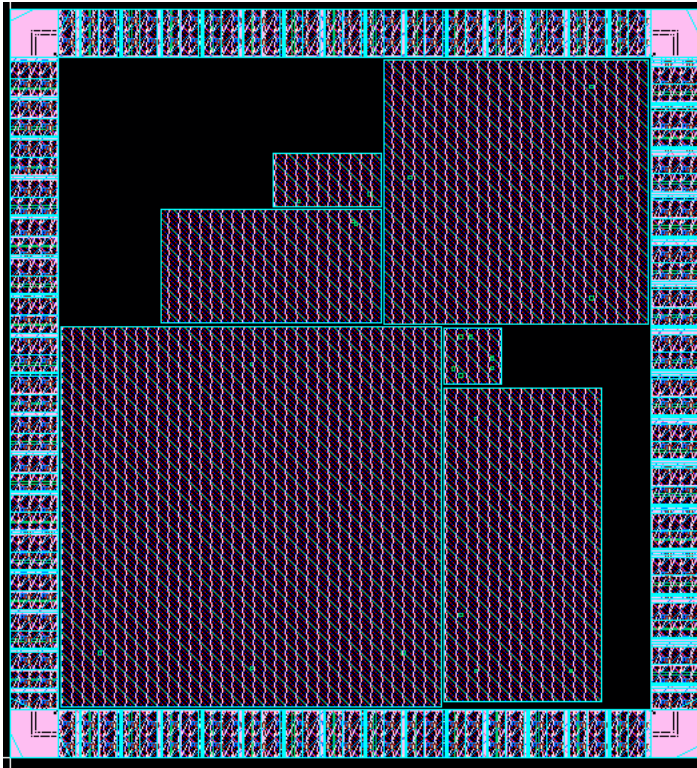
1. To set up the Block Placer form:
 - a. Set *Minimum I/O to Core Distance* to **10**.
 - b. Set *Default Minimum Distance between Instances* to **10**.
 - c. Set the level of *Effort* to **3**.
 - d. Set the *Number of Solutions* to **3**.
 - e. Deselect **Optimize Top-Level Pins**.

2. Click **OK** to start the block placer.

The run might take few minutes. The result shown is one possible result.



3. Press **Shift+F** to see the actual blockages in the abstracts.



Note: The output is looking more like a routable solution.

Refer to the CIW window for the Block Placer run details. You can refer to the bpLogs directory for the Block Placer log details.

4. Close all the cellviews without saving the changes and exit the Virtuoso session.

Summary

In this lab, you learned:

- ◆ How to create a top-down floorplan.

